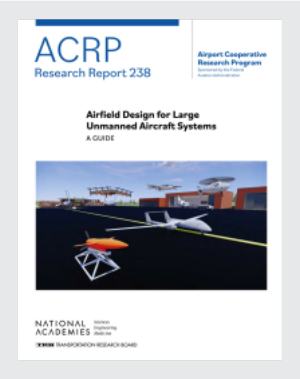
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SUGGESTED CITATION

National Academies of Sciences, Engineering, and Medicine. 2022. *Airfield Design for Large Unmanned Aircraft Systems—A Guide*. Washington, DC: The National Academies Press. https://doi.org/10.17226/26839.

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AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP RESEARCH REPORT 238

Airfield Design for Large Unmanned Aircraft Systems

A GUIDE

Thomas Mackie
Woolpert, Inc.
Dayton, OH

Subscriber Categories
Aviation • Operations and Traffic Management • Terminals and Facilities

Research sponsored by the Federal Aviation Administration

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AIRPORT COOPERATIVE RESEARCH PROGRAM

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ACRP RESEARCH REPORT 238

Project 07-18
ISSN 2572-3731 (Print)
ISSN 2572-374X (Online)
ISBN 978-0-309-68741-6
Library of Congress Control Number 2022946396

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Transportation Research Board
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AUTHOR ACKNOWLEDGMENTS

The research reported herein was performed under ACRP Project 07-18 by Woolpert, Inc.; Ricondo & Associates, Inc.; and Faith Group, LLC. Woolpert, Inc., was the contractor and fiscal administrator for this work.

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FOREWORD

By Theresia H. Schatz
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ACRP Research Report 238: Airfield Design for Large Unmanned Aircraft Systems—A Guide provides insight into airfield design issues, considerations, and challenges for the unique operational needs of large unmanned aircraft systems (L-UAS; weighing 55 lbs or more), considering safety and capacity at existing airfields of different types and sizes.

The guide addresses integration vs. segregation of operational areas at airfields as well as considerations for different L-UAS categories and capabilities, integration of technology (e.g., command and control systems, detect and avoid systems, cybersecurity, and infrastructure and utilities). It also addresses airport master planning, including economic and cost considerations, L-UAS support infrastructure (e.g., maintenance facilities, hangars, terminal, and fueling), environmental impacts (e.g., hazmat, noise, and battery storage), and approach surfaces and terminal airspace. The project did not address accommodations for space vehicles and spaceports.

L-UAS activity for civilian purposes continues to grow and expand as operators use L-UAS not only for surveillance, aerial photography, and infrastructure inspection, but also for disaster relief and commercial operations.

As airports consider introduction of L-UAS that need to use airports, a number of questions arise. It is unknown whether current airfield surfaces are adequate for use by L-UAS; whether there is a design impact on capacity and safety; whether airports should seek to integrate them with or segregate them from traditional aircraft operations; and, in communities with multiple airports, whether communities should integrate L-UAS at the primary commercial service airport or design another airport for L-UAS.

As indicated in *ACRP Research Report 212: UAS and Airports, Volume 2, Incorporating UAS into Airport Infrastructure Planning*, airport planners and engineers are considering whether and how to update master plans/airport layout plans for UAS. As technologies advance, UAS operators will find airfield design guidelines useful to their business and operational decisions. Many L-UAS operate in a manner similar to traditionally piloted aircraft and, therefore, need airfield facilities for their safe and efficient operation, but airfield facility planning guidance specifically tailored to address the unique needs of L-UAS is limited.

ACRP Project 07-18 was led by Woolpert, in association with Ricondo & Associates, and Faith Group, LLC. The research was conducted over a period of 18 months to address the challenges, issues, and concerns associated with introducing new L-UAS into the airport environment. The primary audience for this research consists of airport managers, planners, UAS operators, and other stakeholders.

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In addition to the research and technology studies, the research team collaborated with facilities and UAS industry experts in the United States who are actively operating L-UAS operations at airports. Interviews with a selected group of airports, organizations, UAS operators, and various subject matter experts from across the UAS and aviation industries were conducted. With the limited number of existing L-UAS operations at U. S. airports, and the number of uniquely designed L-UAS, it was crucial to call on those stakeholders who have experience within the L-UAS industry.

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SUMMARY

Airfield Design for Large Unmanned Aircraft Systems—A Guide

Introduction

The goal of this research was to provide insight into the challenges, issues, and concerns associated with introducing new large unmanned aircraft systems (L-UAS) aircraft types into the airport environment and to identify any associated gaps in Federal Aviation Administration (FAA) airport design standards. An overview of L-UAS types was completed using the Association for Unmanned Vehicle Systems International (AUVSI) Unmanned Systems and Robotics Database (USRD) as it existed in December of 2020. Interviews were also conducted with stakeholders and airports accommodating L-UAS flights to obtain an understanding of their experience and any lessons learned. The information from the USRD and the interviews was then examined in relation to FAA airport design standards to identify gaps, issues, and considerations for integration versus segregation of these aircraft at an airport and to identify transferable standards as appropriate where specific standards were not found. This section provides a macro-level summary of the findings and conclusions drawn from this research.

Airport Design

The primary guiding criteria for the design of facilities that accommodate aircraft and helicopters can be found in FAA AC 150/5300-13, *Airport Design*, and AC 150/5390-2C, *Heliport Design*. Most of the design standards for aprons, taxiways, and runways found in these documents are based on aircraft approach category (AAC), aircraft design group (ADG), taxiway design group (TDG), and specific aircraft dimensions. The USRD lacks some of this information, so categorizing L-UAS by all the appropriate design measures is difficult without direct communication with the original equipment manufacturer. L-UAS are extremely diverse in their performance and dimensional characteristics and, therefore, may not always conform to established FAA airport design standards.

L-UAS Categories and Capabilities

L-UAS vary in categories and capabilities and are ever evolving. The major categories are fixed wing (operate like traditional piloted aircraft); rotary wing (rotorcraft that operate like a helicopter); hybrid (lift-generating rotor to achieve vertical takeoff and landing and fixed wings to sustain flight); and other (e.g., lighter-than-air). These platform categories are important for understanding the aircraft's ability to integrate into the existing airport infrastructure and how existing design criteria will apply.

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L-UAS can be further categorized based on the following takeoff and landing capabilities:

- **Vertical takeoff and landing (VTOL).** Takeoff and landing are performed vertically without the use of a runway.
- **Horizontal takeoff and landing (HTOL).** Takeoff and landing are performed horizontally on a runway or strip of land.
- **Assisted.** Takeoff performed with assistance from an auxiliary device such as a pneumatic launcher, catapult, or rocket-assisted takeoff (RATO); landing done with parachutes or nets.
- **Combination.** Takeoff and landing using different or multiple methods or mechanisms.

Understanding the takeoff and landing mechanism and capabilities of L-UAS is essential because it helps determine if the aircraft will use the airport's runway or helipad or needs to be segregated because it requires specialized infrastructure.

L-UAS can operate with various types of propulsion systems (e.g., combustion engine, electric motor, turbojet engine, jet engine, or ducted fan) that are driven by different energy sources (e.g., aviation fuel, gasoline, battery, heavy fuel, diesel, kerosene, batteries, or hydrogen). Understanding L-UAS propulsion mechanisms and fuel types allows the airport sponsor to identify the infrastructure and safety measures required on the airfield to support aircraft refueling and storage.

Airport Component Analysis

Lighting, Marking, and Signage. Existing FAA standards on marking, lighting, and signage for airports and helipads are transferable to L-UAS. L-UAS will require case-by-case analysis to evaluate their situational awareness capability as well as their ability to detect the airfield or heliport marking, lighting, and signage and detect other aircraft operating on the surface. If they cannot, some means of augmented situational awareness may be required before integration is possible. There is a gap in guidance on sensor acuity requirements for L-UAS.

Hangars and Support Facilities. Existing FAA standards for building setbacks and safety areas are transferable to L-UAS aircraft as they relate to wingtip and tail height clearances. The ADG applied to the airport's existing hangar taxilanes is a major limiting factor for storing most L-UAS. Additionally, the hangar door widths and heights also determine what L-UAS sizes/types can be stored on an airport. Gaps exist in guidance for the storage of different energy sources that are or may be used for different L-UAS platforms.

Pavement Strength. L-UAS maximum takeoff weight (MTOW) is a key factor when considering integration into an airport environment. Most L-UAS on the market today have MTOWs that fall below those defined for large aircraft; therefore, existing pavement design standards should be transferable when considering the integration of L-UAS. Airport operators need to take additional factors into consideration when accommodating L-UAS over 12,500 lb. Another consideration is that the precision of L-UAS landings can result in a very consistent landing location that may make one pavement area wear more quickly than experienced with traditionally piloted aircraft, which have a larger scatter during landing. The FAA's pavement design software, FAARFIELD, does not include data for L-UAS aircraft, so landing gear configuration and MTOW need to be evaluated for the specific L-UAS operating at the airport and entered directly into the software.

Airfield Surface Infrastructure. There is a gap in guidance on the AAC classifications for L-UAS, along with TDG and maximum allowable crosswind components as they relate to airfield design. Existing runway design standards apply to L-UAS that require an HTOL

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surface. However, the basis of runway design derives from the AAC and ADG, where ADG is based on the aircraft wingspan and tail height and AAC is based on the approach speed. Approach speed data on L-UAS are not readily available and will likely need to come directly from the original equipment manufacturer (OEM). If an engineered material arresting system (EMAS) is installed on an airport, it should be evaluated for its ability to arrest an L-UAS.

Taxiway design standards are tied to the TDG and ADG of an aircraft. OEM landing gear configurations will need to be considered, along with onboard sensor capabilities and operational characteristics, as they may influence if and how an L-UAS uses the taxiway. The dimensions of L-UAS should be considered when parking these vehicles, especially if they are bigger or taller than the existing design aircraft.

While not helicopters, L-UAS using a VTOL platform fly similarly, so FAA heliport standards are transferable to some degree with certain adjustments for aircraft size.

Navigational Aids. The primary source of navigational guidance for L-UAS is space-based navigation from Global Positioning System (GPS) and Inertial Navigation Systems (INS). A review of the USRD and responses captured during the stakeholder interviews revealed no impacts to the necessity, design, location, or application of traditional ground-based navigation systems.

Communication and Data Systems. The current criteria outlining the requirements for compliant communication and data link systems do not present gaps when introducing L-UAS because the aircraft and operators are expected to use the same networks as traditionally piloted aircraft. The airport sponsor should coordinate closely with FAA air traffic control (ATC) to understand the systems and protocols unique to the various L-UAS to ensure appropriate infrastructure exists at the airfield to support their missions. Furthermore, some L-UAS missions rely on additional support from internet-based systems, so the airport's broadband utility infrastructure may be affected.

Fueling and Charging. The fuel source for L-UAS does not present an issue with integrating the operation of L-UAS into the traffic flow of the runway or the taxiway. However, segregation on the ramp for servicing or refueling and in hangars for storing may be warranted based on power sources, servicing needs, and thermal runaway potential. There is a gap in guidance on the handling of alternative fuel sources on an airport, fire protection response, and electromagnetic interference as it relates to alternative fuels, especially lithium-based battery packs and their charging.

Obstacles and Airspace. The existing standards for airports and heliports prescribed in 14 CFR Part 77—Safe, Efficient Use, and Preservation of the Navigable Airspace allow L-UAS to interact with the airport like traditional aircraft if they meet the same performance capability. If they do not, alternative takeoff and landing areas may be necessary. Specific references to L-UAS obstacle identification, protection, and evaluation are not in FAA design standards; however, the need for them to be specified is not considered critical.

Noise and Environmental Impacts. Whether integrated or segregated, L-UAS are not hindered from operating at existing airports based on environmental factors any more than traditionally piloted aircraft. While there is a gap in information available to assess L-UAS aircraft in FAA's *Aviation Environmental Design Tool* (AEDT) model, FAA Office of Environment and Energy can provide guidance to an AEDT user on an appropriate FAA-approved substitution aircraft. L-UAS that use alternative energy may present new

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and unique environmental concerns, like a hazardous waste spill if an aircraft's lithium-ion battery pack were to leak.

Airport Passenger Terminals. A clear need for L-UAS to have access to a commercial airport terminal was not uncovered in this research; however, integrating them into the airport terminal could be a consideration. The need for terminal connectivity occurs mainly in the apron/ramp area for preflight and postflight procedures or loading and unloading. While there is a gap in information for passenger-carrying L-UAS, this is primarily because there are no such aircraft with FAA approval. The information does exist in terminal development related to traditional passenger aircraft that can provide guidance. This information should be paired with detailed aircraft specifications from an L-UAS manufacturer for any future passenger-carrying UAS.

Air Traffic Control. L-UAS operations conducted at airports in the national airspace system (NAS) are integrated using existing infrastructure types, procedures, and regulations governing traditional flight operations. Airports will need to remain vigilant and knowledgeable about the advancing uncrewed traffic management solutions, L-UAS aircraft technology, and changes to FAA policies and regulations required to maintain the level of safety ATC facilities provide.

Aircraft Rescue and Firefighting. A gap exists in guidance specific to non-traditional aviation fuel sources; when ignited, they act much differently than typical aviation fuels. Therefore, guidance specific to L-UAS and their internal safeguards, battery composition, and fuel sources is appropriate, along with training all first responders on the firefighting techniques needed for any new fueling sources introduced by these aircraft. Guidance can be found from other industries until FAA provides aviation-specific recommendations.

Conclusions on Integration Versus Segregation

Fixed wing L-UAS are the most straightforward to integrate if they operate like traditional airplanes and require a runway for takeoff and landing. Their integration heavily depends on the aircraft size/weight and its need to use existing runway/taxiway surfaces for movement, with the heavier aircraft integrating more easily than the lighter ones. Aircraft that operate with rotors during all phases of flight do not necessarily need a runway for takeoff and landing. Their taxi capabilities vary, so integration with the runway and taxiway system may be limited. Hybrid aircraft vary between fixed wing and rotorcraft during either takeoff or landing and their need for a runway or ability to use a taxiway will impact their integration ability. Assisted Launch and Recovery aircraft will not likely be able to use the existing runways, taxiways, and ramps because they require a special apparatus for launch and recovery. Segregation is probable for these aircraft. The uniqueness of the various L-UAS vehicles will require the airport sponsor to understand their specific requirements to determine how existing infrastructure can support their operations.

L-UAS have various propulsion system types using liquid fuels and renewable energy. L-UAS that use other than traditional aviation fuels may require additional consideration to determine the appropriate safety measures to support these aircraft. Electrically propelled L-UAS primarily impact airport infrastructure with their need for high-voltage electrical power and aircraft-specific charging systems. Additional risks are introduced by the hazardous materials their batteries contain, which can leak, explode, or catch fire.

Each airfield surface layout and L-UAS type poses unique issues for determining the level and locations of integration on the airport surface. Airports with multiple movement surfaces

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will increase integration potential by allowing the best option for each mission according to the user as needed. Since there is no pilot on board, L-UAS have visual capabilities that differ from those of traditionally piloted aircraft. Properly equipped L-UAS should be able to use an airport's existing systems to attain the situational awareness necessary to operate or have equivalent mechanisms of attaining it.

If the airport demand is high while capacity is low, integrating L-UAS will be more difficult, and therefore, segregation may be needed. An L-UAS equipped with the proper avionics (e.g., VHS communication radio capability, transponder, automatic dependent surveillance-broadcast [ADS-B], and GPS) should be able to operate and communicate with ATC in the same way as traditionally piloted aircraft. The need for on-field ATC services is dependent on the level of traditional aviation service at each airport and not driven by L-UAS specifically. If an L-UAS does not have detect and avoid (DAA) capability, integration is easier with an air traffic control tower (ATCT) that can provide aircraft separation services.

The precision of L-UAS landings can result in a very consistent landing location that may make one pavement area wear more quickly than experienced with traditionally piloted aircraft, which have a larger scatter diagram during landing. The dimension of L-UAS should be considered when parking these aircraft, especially if they are bigger or taller than the existing design aircraft.

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CHAPTER 1

Introduction

The transportation system within the United States provides a safe, efficient, and reliable means to move people and goods between locations with mature, socially acceptable, and connected modes (i.e., surface, marine, and airborne networks). Within each mode of transportation, government agencies and the respective industries continually explore technology and methods to increase throughput, mitigate safety risks and environmental impacts, and expand the functionality of the existing infrastructure. The explosion of unmanned aircraft systems (UAS) technology, and its seemingly endless applications, offers society a broad spectrum of enhancements and solutions to many problems.

Over the past several years, FAA has focused on integrating small UAS (less than 55 pounds in weight) into the national airspace system (NAS) to conduct and support routine tasks or develop new and inventive ways to tackle work. Technology, procedures, and regulations in this area have matured through countless research and development activities conducted by various agencies [e.g., Departments of Transportation (DOTs), the National Aeronautics and Space Administration (NASA), the Department of Defense (DOD), academia, and private industry]. Meanwhile, the emerging market of advanced air mobility (AAM) has simultaneously pushed the development and integration of aviation innovation (e.g., small drones, electric aircraft, and new air traffic management systems) into existing air transportation networks, impacting aviation's role in the movement of goods and people. The next stage, uncrewed systems weighing 55 lb or more, known as large UAS (L-UAS) (Figure 1-1), is now upon us as these aircraft begin to find their way to America's airports.

This guide provides airport sponsors with a fundamental understanding of L-UAS and their typical supporting infrastructure needs to assist with determining potential enhancements to support L-UAS operations at their facilities. It will help airport sponsors to identify the challenges L-UAS may present to their airports, gaps in current airport design standards related to these aircraft, and potential transferable guidance that can be followed until these gaps are filled by the regulatory agencies.

Research Goals and Objectives

The goals of this research were to provide insight into the challenges, issues, and concerns associated with introducing new L-UAS types into the airport environment and to identify any associated gaps in FAA airport design standards. The guide is directed at airport sponsors, managers, planners, UAS operators, and other airport/UAS stakeholders involved in the operation of L-UAS and addresses key considerations and gaps in existing design criteria in the following areas (Figure 1-2):

- Integration versus segregation of L-UAS on operational areas of the airfield,
- UAS categories and capabilities,
- Integration technology,

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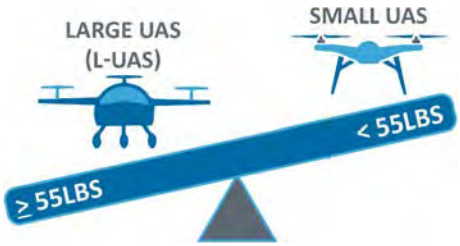


Figure 1-1. Large UAS (L-UAS) and small UAS.

- Airport planning and design issues for federally obligated airports in the National Plan of Integrated Airports,
- UAS support infrastructure,
- Environmental concerns, and
- Approach surfaces and terminal airspace issues.

UAS Industry—Current State

The UAS industry is rapidly advancing and evolving and is increasingly able to enhance the operations of many industries and support many applications. The small UAS (aka drones) are the most understood uncrewed aircraft systems, and they have been adopted by many industries at an unprecedented rate, providing positive impacts to safety, quality, efficiency, and security. Growth in the small UAS market may be setting a precedent for growth in the L-UAS market. The global UAS civil market is estimated to grow from $5 billion in 2019 to $14.5 billion by 2028 (Finnegan 2019). This forecast demonstrates continued focus on the research, development, and manufacturing of L-UAS technology. The forecasted growth accounts for aircraft, control systems, sensor technology, surveillance, and traffic management systems.

The forecasted metrics available for revenue and volume of UAS operations primarily focus on small UAS but likely include several aircraft and applications for L-UAS that narrowly breach the weight threshold of 55 pounds. Without available statistics or studies stratifying the



Figure 1-2. Key considerations addressed in the guide.

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entire UAS market into more finite levels, it can only be assumed that most of the available data address expected growth for a class of aircraft generally not requiring significant infrastructure to function; they can be considered self-contained operations. Regardless, the indicators are clear that UAS, their payloads, applications, and the technology surrounding their operations have arrived and will only continue to grow (Cohn 2017).

With the success of the lighter L-UAS, the focus is shifting to applications for and integration of heavier aircraft into the NAS. Those that need significant infrastructure to take off and land are the ones most likely to operate at an airport. In addition to L-UAS designed to be uncrewed, existing fixed wing aircraft are being retrofitted and tested to operate uncrewed. These L-UAS are being designed for new applications but also to replace existing traditional aircraft routes (i.e., cargo feeder vehicles).

Other segments of the industry gaining some of the greatest traction now are the advanced air mobility (AAM), urban air mobility (UAM), and electric vertical and takeoff (eVTOL) sectors of L-UAS. These sectors are driving forecasts and account for a significant influx of aircraft, which will drive the use of airspace and ground facilities (i.e., vertiports) in locations where aviation infrastructure has not previously been required, as demonstrated in the NASA UAM Vision Concept of Operations (NASA 2020). This guide does not cover new and unique facilities to support those sectors but directs attention to facilities and airspace that are common to and protected for aviation use—i.e., airports.



Figure 1-3. Focus of the guide.

Focus of the Guide—L-UAS

This guide focuses on where gaps and challenges reside related to the introduction of L-UAS into the airport operating environment. Airport infrastructure has traditionally been planned and designed according to criteria based on a known fleet of fixed wing aircraft classified by aircraft approach category (AAC), aircraft design group (ADG), and various aircraft dimensions. When new aircraft enter the market for traditionally piloted operation, they usually fit within one of the categories and groups where defined criteria exist. This research studied a fleet of new aircraft differentiated from the established fleet of traditional aircraft in speed, configuration, and the key characteristic of being uncrewed. Examples of these L-UAS aircraft and their capabilities are summarized in Chapter 2 of this guide, with more specific details provided in Appendix A.

This guide focuses on federally obligated airports that have received FAA Airport Improvement Program funds. These airports are obligated to follow FAA criteria for airport design and operation. This guide addresses if and how L-UAS can be integrated or segregated at these facilities. It focuses on the abilities of and challenges for these aircraft to intermingle with existing aircraft and points out where segregation may be the best approach, with the following emphases (Figure 1-3):

- Unique characteristics of L-UAS and their supporting technology,
- Levels of impacts on applicable airfield infrastructure standards,
- Key indicators for integration versus segregation, and
- Important considerations around airfield assets.

Roles of Airports and Stakeholders in Integration of L-UAS

The integration of L-UAS into traditional airport operations requires several stakeholders to understand UAS technology, operating models, regulation/policy, and infrastructure. Determining when an airport is essential to support L-UAS operations is a function of this understanding.

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Each stakeholder plays an important role in the successful integration of L-UAS into the airport environment, with each being responsible for its specific function regarding its operation.

- **Role of the Airport.** Airports offer the aviation industry the ideal infrastructure, facilities, and services to enable aircraft operations. Airports will likely become the central focus for many L-UAS operations because of their elementary capability to provide runways, taxiways, parking, maintenance, fueling, emergency services, air traffic management, surveillance, and airspace protection that help to ensure safe and secure operations. In this light, airport sponsors are facility "suppliers" who should evaluate their facilities for L-UAS operations as "demand" warrants.
- **Role of the UAS Manufacturer.** Current interactions between L-UAS manufacturers and airports are minimal, essentially occurring only when coordination is required for testing or specific vehicle operations (e.g., military operations). As the industry advances and operations at airports increase, manufacturers should coordinate with airport operators on any unique aspects of their aircraft that require special support to operate in the airport environment.
- **Role of the UAS Operator.** Operators are responsible for the safe and efficient operation of L-UAS, routinely coordinating with the airport sponsor on the characteristics of their operations and seeking related improvements. Communication on specific vehicle and system facility support needs, air traffic management coordination and authorizations, and the appropriate emergency service capabilities must occur between the operator and airport sponsor to confirm the ability of the various L-UAS to operate at a specific airport.
- **Role of the Planner or Designer.** Professionals supporting airport planning and design will need to be fluent in pertinent criteria and guidance required to maintain compliance with FAA regulations and to address L-UAS infrastructure needs and environmental impacts at each specific airport.

Guide Structure

One of the primary purposes of this guide is to identify the FAA airport design criteria applicable to L-UAS and recommend when integration versus segregation of their operations is appropriate. Accordingly, this guide is structured to cover the following material:

- L-UAS technologies and their operational components in the marketplace,
- Components of an L-UAS related to airport infrastructure or operational considerations,
- Effects and considerations to determine integration versus segregation of L-UAS at an airport,
- Overview and references to the foundational components of airport designs, and
- Summary of the most impactful findings for airport sponsors to consider as they relate to L-UAS integration into the airport environment.

The organization of this guide is as follows (Figure 1-4):

Summary. The summary provides an overview of the key findings and challenges discovered for the applicability of FAA airport design standards to L-UAS on the market at the time of the study.

Chapter 1: Introduction. This chapter introduces the goals and objectives of the research, the role each stakeholder plays in L-UAS integration into the airport environment, and the structure of this guide.

Chapter 2: L-UAS Technology Review. This chapter offers an overview and summary of L-UAS on the market at the time of this study, grouping them by platform, launch and recovery mechanism, and propulsion type. It ends with a comparison of this technology to the FAA's "critical aircraft" groupings used in airport design.

Chapter 3: Airfield Component Impact Analysis. This chapter presents an analysis of the gaps in the existing design criteria's ability to accommodate L-UAS. The chapter is arranged by

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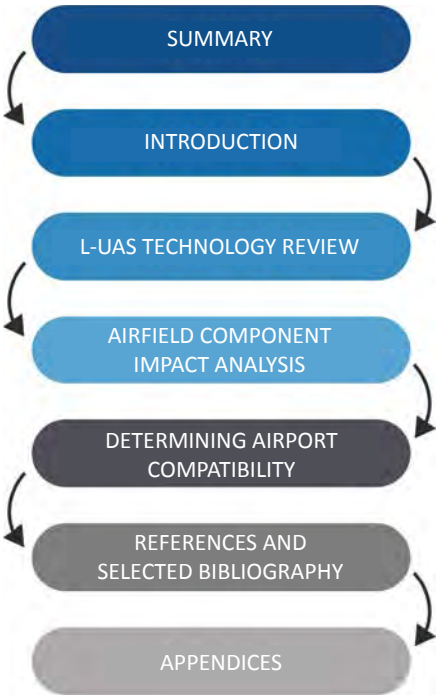


Figure 1-4. Organization of this guide.

the major components or functions of the airport and includes information on the integration potential of L-UAS to each specific airport component and references to the associated or transferable standards.

Chapter 4: Determining Airport Compatibility. This chapter provides high-level guidance on the complexity of the decision to integrate or segregate L-UAS from traditionally piloted traffic and the possibilities and advantages of both approaches.

References and Selected Bibliography. This section offers references to support the findings and provide additional guidance for assessing the incorporation of L-UAS into an airport's operating environment.

Appendices. This section includes a selected inventory of the L-UAS listed in the Association for Unmanned Vehicle Systems International (AUVSI) Unmanned Systems and Robotics Database (USRD) and reviewed for this project, a summary of the interviews conducted with stakeholders and airports accommodating L-UAS flights, acronyms and abbreviations, and a glossary.

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CHAPTER 2

L-UAS Technology Review

Introduction

L-UAS, which generally operate in the airspace above 400 feet in altitude and weigh 55 pounds or more, are entering service and could affect the air transportation system. L-UAS can support cargo shipping, disaster response, geospatial mapping, security and surveillance, passenger transportation, and other civilian, commercial, and military applications. Beyond the highly publicized air traffic-related aspects of integrating L-UAS into airport operations, the airfield's infrastructure design requirements are critical considerations for airport sponsors. This chapter includes a review of L-UAS configurations and capabilities and a comparison of these to the traditional aircraft design groupings (AAC and ADG) and runway design codes used in airport design included in AC 150/5300-13, *Airport Design*.

L-UAS Categories and Capabilities

The research team relied on AUVSI to capture the sheer volume of L-UAS that operate today. AUVSI is the world's largest nonprofit organization dedicated to the advancement of uncrewed systems and robotics. This organization represents corporations and professionals from more than 60 countries. Members are involved in defense, industries, government, and academia.

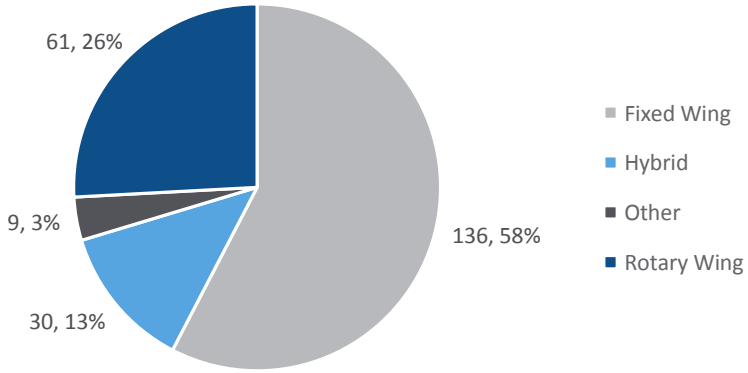
AUVSI maintains the USRD containing performance characteristics and dimensional data for almost 3,740 unique UAS platforms. The research team relied on this expansive database to capture the various types of L-UAS in existence at the time of this study (December 2020). From the database, the research team isolated 236 unique UAS platforms weighing 55 lbs or more listed as "active" in the United States. These 236 unique platforms represent L-UAS in or anticipated to enter the NAS. Available performance and dimensional data for these L-UAS are presented in Appendix A. These L-UAS platforms are used as the basis for this research.

An analysis of L-UAS active in the U.S. as identified in the USRD reveals a wide variety of UAS types. Based on operating characteristics, these L-UAS platforms fall into four main categories: fixed wing, rotary wing, hybrid, and other, as shown in Figure 2-1. The L-UAS reviewed in the USRD are binned by industry application as in Figure 2-2. These platforms are important for understanding the aircraft's ability to integrate into the existing airport infrastructure and how existing design criteria will apply.

Fixed Wing Platforms

Fixed wing L-UAS resemble traditional-looking airplanes having a fuselage, wings, propulsion source, and stabilizers. Fixed wing UAS platforms are designed to fly for long durations with increased stability over long distances. They take various configurations for wing shape

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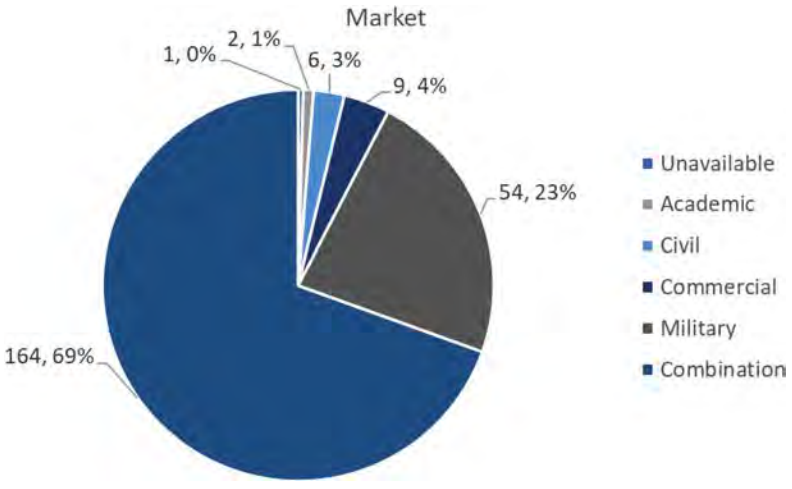
Source: AUVSI USRD, Accessed 10/2020. Prepared by Woolpert.

Figure 2-1. L-UAS categories.

including rectangular, tapered, or rounded straight wings; high, medium, or low swept; and delta wing designs. Of the 236 L-UAS identified in the USRD, 136 (58%) fall into the fixed wing category. Analysis of these aircraft identified wingspan dimensions as small as 2 feet and as wide as 243 feet, lengths up to 50 feet, heights as tall as 17 feet, and MTOWs as heavy as 61,800 pounds. These variances are illustrated in Figures 2-3 through 2-6, below.

Rotary Wing Platforms

Rotary wing L-UAS platforms are equipped with lift-generating rotors to provide vertical takeoff and landing (VTOL) capabilities to sustain flight. In principle, rotary wing platforms operate like helicopters and can have a single, multiple, co-axial, or tilt-rotor configuration. These platforms tend to operate in much more confined areas because they can hover. Sixty-one of the aircraft (26%) in the USRD fall into the rotary wing category with wingspans, or total rotor diameters, ranging between 5 feet and 49 feet. The lengths for these aircraft are up to 57 feet, while heights are as tall as 15 feet and MTOWs as heavy as 12,000 pounds. Variances



Source: AUVSI USRD, Accessed 10/2020. Prepared by Woolpert.
Note: Combination category represents a combination of civil, academic; civil, commercial; civil, commercial, academic; civil, commercial, military; civil, commercial, military, academic; civil, military; civil, military, academic; and commercial, military.

Figure 2-2. L-UAS market breakdown.

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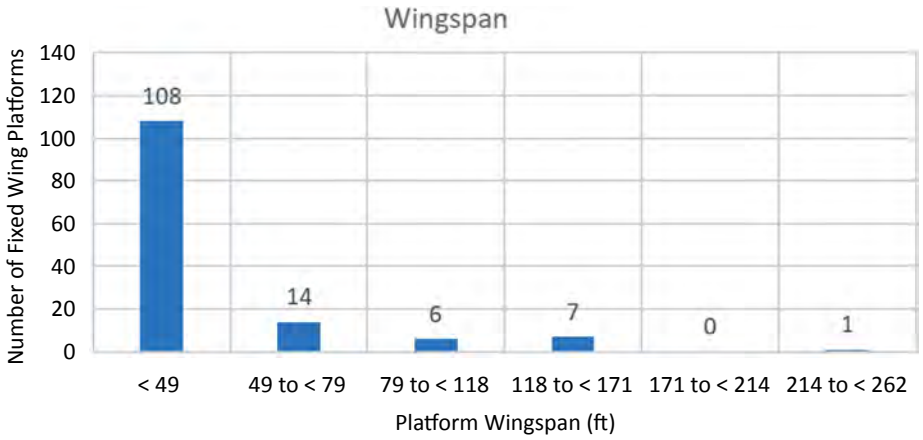


Figure 2-3. Wingspan of fixed wing L-UAS platforms.

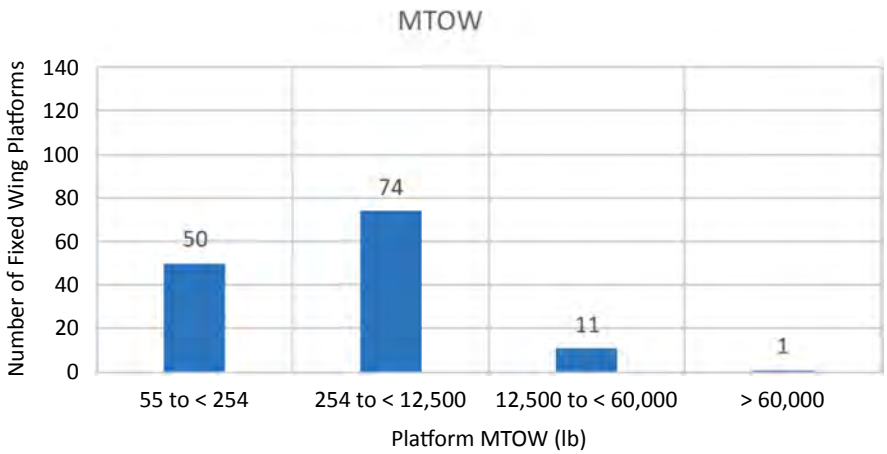


Figure 2-4. MTOW of fixed wing L-UAS platforms.

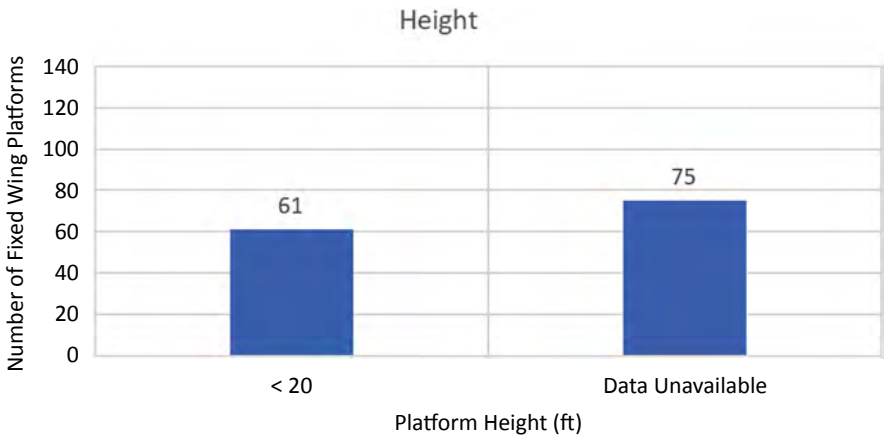
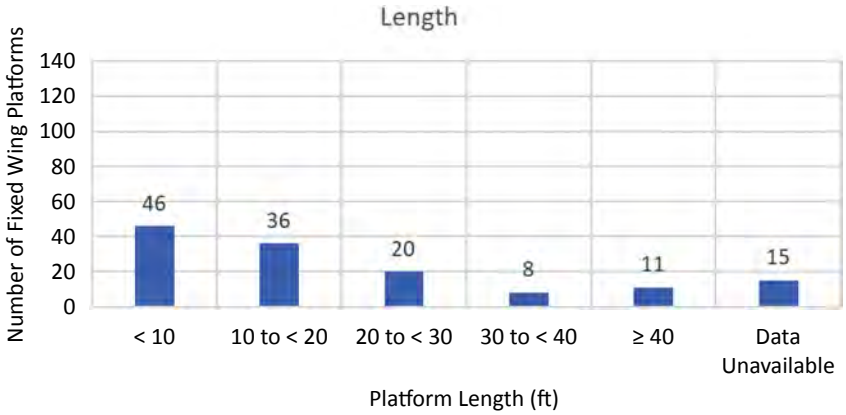


Figure 2-5. Height characteristics of fixed wing L-UAS platforms.

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Source: AUVSI USRD, accessed 10/2020. Prepared by Woolpert.
Note: Total number of fixed wing platforms is 136.

Figure 2-6. Length characteristics of fixed wing L-UAS platforms.

found in the dimensions and performance data for rotary wing L-UAS platforms are illustrated in Figures 2-7 through 2-10, below.

Hybrid Platforms

Hybrid platforms encompass elements of both fixed wing and rotary wing platforms. These systems typically use a lift-generating rotor to achieve VTOL and fixed wings to sustain flight. Operational characteristics for these systems are much more flexible, as a hybrid platform eliminates the need for a horizontal takeoff or landing surface. Thirty platforms (13%) in the USRD fall into the hybrid category. Their wingspans range from 4 feet to 65 feet, their overall lengths are up to 28 feet, and their MTOWs are as heavy as 29,500 pounds. Variances are shown in Figures 2-11 through 2-14, below.

Other Platforms—Statistical Outliers Removed from Analysis

Analysis of all L-UAS platforms identified in the USRD revealed various outliers classified in the "Other" category. These platforms do not fit into the fixed wing, rotary wing, or hybrid categories shown in Figure 2-1. Included in this "Other" category are systems such as lighter-than-air concepts with limited applications for military, civil, and commercial users. Only nine of the platforms (3%) in the USRD fall into the "Other" category.

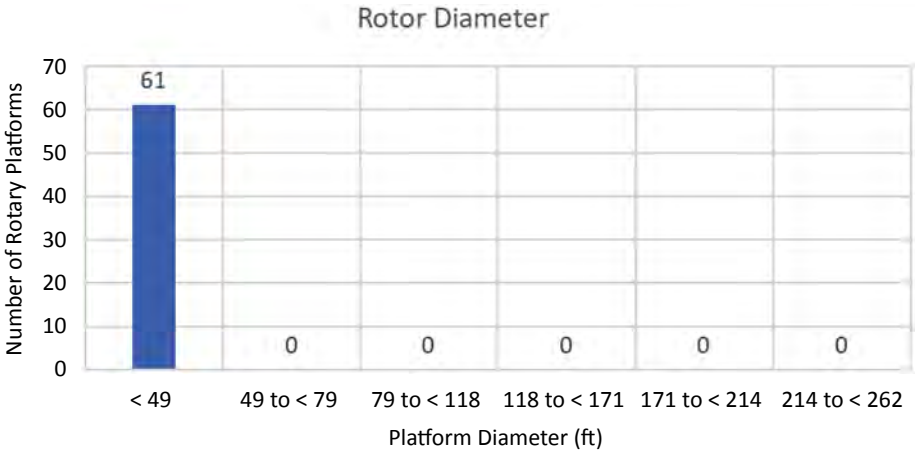


Figure 2-7. Diameter of rotary wing L-UAS platforms.

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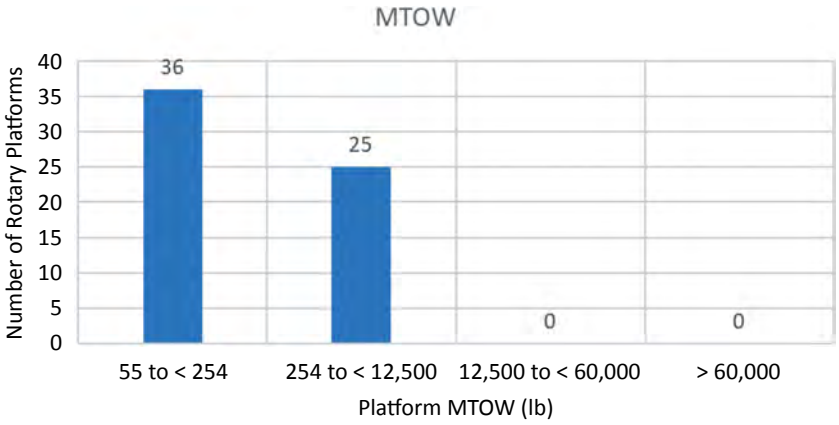


Figure 2-8. MTOW of rotary wing L-UAS platforms.

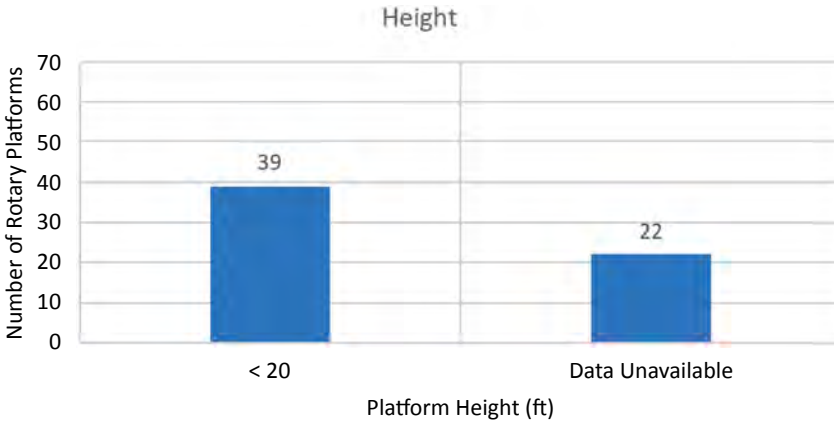
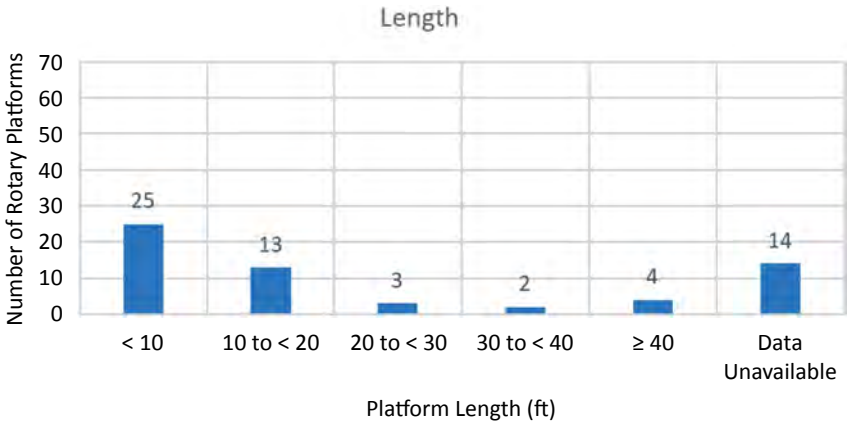


Figure 2-9. Height of rotary wing L-UAS platforms.



Source: AUVSI USRD, accessed 10/2020. Prepared by Woolpert.
Note: Total number of rotary wing platforms is 61.

Figure 2-10. Length of rotary wing L-UAS platforms.

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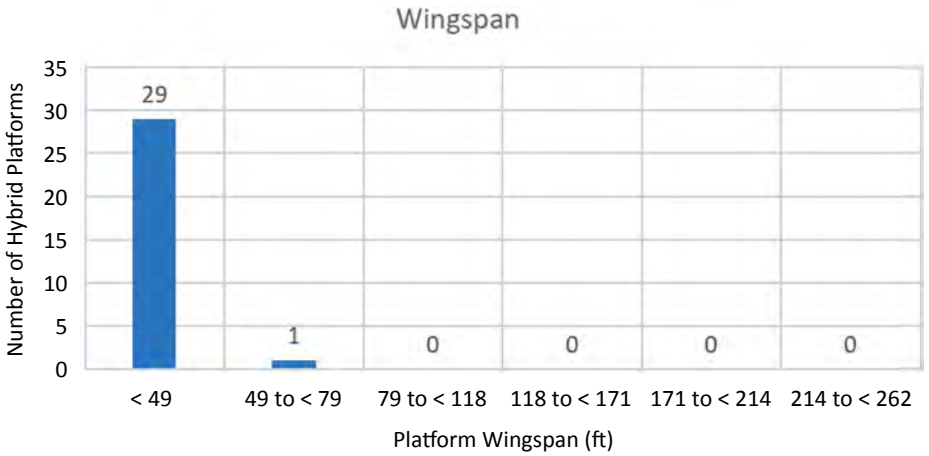


Figure 2-11. Wingspans of hybrid L-UAS platforms.

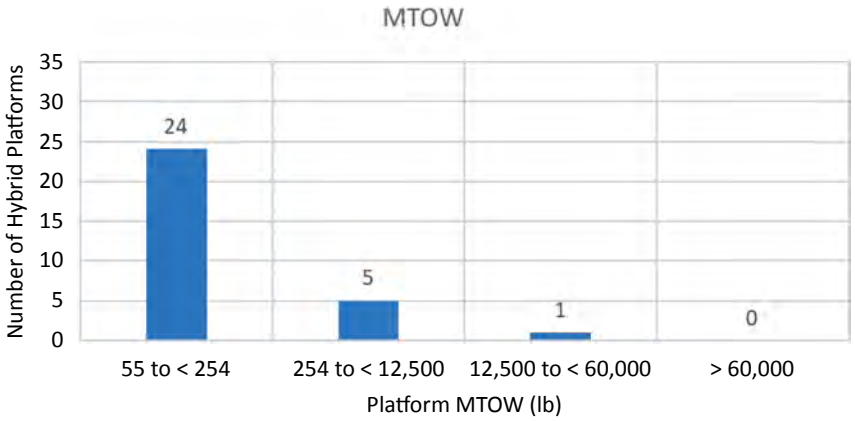


Figure 2-12. MTOWs of hybrid L-UAS platforms.

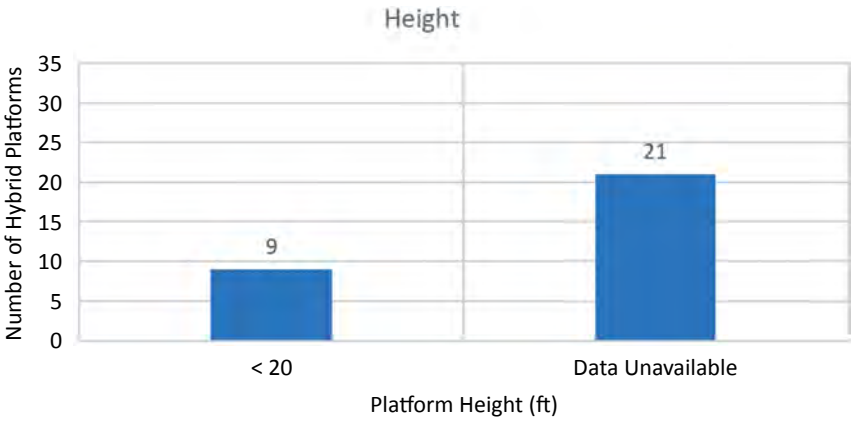
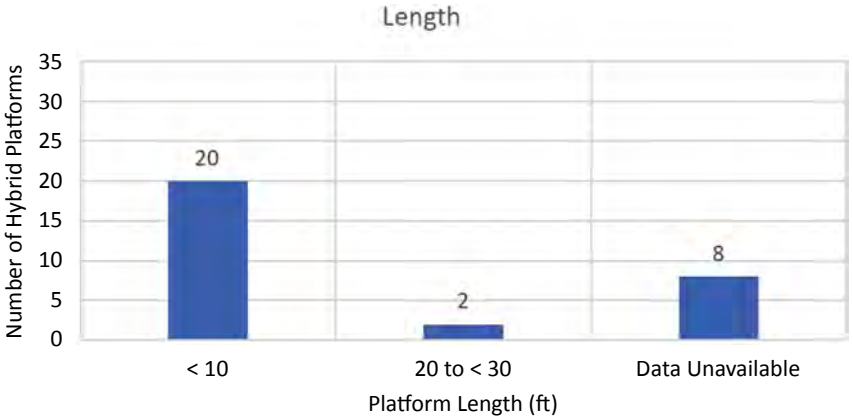


Figure 2-13. Heights of hybrid L-UAS platforms.

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Source: AUVSI USRD, accessed 10/2020. Prepared by Woolpert.
Note: Total number of hybrid platforms is 30.

Figure 2-14. Lengths of hybrid L-UAS platforms.

Platforms included in the "Other" category have wingspan dimensions ranging from 4 to 110 feet, lengths up to 200 feet, and MTOWs up to 10,000 pounds. These platforms support niche applications and do not typically operate in an airport environment. Therefore, platforms in this category were excluded from this research. Examples of these outliers included the AB5 JetQuad, a four-jet engine-propelled rectangular platform manufactured by FusionFlight, Inc., and the HiSentinel 80, a lighter-than-air uncrewed rigid airship built by Aerostar/Southwest Research Institute.

Launch and Recovery Mechanisms

In addition to the platform categorization of fixed wing, rotary wing, hybrid, or other, the USRD includes information on the L-UAS takeoff and recovery mechanisms. This information is important because it helps determine if the aircraft will use existing airfield infrastructures (such as the runway or helipad) or be segregated owing to their need for specialized infrastructures. Typically, the takeoff or recovery mechanisms for L-UAS fall into four categories (see Table 2-1).

Launch Mechanisms

Figure 2-15 provides a breakdown of the various types of launch mechanisms as identified from the analysis of the USRD. The majority fall within the VTOL category (42%), followed

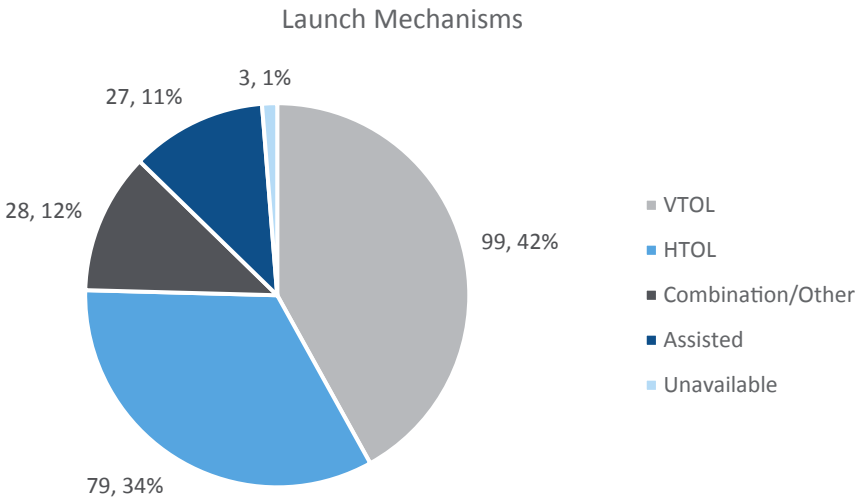
Table 2-1. Categories for L-UAS launch and recovery mechanisms.

Vertical Takeoff and Landing (VTOL)	Takeoff and landing are performed vertically without the use of a runway.
Horizontal Takeoff and Landing (HTOL)	Takeoff and landing are performed horizontally on a runway or strip of land.
Assisted	Takeoffs are performed with assistance from an auxiliary device such as a pneumatic launcher, catapult, or rocket-assisted takeoff (RATO); landings are performed with parachutes or nets.
Combination	Takeoff and landing using different or multiple methods or mechanisms.

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Source: AUVSI USRD, accessed 10/2020. Prepared by Woolpert.

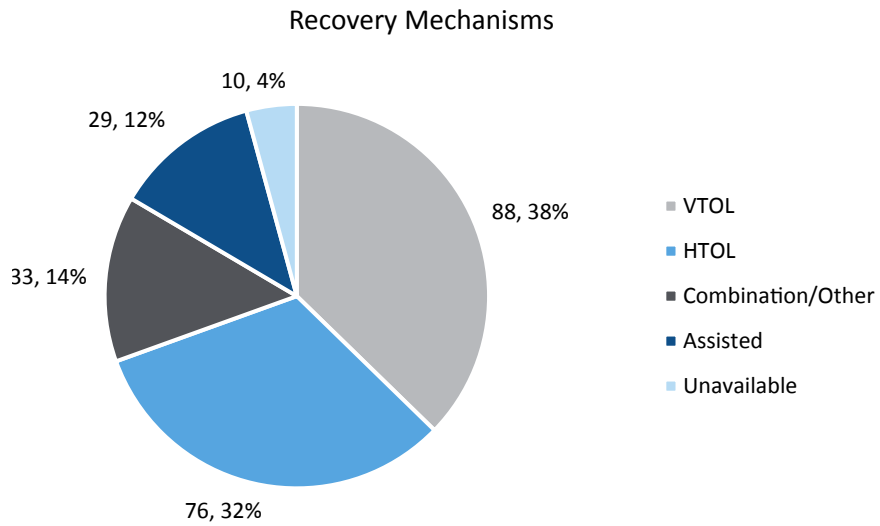
Figure 2-15. Summary of L-UAS launch mechanisms.

closely by the HTOL category (34%). Accordingly, most can likely use an airport's existing airfield infrastructure.

Recovery Mechanisms

Figure 2-16 provides a breakdown of recovery mechanisms used by 236 L-UAS platforms included in the analysis. As in the breakdown for launch mechanisms, the majority fall within the VTOL category (38%), followed closely by the HTOL category (32%). Accordingly, most can likely use an airport's existing airfield infrastructure including runways and helipads. A sampling of different launch and recovery mechanism types from the USRD is provided in Table 2-2.

Table 2-3 provides a summary of the launch and recovery mechanism review completed in conjunction with this research. This table summarizes the number of platforms included in

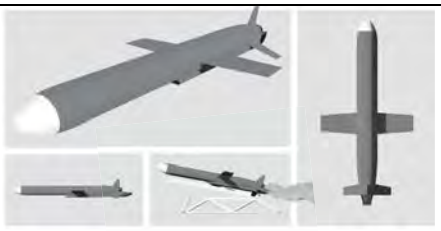


Source: AUVSI USRD, accessed 10/2020. Prepared by Woolpert.

Figure 2-16. Summary of L-UAS recovery mechanisms.

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Table 2-2. Examples of launch and recovery mechanisms.

	Launch	Recovery
VTOL	 **GL-10 Greased Lightning**, a 10-rotor tiltwing hybrid platform with a 10-ft wingspan and an MTOW of 62 lb developed by NASA as part of research to develop efficient hybrid-electric VTOL aircraft.	 **Dragon Warrior/Vantage**, with a wingspan of 9.08 ft, a length of 9.33 ft, and a MTOW of 380 lb was manufactured as a prototype by U.S. Naval Research Laboratory for research purposes across various applications (e.g., disaster response, imaging, inspection, and precision agriculture).
HTOL	 **Northrop Grumman Firebird** platform. With a wingspan of 65 ft, length of 9.8 ft, and a 5,000 lb MTOW, this platform is primarily used to gather intelligence.	 **Viper M14** by Saxon Remote Systems. This system has a length and width of 14 ft, an MTOW of 55 lb, and is used for mapping, surveying, target acquisition, and other operations.
Combination/Other	 **BQM-74C Chukar III** manufactured by Northrop Grumman. This system has a wingspan of 5.8 ft, a length of 12.9 ft, an MTOW of 1,455 lb, and uses rocket-assisted takeoff (RATO) to launch. This system can be air-launched as well as ground-launched and is used for aerial targeting and training by the U.S. Navy.	 **Aerosonde Mark 4.7** manufactured by Textron Systems. With a 12-ft wingspan and an 8-ft length, this 80 lb system is capable of VTOL flight, catapult assisted flight, or net recovery flight and is used in expeditionary land- or sea-based operations.
Assisted	**Air Launch**  **Airborne Systems North America's DragonFly**. This is a system with a 110 ft wingspan and a maximum MTOW of 10,000 lb. It is used by the U.S. DOD as a Joint Precision Airdrop System.	**Parachute**  **Apex (Airborne Pursuit and Exploitation)**. This system has a wingspan of 14.1 ft, a length of 6.1 ft, and an MTOW of 61.7 lb. The Apex, manufactured by L3Harris Technologies, uses a parachute recovery system. It is used in civil and military applications.

(continued on next page)

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Table 2-2. (Continued).

	Launch	Recovery
	Catapult/Pneumatic  **Boeing Insitu's ScanEagle2** platform. This platform has a wingspan of 10.2 ft, a length of 5.6 ft, and a total MTOW of 58 lb. It is launched from a pneumatic device for applications in reconnaissance, monitoring, imaging, inspections, and others.	**Skid/Belly**  **MQM-171A BroadSword** manufactured by Griffon Aerospace. This platform has a width of 22.5 ft, a length of 15 ft, and a 551-lb MTOW. Deployed via catapult launch, it performs skid belly recovery and is used for military activities (research, targeting, and training).
	Rocket-Assisted Takeoff (RATO)  **BQM-177A/I** manufactured by Kratos Defense and Security Solutions, Inc. is used by U.S. Navy for aerial target and training. This system has a length of 17 ft and an MTOW of 1,400 lb.	**Net Recovery**  **Northrop Grumman's BAT 12**, an aerial intelligence gathering platform used by the U.S. Armed Forces that relies on a net recovery mechanism. It has a wingspan of 12 ft, a width of 6.3 ft, and a total MTOW of 220 lb.

Table 2-3. Breakdown of launch and recovery mechanisms.

Launch/Recovery Type	Launch/Recovery Mechanism	Number of Platforms Identified (N=236)	
		Launch (%)	Recovery (%)
VTOL	Vertical Takeoff and Landing	99 (42)	88 (37)
HTOL	Horizontal Takeoff and Landing	79 (33)	76 (32)
Assisted	Air Launch	14 (6)	-
	Rocket-Assisted Takeoff	3 (1)	-
	Launcher—Pneumatic, Catapult	10 (4)	-
	Parachute Recovery	-	20 (8)
	Net Recovery	-	3 (1)
	Skid/Belly Recovery	-	6 (3)
Combination, Other	Combination or Other Launch/Recovery	28 (12)	33 (14)
Data Unavailable	Data Unavailable	3 (1)	10 (4)

Source: AUVSI USRD, accessed 10/2020. Prepared by Woolpert.

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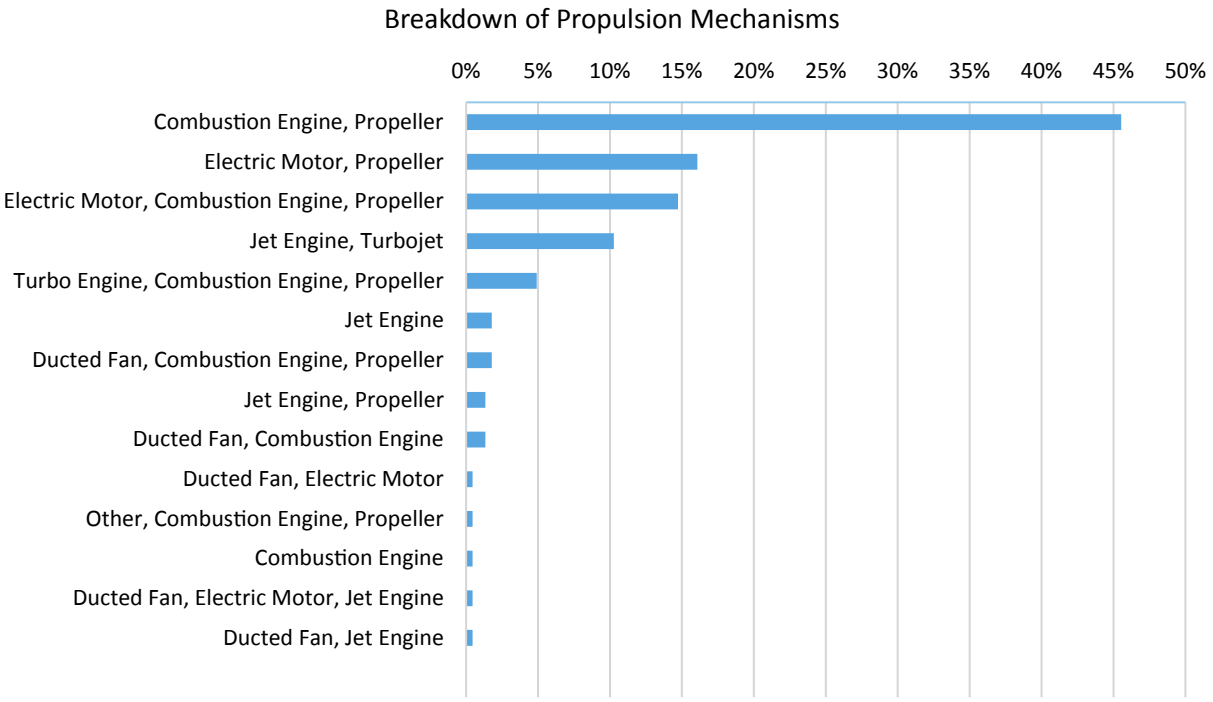
each category (VTOL, HTOL, Assisted, and Combination) for both launch and recovery. In some cases, launch and recovery mechanisms could not be determined; these results are also reflected in Table 2-2.

Propulsion Mechanisms

The various types of propulsion mechanisms (Figure 2-17) identified from the analysis of the 236 platforms identified in this research effort include either one or a combination of the following:

- Combustion engine,
- Electric motor,
- Turbojet engine,
- Jet engine, and
- Ducted fan.

Understanding the L-UAS propulsion mechanisms and the fuel types allows the airport sponsor to identify the infrastructure required on the airfield to support the vehicles. Most of the L-UAS platforms in the USRD are powered by either aviation fuel, gasoline, battery, or a combination thereof. Heavy fuel, diesel, and kerosene were also found in the database to a lesser extent, along with hydrogen. The electric power supply available at the airport can play a role in the airport's ability to serve aircraft using batteries as part of their power source. Additionally, hydrogen introduces a new area of concern for handling and storage at the airport. The other fuel sources are often provided by the L-UAS operator itself, but the airport sponsor should have operating procedures for their dispensing and storage if not part of the airport's fueling services.



Source: AUVSI USRD, accessed 10/2020. Prepared by Woolpert.

Figure 2-17. Summary of L-UAS propulsion mechanisms.

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Comparison with Aircraft Design Standards

Aircraft Classification

Airport infrastructure design funded by the Airport Improvement Program (AIP) is governed by the FAA standards, which are published in the form of advisory circulars (ACs), engineering briefs (EBs), and other FAA orders. Airport facilities are typically designed to accommodate a critical aircraft, which is defined as the most demanding aircraft type, or groups of aircraft with similar characteristics that make regular use of the airport. Regular use, as defined by FAA in AC 150/5000-17, *Critical Aircraft and Regular Use Determination,* is 500 annual operations (takeoff or landing). According to FAA AC 150/5300-13, *Airport Design*, most established design standards for aprons, taxiways, and runways are based on three key components of the aircraft: aircraft approach category (AAC), aircraft design group (ADG), and taxiway design group (TDG). The AAC, ADG, and available instrument approach procedure visibility minimums together make up what is known as the runway design code (RDC) for each runway at an airport.

AAC. A grouping of fixed wing aircraft based on approach speed (also referred to as landing speed or V_{REF}). If V_{REF} is not specified, then 1.3 times stall speed (V_{SO}) at the maximum certificated landing weight of an aircraft is used as the approach speed. Table 2-4 provides FAA's groupings for fixed wing aircraft based on approach speeds.

ADG. A classification of airplanes based on their wingspan and tail height. Table 2-5 provides FAA's groupings for aircraft based on these two dimensions.

TDG. A classification of airplanes based on outer-to-outer main gear width (MGW) and cockpit to main gear distance (CMG) (Figure 2-18).

Table 2-4. Aircraft approach category (AAC).

AAC	Approach Speed
A	Less than 91 knots
B	91 knots or more but less than 121 knots
C	121 knots or more but less than 141 knots
D	141 knots or more but less than 166 knots
E	166 knots or more

Source: FAA AC 150-5300-13, Airport Design. Table 2-1. Prepared by Woolpert.

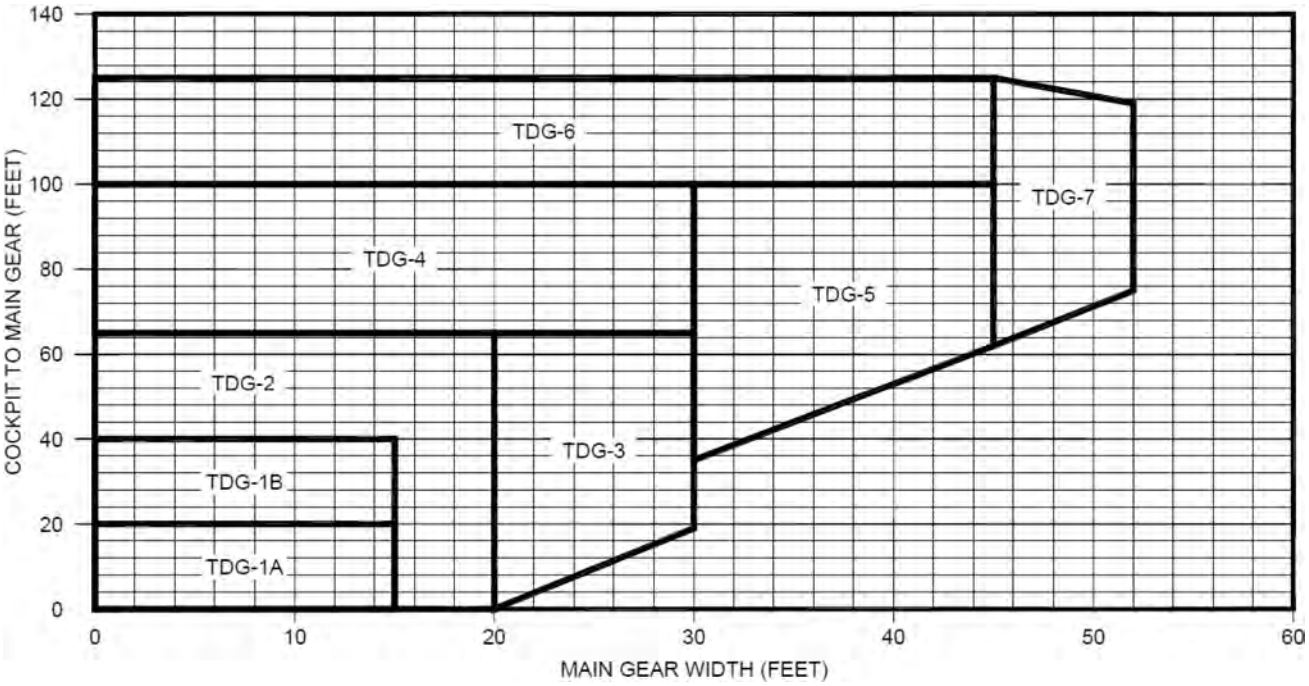
Table 2-5. Aircraft design group (ADG).

ADG	Tail Height (ft)	Wingspan (ft)
I	< 20	< 49
II	20 to < 30	49 to < 79
III	30 to < 45	79 to < 118
IV	45 to < 60	118 to < 171
V	60 to < 66	171 to < 214
VI	66 to < 80	214 to < 262

Source: FAA AC 150/5300-13, Airport Design. Table 2-2. Prepared by Woolpert.

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Source: FAA AC 150/5300-13, Airport Design

Figure 2-18. Taxiway design group.

The certificated MTOW of an aircraft is also a consideration in certain aspects of airport design. AC 150/5300-13, *Airport Design*, includes two weight categories: large and small. Large aircraft are those with an MTOW of more than 12,500 lb; small aircraft are those with an MTOW of 12,500 lb or less. A sub-group of the small category is ultralights, which have an MTOW of less than 254 lb.

AACs, ADGs, and TDGs consider speeds and dimensions based on traditional aircraft. L-UAS are extremely diverse in their performance and dimensional characteristics and, therefore, may not always conform to established FAA airport design standards. As previously stated, airfield design standards in FAA AC 150/5300-13, *Airport Design*, are based on the critical aircraft's AAC, ADG, and TDG.

Based on FAA guidance and available data, the L-UAS in the USRD could be classified by ADG and MTOW, but not AAC or TDG. The USRD does not provide data on landing gear configurations or approach speed. Most entries include cruise speed, but only a few include aircraft maximum descent speed or stall speed. While approach speed can generally be estimated to be 1.3 times the stall speed, very few entries included these data, and FAA has does not have an estimation formula to convert maximum descent speed or cruise speed to approach speed. Further detailed investigation into a few of the L-UAS platforms also revealed that approach speeds were not readily available from any other sources. One reason for this may be because the automation of these aircraft makes this speed less critical to the operator as opposed to aircraft with the pilot on board. Therefore, the comparison to the traditional aircraft approach speed categories could not be included as part of this analysis. A breakdown of the L-UAS platforms compared with existing FAA aircraft design group standards is provided in Table 2-6. For the analysis of the 236 L-UAS identified in the USRD, nine platforms were removed from the analysis as outliers because they did not belong in fixed wing, rotary wing, or hybrid categories. The nine outliers were lighter-than-air concepts (balloons or blimps), or adequate platform information was not made available by the manufacturer for their tail height or wingspan.

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Table 2-6. Aircraft design groups for L-UAS.

ADG	Number of Platforms Identified (N=227)	Percentage of Total
I	198	88%
II	15	6%
III	6	3%
IV	7	3%
V	0	0%
VI	1	0%

Prepared by Woolpert.

FAA AC 150/5300-13, *Airport Design*, along with AC 150/5325-4, *Runway Length Requirements for Airport Design*, also use the MTOW of an aircraft to establish some airport design standards. These ACs generally categorize aircraft by MTOW into the following groups:

- 55 to 254 lb or less (ultralight)
- Over 254 to 12,500 lb or less (small airplanes)
- Over 12,500 to less than 60,000 lb (large airplanes)
- 60,000 lb or more

Most of the 227 L-UAS platforms presented in Table 2-7 fall within the lighter weight groups (MTOWs of 55 pounds to 12,500 pounds) for weight calculations. Platforms were then further delineated by type (category), weight, and average dimensions to develop composite data for representative L-UAS platforms.

Finally, the L-UAS were also summarized according to FAA ADG. The results are presented in Table 2-8.

Table 2-7. L-UAS classifications by category, size, and weight (for research purposes only).

L-UAS Weight Groupings (lb)	Type	Number of Platforms Identified	Average of MTOW (lb)	Average of Wingspan (ft)	Average of Width (ft)	Average of Length (ft)	Average of Height (ft)
55 to ≤ 254		110	110	13	5	8	3
	Fixed	50	127	17	14	8	2
	Hybrid	24	84	12	7	7	2
	Rotary	36	104	8	4	8	3
> 254 to ≤ 12,500		103	2,297	32	20	20	6
	Fixed	73	2,163	35	51	19	6
	Hybrid	5	1,320	28	-	20	7
	Rotary	25	2,886	22	8	23	7
> 12,500 to < 60,000		13	29,752	101	-	45	15
	Fixed	12	29,773	104	-	45	15
	Hybrid	1	29,500	65	-	-	-
≥ 60,000		1	61,796	38	-	30	16
	Fixed	1	61,796	38	-	30	16

Source: AUVSI USRD, accessed 10/2020. Prepared by Woolpert.

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Table 2-8. L-UAS ADG and classification summary (for research purposes only).

ADG	Characteristics		Total
I	Propulsion	Combustion	127
		Electric	65
		Combination/Other	6
	Launch	VTOL	98
		HTOL	54
		Assisted	22
		Combination	24
	Recovery	VTOL	83
		HTOL	51
		Assisted	22
		Combination	42
	MTOW (lb)	55 to ≤ 254	108
		> 254 to ≤ 12,500	88
		> 12,500 to < 60,000	1
		≥ 60,000	1
II	Propulsion	Combustion	14
		Electric	1
		Combination/Other	0
	Launch	VTOL	1
		HTOL	13
		Assisted	0
		Combination	1
	Recovery	VTOL	1
		HTOL	12
		Assisted	1
		Combination	1
	MTOW (lb)	55 to ≤ 254	0
		> 254 to ≤ 12,500	12
		> 12,500 to < 60,000	3
		≥ 60,000	0
III	Propulsion	Combustion	4
		Electric	2
		Combination/Other	0
	Launch	VTOL	0
		HTOL	5
		Assisted	0
		Combination	1
	Recovery	VTOL	0
		HTOL	5
		Assisted	0
		Combination	1
	MTOW (lb)	55 to ≤ 254	2
		> 254 to ≤ 12,500	0
		> 12,500 to < 60,000	4
		≥ 60,000	0

(continued on next page)

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Table 2-8. (Continued).

ADG	Characteristics		Total
IV	Propulsion	Combustion	6
		Electric	0
		Combination/Other	1
	Launch	VTOL	0
		HTOL	7
		Assisted	0
		Combination	0
	Recovery	VTOL	0
		HTOL	7
		Assisted	0
		Combination	0
	MTOW (lb)	55 to ≤ 254	0
		> 254 to ≤ 12,500	2
		> 12,500 to < 60,000	5
		≥ 60,000	0
V	No vehicles were classified as ADG-V.		
VI	Propulsion	Combustion	0
		Electric	1
		Combination/Other	0
	Launch	VTOL	0
		HTOL	0
		Assisted	0
		Combination	1
	Recovery	VTOL	0
		HTOL	1
		Assisted	0
		Combination	0
	MTOW (lb)	55 to ≤ 254	0
		> 254 to ≤ 12,500	1
		> 12,500 to < 60,000	0
		≥ 60,000	0

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CHAPTER 3

Airfield Component Impact Analysis

Design Criteria Impact Summary

This chapter identifies the challenges and issues in current airfield design criteria as they relate to incorporating L-UAS. It also provides guidance on the integration potential of L-UAS with existing design criteria for an airport's major components. It is organized by the following key airfield attributes:

- Lighting, marking, and signage
- Hangar storage facilities
- Pavement strength
- Airfield surface infrastructure
- Navigational aids
- Communications and data systems
- Fueling and charging
- Obstacles and airspace
- Environmental impacts
- Airport passenger terminal
- Air traffic control
- Aircraft rescue and firefighting support

Each component begins with an impact overview that describes the topic being covered, followed by key considerations for airport sponsors when introducing L-UAS operations into that component of the airfield. The next section is a synopsis of the relative design criteria gaps, the potential for integration of L-UAS into the component, and any existing applicable standards that may be transferable. Each component description includes the relative level of considerations depending on the significance of the impact or its complexity.

Finally, each component ends with a quick reference table summarizing the section. The "integration potential" for each component is based on the combination of the impact level, number of unique considerations, and applicability of existing design standards. Table 3-1 defines the terms used to describe "integration potential."

Airfield Components

Lighting, Marking, and Signage

Impact Overview

The airfield marking, lighting, and signage standards for airports are prescribed in various ACs published by FAA. Since these standards vary for airports and heliports, their applicability depends on the category of L-UAS to be served at a facility, as well as on the launch and recovery methods.

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Table 3-1. Integration potential and impact level on existing design standards.

Integration Potential	Impact Level
High	The impact from L-UAS is considered low and departure from existing airport design criteria is not significant (or non-existent). Integration of L-UAS using existing criteria is relatively unproblematic.
Medium	A comprehensive review of L-UAS type and operational characteristics is warranted when considering integration. Integration is possible, but gaps and impacts exist and careful review of the specific needs of the L-UAS should be cross-referenced with existing design criteria.
Low	L-UAS introduce challenges in using existing design standards, and their applicability comes with certain problems and gaps. Integration of L-UAS using existing airport design criteria requires further investigation into the specific aircraft's operational needs. Existing criteria likely do not directly apply.

While most L-UAS in the fleet today consist of fixed wing aircraft (58%; AUVSI 2020), the L-UAS most likely to use airport runways, taxiways, and aprons operate similarly to traditionally piloted aircraft. Therefore, the marking, lighting, and signage standards for runways, taxiways/taxilanes, and/or aircraft parking aprons apply to fixed wing L-UAS aircraft operations at the airfield. The major AC in this area is AC 150/5340-1M, *Standards for Airport Markings*.

Rotary wing and hybrid UAS also represent a large share of the UAS fleet, according to USRD data. Like traditionally piloted helicopters and VTOLs, rotary wing and hybrid UAS can also approach and depart from runways and operate on the taxiway system to transition to/from the aircraft parking apron. However, high activity airports may establish a dedicated helipad touchdown and liftoff area (TLOF) to segregate these UAS activities from fixed wing aircraft operations. A designated helipad on an airport can improve the throughput capacity of the airfield and enhance operational safety by segregating rotary wing and/or hybrid aircraft from fixed wing operations when aircraft speed and operational volume are an issue. In such a case, the marking, lighting, and signage standards for heliports are likely transferable (AC 150/5390-2C, *Heliport Design*).

Key Considerations

The primary concern for accommodating L-UAS on the airfield and/or heliport is ensuring adequate visibility of the markings, signage, and lighting systems by the aircraft's guidance system. L-UAS come with various guidance systems, including camera, GPS, and ground control stations. Camera guidance systems typically include infrared cameras that transmit images of the airfield environment to the operator in real time. The location of the cameras on an L-UAS may not ensure an unobstructed line-of-sight (LOS); depth perception to differentiate objects; or identification of markings, lights, and airfield signage. L-UAS will require case-by-case analysis to evaluate their situational awareness capability as well as their ability to detect the airfield or helipad marking, lighting, and signage and other aircraft operating on the surface. If they cannot, some means of augmented situational awareness on the field may be required before integration is possible.

Most airfield and helipad marking, lighting, and signage systems prescribed by FAA are easily identifiable through infrared camera guidance systems available to L-UAS. There may be a rare occasion when enhancing the reflectivity and/or contrast of airfield pavement markings may be necessary to permit operations during low visibility conditions, which could be accomplished by applying for a "modification of standards" with FAA. Additional research to determine whether alternate paint materials, such as paints with higher reflectivity, may be required under certain conditions and to ensure they do not adversely affect traditional aircraft operation.

Gap Analysis and References

Since there are no pilots on board, L-UAS have visual capabilities that differ from traditionally piloted aircraft. The existing standards for marking, lighting, and signage apply to these aircraft

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but a gap in knowledge exists on the various L-UAS visual detection performance capabilities via onboard camera or sensor systems for flight and taxiing beyond visual line-of-sight (BVLOS). The following documents can provide guidance and standards for airfield and helipad marking, lighting, and signage (Table 3-2).

Hangar Storage Facilities

Impact Overview

UAS aircraft can be either integrated with or segregated from traditional aircraft for storing. When integrating, the major areas of concern are wingspan, tail height, and energy

Table 3-2. Airfield lighting, marking, and signage.

AIRFIELD COMPONENT	AIRFIELD LIGHTING, MARKING, AND SIGNAGE
Integration Potential	High.
Gap Areas	Lack of guidance on the delineation of movement area boundaries that segregate crewed and uncrewed operations or rotary/hybrid L-UAS from fixed wing UAS. Lack of guidance on UAS sensor acuity requirements to allow a remote pilot to properly comprehend airfield signage and markings.
Potential Issues	Remote pilot's ability to see visual cues on an airfield due to camera location on aircraft resulting in a limited field of view. Large variance in L-UAS sensor capabilities and existing Airfield Markings Standards.
APPLICABLE STANDARD FOR GUIDANCE	**TITLE**
AC 150/5340-1M	*Standards for Airport Markings*
AC 150/5340-5D	*Segmented Circle Airport Marker System*
AC 150/5340-18G	*Standards for Airport Sign Systems*
AC 150/5340-23	*Supplemental Wind Cones*
AC 150/5340-26C	*Maintenance of Airport Visual Aid Facilities*
AC 150/5340-30J	*Design and Installation Details for Airport Visual Aids*
AC 150/5345-12	*Specification for Airport and Heliport Beacon*
AC 150/5345-27	*Specification for Wind Cone Assemblies*
AC 150/5345-28H	*Precision Approach Path Indicator Systems (PAPI)*
AC 150/5345-39	*FAA Specification L-853, Runway and Taxiway Centerline Retroreflective Markers*
AC 150/5345-42	*Specification for Airport Light Bases, Transformer Houses, Junction Boxes and Accessories*
AC 150/5345-43J	*Specifications for Obstruction Lighting Equipment*
AC 150/5345-44K	*Specification for Runway and Taxiway Signs*
AC 150/5345-45	*Lightweight Approach Light Structure*
AC 150/5345-46	*Specification for Runway and Taxiway Light Fixtures*
AC 150/5345-49D	*Specification L-854, Radio Control Equipment*
AC 150/5345-50	*Specification for Portable Runway Light*
AC 150/5345-51	*Specification for Discharge-Type Flasher Equipment*
AC 150/5345-52A	*Generic Visual Glideslope Indicators*
AC 150/5345-53	*Airport Lighting Equipment Certification Program*
AC 150/5390-2C	*Heliport Design*
Joint Order 6850.2B	*Visual Guidance Lighting Systems*

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source for propulsion. For wingspan and tail height, FAA's guidance in AC 150-5300-13, *Airport Design* categorizes conventional fixed wing aircraft by design group based on tail height, which reaches up to 80 feet, and wingspan, which reaches up to 262 feet. The largest wingspan and tail height found among L-UAS platforms analyzed for this research project were 243 feet and 16.4 feet, respectively, according to USRD, which fall within the FAA's traditional aircraft design groups. Accordingly, the existing FAA standard for building setbacks and safety areas are transferable to the fleet of L-UAS aircraft included in this research project as it relates to wingtip and tail height clearance. Therefore, the ADG applied to the airport's existing hangar taxilanes is a major limiting factor for storing most L-UAS. Additionally, the hangar door widths and heights also determine what L-UAS sizes/types can be stored at an airport.

Key Considerations

Many of the L-UAS reviewed in this analysis have high integration potential because they operate like traditional aircraft. If an L-UAS taxis under its own power with wheels, integration is easier. The airport operator should take into consideration the wheelbase of an L-UAS and its taxi process to ensure it has appropriate access through the necessary taxilanes to the hangar area. If the aircraft hover-taxis or lands differently than conventional aircraft, then it will likely need to be tugged on a platform to the hangar.

Segregation of L-UAS for storage may be warranted based on a particular aircraft's propulsion energy source, such as batteries or hydrogen. Any stored energy is a source of risk. If L-UAS use batteries for propulsion, they introduce a set of complexities not associated with traditionally powered aircraft. Lithium-ion battery packs, for example, undergo substantial thermal stress and must be maintained within certain temperature limits, and they are susceptible to short-circuiting and thermal runaway. Suppressing a fire whose source is lithium-ion batteries is different from suppressing one whose source is Jet-A, aviation gasoline (Avgas), or hydrogen. Additionally, hazardous material storage guidelines (quantities and locations) for different chemicals and fuels should also be considered before storing different fuel materials near one another.

Charging L-UAS that operate with batteries should also be given careful consideration for aircraft storage. As our automobile cousins have taught us, charging inside a building may not be safe because there have been occasions when lithium-ion batteries have exploded and/or caught fire while a car was charging in a garage.

Gap Analysis and References

Gaps exist on guidance for the storage of the different energy sources that are or may be used for different UAS platforms (see also Fueling and Charging, below). While there is no direct guidance on L-UAS storage in traditional hangar facilities, the following sources can provide direction and recommendations on this topic (Table 3-3).

Pavement Strength

Impact Overview

A facility's existing pavement strength has usually been designed with the maximum anticipated aircraft MTOW that will regularly operate on it in mind. It is important to note that pavement design not only needs to withstand an aircraft's MTOW but also takes into consideration repetitive loading (e.g., aircraft landing) to ensure pavement longevity. Landing gear configurations and geometry are also taken into consideration when designing airfield pavements in FAA's FAARFIELD pavement design software.

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Table 3-3. Hangars.

AIRFIELD COMPONENT	HANGARS
Integration Potential	High.
Gap Areas	Lack of guidance on storage requirements for alternative fuel sources, specifically at airports. Lack of guidance on the variety of unique ARFF considerations for L-UAS.
Potential Issues	Lithium-ion batteries pose a fire hazard and may require unique fire suppression or ARFF considerations.
APPLICABLE STANDARD FOR GUIDANCE	**TITLE**
FAA AC 150/5230-4B	*Aircraft Fuel Storage, Handling, and Dispensing on Airports*
International Fire Code (IFC)/ National Fire Protection Association (NFPA) 1 Uniform Fire Code	*Requirements for hydrogen fueling stations, flammable gas, and cryogenic fluid storage*
IEC SC23H	*Plugs, Socket-outlets, and Couplers for industrial and similar applications, and for Electric Vehicles*
IEC TC 69	*Electrical power/energy transfer systems for electrically propelled road vehicles and industrial trucks*
NFPA 855	*Standard for the Installation of Energy Storage Systems (ESS)*
NFPA 55	*Compressed Gas and Cryogenic Fluids Code*
NFPA	*Emergency Field Guide: Hybrid, Electric, Fuel Cell, and Gaseous Fuel Vehicles*
NFPA 2	*Hydrogen Technologies Code*
NFPA 400	*Hazardous Material Code*
NFPA 853	*Standard for the Installation of Stationary Fuel Cell Power Systems*
National Renewable Energy Laboratory (NREL)	*Hydrogen Technologies Safety Guide*
NFPA	*Sprinkler Protection Guidance for Lithium-Ion Based Energy Storage Systems*
FAA AC 150/5300-13	*Airport Design*
NFPA 409	*Standard on aircraft hangars*
Occupational Safety and Health Administration (OSHA) 1910 Subpart H	*Safe storage, use, and handling of hydrogen in the workplace*

Key Considerations

L-UAS MTOW is a key factor when considering integration into an airport environment. The technology review performed for this research project indicates that most L-UAS MTOWs fall below those defined for large aircraft; therefore, existing pavement design standards should be transferable when considering the integration of L-UAS. Airport operators need to take additional factors into consideration when accommodating L-UAS over 12,500 lbs. The technology review also identified that L-UAS can have various landing gear configurations and can have unique takeoff and landing mechanisms, such as catapult launch or skid belly landings. These variances in L-UAS landing gear type or takeoff/landing mechanisms will affect pavement design, pavement maintenance, distress evaluation, and rehabilitation standards.

Another consideration is that the precision of L-UAS landings can result in a very consistent landing location that may make one pavement area wear more quickly than with traditionally piloted aircraft, which have a larger scatter during landing.

Finally, the FAA's pavement design software, FAARFIELD, does not include data for L-UAS aircraft. Landing gear configuration and MTOW need to be evaluated for the various L-UAS operating at the airport and input directly into FAARFIELD.

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Gap Analysis and References

A gap exists in L-UAS landing gear configurations data and MTOW, which are missing from the FAARFIELD software. However, the existing pavement design criteria have a high integration potential when considering L-UAS operations once their gear configurations and MTOW are determined (Table 3-4).

Airfield Surface Infrastructure

Impact Overview

For this research, airfield surface infrastructure includes airport movement areas such as runways, taxiways, and aprons. Each of these components has unique design criteria outlined in FAA AC 150/5300-13. The L-UAS may interact with these components differently based on its platform and configurations. As outlined in Chapter 2, airfield infrastructure is designed to be able to accommodate all operations for a particular critical aircraft. The critical aircraft can be either a specific aircraft model or a grouping of aircraft that are expected to perform regular operations (FAA AC 150/5000-17, *Critical Aircraft and Regular Use Determination*). If an airport sponsor is using an L-UAS as the critical aircraft for their facility, the OEM performance and dimensional criteria should be thoroughly reviewed to ensure the airport conforms to the applicable design standards.

Key Considerations

Runways. Existing runway design standards apply to L-UAS that require an HTOL surface. The basis of runway design is AAC and ADG, where ADG is based on an aircraft wingspan and tail height and AAC is based on the approach speed. Approach speed data for L-UAS operating in an airport's fleet mix should be considered to sort the aircraft into the correct RDC. The approach speed data are not readily available for most L-UAS and are critical because they determine the runway and runway safety area (RSA) dimensions. Therefore, approach speeds and performance data will need to be obtained from the OEM. FAA AC 150/5300-13, *Airport Design*, recommends contacting the appropriate FAA Airports District Office personnel for guidance on integrating L-UAS at the airport.

Another consideration for a runway's ability to accommodate L-UAS arises when it is using engineered materials arresting systems (EMAS) to provide for a compliant RSA. In this case, the EMAS should be reviewed for their ability to arrest an L-UAS as compared with aircraft in the airport's typical operating fleet.

Table 3-4. Pavement design.

AIRFIELD COMPONENT	PAVEMENT DESIGN
Integration Potential	High.
Gap Areas	L-UAS landing gear configurations and design types require incorporation into the FAA FAARFIELD aircraft library for pavement design.
Potential Issues	Unique takeoff and landing mechanisms or atypical landing gear configurations may affect pavement design, maintenance, and rehabilitation standards. The precision of repetitive L-UAS landings in the same location may make certain areas wear more quickly than has historically occurred.
APPLICABLE STANDARD FOR GUIDANCE	**TITLE**
FAA AC 150/5320-6G	*Airport Pavement Design and Evaluation*
FAA AC 150/5300-13	*Airport Design*
FAA FAARFIELD	*FAA Rigid and Flexible Iterative Elastic Layered Design software*

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Runway Geometry. FAA AC 150/5325-4B, *Runway Length Requirements for Airport Design*, provides further guidance on runway length. This AC categorizes aircraft based on MTOW and includes factors such as loads, airport elevation, temperature, aircraft approach speed, and available passenger seats. This AC was developed around piloted aircraft that existed before its publication date, which was July 1, 2005. Therefore, it does not apply to L-UAS. The specific aircraft takeoff and landing lengths for any proposed L-UAS should be researched to determine whether the airport's runway length is sufficient.

FAA AC 150/5300-13, *Airport Design*, provides guidance on runway widths generally based on the AAC and ADG of the critical aircraft. When integrating L-UAS that require a horizontal surface to take off and land, refer to the OEM design and performance specifications to properly categorize the aircraft into the correct airplane AAC and ADG to determine the width requirement.

Runway Orientation. According to AC 150/5300-13, there is a maximum allowable crosswind component per RDC, which again, is based on the critical aircraft's AAC and ADG. As stated previously, the AACs for L-UAS are not readily available. Therefore, the OEM crosswind component specifications should be identified to ensure that the runway orientation and prevailing winds are consistent with the aircraft's capabilities.

Taxiway Geometry. According to AC 150/5300-13, taxiway design standards are tied to the taxiway design group (TDG) and aircraft design group (ADG) of an aircraft. The ADG of an L-UAS is easily determined by the aircraft's wingspan and tail height, as outlined in Chapter 2. The TDG, however, is determined by the gear configuration, which is not included in the USRD and is not readily available. Therefore, when introducing L-UAS into an airport's fleet mix, OEM landing gear configurations will need to be considered, along with onboard sensor capabilities and operational characteristics, as they may influence if and how a particular UAS uses the taxiway.

Apron/Ramp. Per AC 150/5300-13, *Airport Design*, and AC 150/5390-2C, *Heliport Design*, adequate wingtip and rotor clearances should be provided for aircraft parking and tiedown areas. Ramp/apron areas should accommodate nearby taxilanes, taxiways, and/or runway geometry, as no part of a parked aircraft on an apron can penetrate a runway approach surface, departure surface, or runway/taxiway object-free area (OFA). The dimensions of the L-UAS should be considered when parking them, especially if they are bigger or taller than the existing design aircraft. AC 150/5300-13 recommends separating different sized aircraft based on aircraft wingspan, allowing taxilane placement and dimensions to be adjusted for the aircraft intended to use the area. Separating L-UAS from large and heavy jet aircraft may help avoid any effects of jet blast. L-UAS tiedown point dimensions should also be considered to ensure the size of the existing tiedown points on the apron area will accommodate these aircraft should they need to be tied down.

Helipad. AC 150/5390-2C, *Heliport Design*, provides dimensions for touchdown and liftoff, final approach and takeoff, and safety areas for helipads on airports that segregate helicopter operations. It also provides guidance on separation requirements between runways and helipads. While not helicopters, L-UAS using a VTOL platform fly similarly, so these standards are transferable to some degree with certain adjustments for vehicle size. The exact performance capabilities of L-UAS in this category are still unknown, so it is currently difficult to determine whether they are adequate. However, they can be generally applied until the FAA publishes new guidance for vertipads and vertiports in an AC or engineering brief (EB). [Note: FAA canceled the guidance on vertiport design (AC 150/5390-3) in 2010].

Gap Analysis and References

Approach speeds and landing gear configurations for L-UAS are not readily available to assist in categorizing them in an AAC or ADG, which in part determines the design requirement

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necessary to accommodate the aircraft. The ability of EMAS to mitigate runway overruns is also unknown. Additionally, the FAA has not published new standards to replace the canceled AC on vertiport design (Table 3-5).

Navigational Aids

Impact Overview

The foundational need for navigational aids (NAVAIDs) at airports is to provide human aircraft operators with appropriate positional awareness and guidance to navigate safely and efficiently into, out of, and on the surfaces of airports and heliports. As demonstrated in Chapter 2, L-UAS designs and characteristics present unique capabilities and constraints depending on the vehicles' operational designs. A review of the USRD and responses captured during the stakeholder interviews revealed no impacts to the necessity, design, location, or application of traditional ground-based navigation systems.

Visual NAVAIDs, such as precision approach path indicator (PAPI), visual approach slope indicator (VASI), and approach lighting system (ALS) will provide L-UAS pilots the same navigation guidance when they are equipped with visual surveillance capabilities. Similarly, electronic NAVAIDs, such as instrument landing system (ILS), VHF omnidirectional range/tactical air navigation system/distance measuring equipment (VORTAC/VOR/DME), and non-directional beacon (NDB) will provide L-UAS pilots the same capabilities crewed aircraft pilots rely on when the aircraft is appropriately equipped and the pilot is trained/certified to use them. As described in the section on marking, lighting, and signage, L-UAS visual guidance systems' ability to distinguish visual NAVAIDs may impact their ability to integrate, as sensor acuity requirements for L-UAS have not been established. However, the primary source of navigational guidance for L-UAS is space-based navigation from GPS and inertial navigation systems

Table 3-5. Airfield infrastructure.

AIRFIELD COMPONENT	AIRFIELD INFRASTRUCTURE
Integration Potential	Medium.
Gap Areas	Lack of guidance on AAC classification for L-UAS. Lack of guidance on TDG classification for L-UAS. Lack of guidance on maximum allowable crosswind components of L-UAS. Lack of guidance on vertiport/vertipad design standards for VTOL aircraft.
Potential Issues	Airfields equipped with EMAS for aircraft overruns should consider optimizing EMAS design to accommodate L-UAS. Aircraft approach speed and land gear configuration data are not available for the L-UAS researched in the USRD, thus making UAS approach and TDG categorization difficult without OEM input. L-UAS performance and design characteristics need to be thoroughly reviewed to ensure they conform to the existing design standards.
APPLICABLE STANDARD FOR GUIDANCE	**TITLE**
AC 150/5320-6F	*Airport Pavement Design and Evaluation*
AC 150/5220-22B	*Engineered Materials Arresting Systems (EMAS) for Aircraft Overruns*
AC 150/5300-13	*Airport Design*
AC 150/5390-2C	*Heliport Design*
AC 150/5423-4B	*Runway Length Requirements for Airport Design*
AC 150/5320-6F	*Airport Pavement Design and Evaluations*
AC 150/5000-17	*Critical Aircraft and Regular Use Determination*

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(INS). While GPS is not considered an airport-specific NAVAID and does not require an airport sponsor's direct investment, GPS provides an alternative navigation service in the air and on the ground that crewed aircraft operators use.

Beyond traditional NAVAID systems, the ability for some L-UAS to integrate into the airport environment requires obtaining and maintaining accurate geospatial data for digital map-based systems to support navigation on the surface of the airport. Whether the L-UAS are remotely piloted or are moving across the airport autonomously without a visual guidance system, the operation includes the use of digital maps to safely navigate so that airports leveraging geospatial data may be able to integrate these aircraft more easily. AC 150/5300-18B, *General Guidance and Specifications for Submission of Aeronautical Surveys to NGS: Field Data Collection and Geographic Information System (GIS) Standards*, provides the data collection and formatting requirements to support standardized airport geospatial data for the FAA Airport GIS program.

Key Considerations

Given that the intent of NAVAIDs (visual or electronic) on the airfield is to provide human pilots guidance to approach and successfully land at an airport, the relevance to their use by L-UAS will be based on the aircraft and its onboard technology. The burden for providing and ensuring successful deployment and use of the NAVAIDs does not usually rest with the airport sponsor; however, understanding what NAVAIDs are usable by the L-UAS entering the airport is the chief consideration to be addressed. In the event the airport does not possess the appropriate ground-based NAVAIDs, coordination with FAA for capturing and installing the needed systems should be completed.

If the airport has GPS-based instrument approach procedures (IAPs), no additional impacts are likely because the appropriate evaluations to support GPS navigation have already been conducted under existing FAA criteria. If the airport does not have GPS-based IAPs, the need for completing the necessary surveys and evaluations to develop GPS-based IAPs should be considered. The airport sponsor should coordinate with FAA to complete the necessary activities for their plan and development.

The importance of FAA standardized airport GIS databases will continue, and airports lacking these data could be less capable of supporting L-UAS. FAA already requires many airports to have these data and funds their development as part of their Airport Improvement Program (AIP). Airports are encouraged to capture appropriate GIS data applicable to each AIP-funded project they undertake and to capture new airport features when needed to consistently maintain their geospatial database. However, it is worth noting that the GPS coordinates used by L-UAS might differ from the ones fed into the GIS databases.

Gap Analysis and References

Gaps in criteria for integrated L-UAS based on visual or electronic NAVAIDs do not exist in the current standards and specifications. Understanding the navigation capabilities of the various L-UAS is important to identify whether existing NAVAIDs at the airport provide sufficient support for L-UAS operations (Table 3-6).

Communications and Data Systems

Impact Overview

Data and communication technologies are critical components of all L-UAS operations to maintain control, communication, and navigation. Like traditional piloted aircraft operations, L-UAS short- and long-range communications systems are the circulatory system conveying voice and data link communications between pilots and ATC to maintain separation and deconflict aircraft. The challenge introduced to the communication and data transfer network

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Table 3-6. Navigational aids.

AIRFIELD COMPONENT	NAVIGATIONAL AIDS
Integration Potential	High.
Gap Areas	Lack of guidance on UAS sensor acuity requirements.
Potential Issues	L-UAS that are not equipped with appropriate technology to use traditional ground-based NAVAIDs will not be able to use them. Airports without GPS IAPs will not be able to integrate L-UAS with GPS-based navigation.
APPLICABLE STANDARD FOR GUIDANCE	**TITLE**
AC 150/5345-28H	*Precision Approach Path Indicator (PAPI) Systems*
AC 150/5340-30J	*Design and Installation Details for Airport Visual Aids*
AC 150/5345-27E	*Specification for Wind Cone Assemblies*
AC 150/5340-5D	*Segmented Circle Airport Marker System*
AC 150/5345-52A	*Generic Visual Glideslope Indicators (GVGI)*
AC 150/5300-18B	*General Guidance and Specifications for Submission of Aeronautical Surveys to NGS: Field Data Collection and Geographic Information System (GIS) Standards*
FAA Order 8260.3D	*United States Standard for Terminal Instrument Procedures (TERPS)*

by L-UAS is the fact the pilot is not in the flight deck, but rather, is operating from a remote location, which creates another node to the communications network.

Communication and data networks and associated infrastructure are not traditionally the responsibility of the airport sponsor to design, install, and maintain, as they are usually FAA systems. The data and communication demands placed on airports center around the need for connectivity, sufficient bandwidth, and data transfer speed to support the conveyance of information between a control station, an aircraft, and ATC. Airport impacts possibly depend on the infrastructure present to support radio or satellite communication to aid different aircraft communication and control station components, as well as to minimize latency in the signal transfer. Additionally, some L-UAS missions (e.g., surveillance for fighting forest fires) rely on additional support from internet-based systems, so the airport's broadband utility infrastructure may impact their ability to conduct missions there.

Key Considerations

Regardless of the aircraft or system, each requires similar connectivity to provide the same functionality to operate safely. L-UAS operating the longest distances will be flying under the control of a stationary pilot, but the aircraft's communication and data link of flight information are the same as those of existing crewed aircraft. FAA's criteria and specifications for voice and data link communications define the requirements for the circle of communication to be completed. Airports should coordinate with FAA to determine whether sufficient service capabilities exist with the infrastructure present at the airport to support L-UAS desiring access to the airport. Similarly, they should coordinate with the L-UAS operators or OEMs to understand each aircraft's communication methods and protocols and to determine whether additional gaps in service exist.

Airport sponsors are familiar with the growing demand to advance information technology (IT) systems to support airport operations, facility management, or enhanced customer service. The growth of UAS (large and small) will continue to increase the demand for airport-related IT. The impact on airport sponsors' future planning and investment will be defined by

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understanding the necessary level of data, communications speed, and bandwidth as they are approached by L-UAS operators.

As the use of radio, satellite, and internet-based communication systems increases, airports should be mindful of the risk of signal interference or interruption increases. The impacts from L-UAS communication and data transfer methods introduce the risk of signal interference with either ground-to-air or ground-to-ground communications.

Gap Analysis and References

The requirements for communications and data link systems at airports are defined in applicable FAA ACs and Orders and do not pose an immediate gap in criteria. However, many existing standards related to communication and data address traditionally piloted (crewed) aircraft. Pilot requirements do not apply to the airport facility and infrastructure and do not address UAS needs. Rather, the criteria address aircraft and air traffic-related communications and data conveyance. (See Table 3-7.)

Fueling and Charging

Impact Overview

The fuel sources for L-UAS do not present an issue with integrating the operation of L-UAS into the traffic flow of the runway or the taxiway. However, segregation on the ramp for servicing or refueling may be warranted based on power sources and servicing needs. Fuel sources for L-UAS most likely will include some types of hazardous material, so they must be handled with associated risks in mind. As detailed in Chapter 2, these issues are addressed on a case-by-case basis with consultation of the appropriate code, guidance, or regulations governing the fuel source being used. Examples include the National Fire Protection Code on the federal

Table 3-7. Communications and data systems.

AIRFIELD COMPONENT	COMMUNICATIONS AND DATA SYSTEMS
Integration Potential	High.
Gap Areas	Gaps in existing criteria not present. Review of L-UAS communication, data link, and command systems may introduce gaps in service capabilities at each airport.
Potential Issues	Insufficient voice and data link transferring infrastructure installed at the airport to support communication. The current high-speed internet and bandwidth connectivity require upgrading and investment to support L-UAS operations.
APPLICABLE STANDARD FOR GUIDANCE	**TITLE**
FAA AC 20-140C	*Guidelines for Design Approval of Aircraft Data Link Communication Systems Supporting Air Traffic Services (ATS)*
FAA AC 20-160A	*Onboard Recording of Controller Pilot Data Link Communication (CPDLC) in Crash Survivable Memory*
FAA AC 20-177	*Design and Installation Guidance for an Airborne System for Non-Required Telecommunication Service in Non-Aeronautical Frequency Bands*
FAA AC 90-66B CHG 1	*Non-Towered Airport Flight Operations*
FAA AC 90-117 (E UPDATE)	*Data Link Communications*
FAA AC 120-70C	*Operational Authorization Process for Use of Data Link Communication System*
FAA AC 150/5300-19	*Airport Data and Information Program*
FAA Order 5010.4A	*Airport Data and Information Management*

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level and the appropriate state or local regulations of the authority having jurisdiction over the airport.

Key Considerations

Regardless of whether L-UAS are integrated or segregated at an airport, the potential for new fire issues related to alternative fuel sources will be present. The airport sponsor may not be familiar with all applicable regulations. Thirty-two percent of the L-UAS reviewed for this research project were either fully or partially electric-driven, according to the USRD. Many use a hybrid power supply that includes multiple power sources because higher energy, power densities, and endurance can be achieved. Lithium batteries are one such source widely being explored and tested because they are generally lighter than other batteries. Many of the eVTOL being designed now are pushing the limits of lithium batteries. Hydrogen fuel cells are also being explored. Regardless of the means, new energy sources bring their own set of complexities to the risks of fires at an airport.

The Energy Policy Act of 1992 defines the following as alternative fuels: pure methanol, ethanol, and other alcohols; blends of 85% or more of alcohol with gasoline; natural gas and liquid fuels domestically produced from natural gas; propane; coal-derived liquid fuels; hydrogen; electricity; pure biodiesel (B100); fuels, other than alcohol, derived from biological materials; and P-Series fuels (non-petroleum that can substitute for gasoline) (U.S. DOE n.d.). However, the existing standards for aircraft firefighting are based on conventional liquid fuel for internal combustion engines and the dangers and risks associated with this type of fuel.

Thermal runaway is a particularly dangerous situation that can result from lithium-based batteries, especially lithium-ion. This happens when heat is created inside the battery at a faster rate than the heat can be dissipated. Overheating in one battery cell causes the temperature in other cells to increase and the runaway effect occurs. The resulting fires are difficult to extinguish and often reignite long after they have been extinguished. Hydrogen fuel cells present another set of dangers, especially because hydrogen ignites more easily than many other fuels. Hydrogen also presents a hazard of "cold burn" because it requires substantial cooling to be stored in a liquid form.

Gap Analysis and References

There is a gap in FAA guidance on the storage or charging of batteries and the potential for electromagnetic interference with traditional NAVAIDs they may present. Also, the researchers found no FAA guidance during the literature review for addressing fires from alternative fuels or a fire related to the charging system for an electric-powered aircraft battery. This also represents a gap. However, information for addressing fires from alternative energy sources can be obtained from the National Fire Protection Association (NFPA), Society of Automotive Engineers (SAE), and the National Transportation Safety Board (NTSB). This guidance can be followed until FAA provides further guidance.

While this is not an exhaustive list, the following documents can provide guidance for traditional and alternative fuel source safety (Table 3-8).

Obstacles and Airspace

Impact Overview

As established through this research, L-UAS operations are being conducted at multiple traditional airport facilities across the country. These aircraft have been shown to perform like traditional aircraft on approach and departure from an airport. The major exception in similarity comes with those aircraft launching or landing by means other than a runway or helipad.

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Table 3-8. Fueling and charging.

AIRFIELD COMPONENT	FUELING AND CHARGING
Integration Potential	Medium.
Gap Areas	Lack of guidance on handling of alternative fuel sources at an airport, fire protection response, and electromagnetic interference.
Potential Issues	Thermal runaway; hazardous material storage separation requirements; firefighting requirements for different fire sources; potential electromagnetic interference between battery charging and traditional NAVAIDs.
APPLICABLE STANDARD FOR GUIDANCE	**TITLE**
FAA AC 150/5230-4B	*Aircraft Fuel Storage, Handling, Training, and Dispensing on Airports*
International Fire Code (IFC)/ National Fire Protection Association (NFPA) 1 Uniform Fire Code	*Requirements for hydrogen fueling stations, flammable gas, and cryogenic fluid storage*
IEC SC 23H	*Plugs, Socket-outlets, and Couplers for industrial and similar applications, and for Electric Vehicles*
IEC TC 69	*Electrical power/energy transfer systems for electrically propelled road vehicles and industrial trucks*
NFPA 855	*Standard for the Installation of Energy Storage Systems (ESS)*
NFPA 70	*National Electrical Code (NEC)*
NFPA 55	*Compressed Gas and Cryogenic Fluids Code*
NFPA	*Emergency Field Guide: Hybrid, Electric, Fuel Cell, and Gaseous Fuel Vehicles*
NFPA 2	*Hydrogen Technologies Code*
NFPA 400	*Hazardous Material Code*
NFPA 853	*Standard for the Installation of Stationary Fuel Cell Power Systems*
NREL	*Hydrogen Technologies Safety Guide*
Occupational Safety and Health Administration (OSHA) 1910 Subpart H	*Safe storage, use, and handling of hydrogen in the workplace*
SAE J2990/1	*Hydrogen and Fuel Cell Vehicle First and Second Responder Recommended Practice*
SAE J1772	*Surface Vehicle Standard–SAE Electric Vehicle and Plug in Hybrid-Electric Vehicle Conductive Charge Coupler*
NTSB	*Safety Risks to Emergency Responders from Lithium-Ion Battery Fires in Electric Vehicles*

Obstacle identification and evaluation for an airport are generally associated with departures, approaches, and landings. Obstacle evaluation typically occurs during the airport layout plan (ALP) update/revision process and/or when a Notice of Proposed Construction or Alteration (FAA Form 7480) is filed with FAA. The existing standards for airports and heliports prescribed in 14 CFR Part 77—Safe, Efficient Use, and Preservation of the Navigable Airspace allow L-UAS to interact with the airport like traditional aircraft if they meet the same performance capability. If they do not, alternative takeoff and landing areas may be necessary.

Key Considerations

Obstructions should not hinder L-UAS from operating at airports unless the aircraft are landing or departing from a surface other than the existing runways or helipads associated with the facility. Existing public-use landing surfaces are analyzed for obstructions, and 14 CFR Part 77 helps protect public-use airports and instrument-capable heliports from new obstructions. However, unless an official takeoff and landing surface is declared, FAA has no way of evaluating

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it for obstructions. Therefore, using existing infrastructure to the extent possible will help to ensure the same obstruction protection for L-UAS as for other aircraft.

If the UAS use a launch and recovery platform other than horizontal or vertical (i.e., they do not use an existing designated runway or helipad) or need to be segregated for another reason, an update of the ALP is required for federally obligated airports to ensure safety is achievable for this segregated area. For non-federally obligated airports, filing an FAA Form 7480 is required under 14 CFR Part 157—Notice for Construction, Alteration, Activation, and Deactivation of Airports rather than only updating the ALP. Once an official landing and takeoff area is established via the ALP or Form 7480, approach and departure obstacle clearance surfaces can be evaluated for any new runways and helipads, and obstruction protection will be offered through the 14 CFR Part 77 process as appropriate.

Like 14 CFR Part 77, FAA AC 150/5300-13, *Airport Design*, details additional airspace at the airport, including instrument departure and threshold siting surfaces, and standards for terminal instrument procedures (TERPS) that are evaluated for obstructions as part of an airport sponsor's ALP. It also details surfaces that must be clear on the ground (e.g., runway and taxiway safety areas and runway protection zones). If the L-UAS do not interfere with these surfaces, integration is possible. Otherwise, segregation may be necessary. The dimensions of proposed L-UAS should be closely evaluated before this decision is made. If an L-UAS is in a different ADG or AAC than the existing design aircraft, new design surfaces may be necessary if the volume of operations warrants it.

As covered earlier in this section, aircraft should launch and recover from the airport's existing runways and helipads as much as possible to avoid the need to establish another landing surface. Establishing a runway or helipad at an existing airport is relatively straightforward; however, FAA has not established design standards for landing areas specifically designated for L-UAS. The agency is researching design standards needed for aircraft certifying in the eVTOL category that would operate on a vertipad or vertiport, which may be different from a helipad or heliport. FAA AC 150/5390-2C, *Vertiport Design*, was canceled in 2010. While an updated vertiport design AC is anticipated, it will not likely be published for a few years. The existing heliport design and airspace surfaces are transferable to VTOL to some degree, but the exact performance capabilities of L-UAS in this category are still unknown, so it is currently difficult to determine whether the heliport approach and design surfaces are adequate.

FAA published a memorandum on June 22, 2021, entitled *Process for Submitting and Reviewing Proposed Landing Pads and Supporting Equipment for Advanced Air Mobility and Electric Aircraft*. This document provides some guidance on designated landing areas for VTOLs. As stated earlier, the airport ALP must be updated to show any new infrastructure or fixed equipment (e.g., catapult launch mechanism). Additionally, all applicable National Environmental Protection Act (NEPA) analyses may be required. If this process is followed and an FAA airspace determination is made approving the location for a VTOL landing area at the airport for L-UAS in this category, "FAA reserves the right to provide future input/conditions on this proposal and to inspect the location at a later date and make recommendations for design and safety standards" (FAA 2021, 5). Therefore, the process to establish a landing area, other than a runway or helipad, is more complicated and likely to take longer. Additionally, the area may have to be modified when new standards are developed.

Gap Analysis and References

FAA has design standards only for airports and heliports that cover conventionally piloted fixed wing aircraft and helicopters. A gap exists for L-UAS that cannot land on a traditional runway or helipad. Currently, there is no guidance on approach and departure airspace or obstacle consideration needed for these aircraft. (See Table 3-9.)

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Table 3-9. Obstacles and airspace.

AIRFIELD COMPONENT	OBSTACLES AND AIRSPACE
Integration Potential	High for operations on existing runways or helipads. Medium for operations in other areas.
Gap Areas	Gap area does not exist as research and case studies demonstrate L-UAS operate in flight similarly to existing crewed aircraft. Specific references to UAS obstacle identification, protection, and evaluation do not exist; however, the need for them to be specified is not considered critical. Aircraft approach speeds and dimensions of L-UAS may impact airspace considerations, and L-UAS information on approach speeds is not readily available. No design standards exist for facilities serving aircraft that do not land or take off from a runway or helipad.
Potential Issues	Launch or recovery from areas other than runways or helipads could be hazardous because they have not been evaluated for safety at the airport. Establishing a new landing area for L-UAS that do not use traditional runways or helipads will likely take longer and be more complicated.
APPLICABLE STANDARD FOR GUIDANCE	**TITLE**
14 CFR Part 77	*Safe, Efficient Use, and Preservation of the Navigable Airspace*
14 CFR Part 91	*General Operating and Flight Rules*
FAA AC 150/5300-13	*Airport Design*
FAA AC 150/5300-18B	*General Guidance and Specifications for Submission of Aeronautical Surveys to NGS: Field Data Collection and Geographic Information System (GIS) Standards*
FAA 150/5390-2C	*Heliport Design*
FAA Order 7200.2	*Procedures for Handling Airspace Matters*
FAA Memorandum, June 22, 2021	*Process for Submitting and Reviewing Proposed Landing Pads and Supporting Equipment for Advanced Air Mobility and Electric Aircraft*

Additionally, to determine whether an L-UAS is compatible with the airport's design surfaces, its AAC must be known. This previously identified gap also affects this area.

Environmental Impacts

Impact Overview

Environmental, noise, and emissions planning for airports must comply with NEPA, implementing regulations by the Council of Environmental Quality or any special purpose laws. The FAA has a series of orders, ACs, and other guidance for conducting environmental reviews of airport improvements that can be used when assessing UAS operations. This research did not identify any environmental impact categories that require unique analysis for UAS aircraft as compared to crewed aircraft operations. Additionally, the interviews conducted with airport sponsors (see Appendix B) did not identify any specific issues related to compliance with NEPA. However, noise and emissions modeling capabilities do not include the current fleet of L-UAS, so some adjustments and assumptions are needed for their assessment.

In general, the most common environmental challenge for airports is related to noise. Aircraft noise certification metrics and procedures are provided in 14 CFR Part 36. These metrics are then used in FAA's *Aviation Environmental Design Tool* (AEDT) software for evaluating noise levels for projects at federally obligated airports. The ability to readily assess the operational impacts of existing or future L-UAS aircraft is limited and requires using fixed wing or

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helicopter substitutions in FAA's AEDT. The software does not include a database of sound-level information for L-UAS or a generic L-UAS model. The same issue exists for aircraft emission; AEDT does not include emission information for L-UAS aircraft in its database. Accordingly, the degree of noise and emissions exposure of new UAS are not known at a level that can be readily modeled to assess potential environmental exposure and impact, so assumptions must be made.

Since some L-UAS operate with batteries as their power source, the potential for a chemical spill exists from a fire involving these types of aircraft. Lithium is often found in aircraft powered by batteries because it is lightweight, has a high energy density, and is receptive to recharging. A damaged lithium-based battery can leak and is considered hazardous because it can catch fire and explode.

Key Considerations

Whether integrated or segregated, L-UAS are not hindered from operating at existing airports based on environmental factors any more than traditionally piloted aircraft. Concerns may be raised from the public based on the "drone" nature of the operations and an individual's right to privacy, so community outreach may be warranted.

Noise analysis may be required under NEPA for certain projects at airports if they change the runway configurations, aircraft fleet, or flight characteristics in a way that affects existing and future noise. The operation of some L-UAS may be segregated from other operations using unique routes at low altitudes not traditionally used for flying. If an L-UAS increases noise above existing levels or changes flight paths, an analysis may be warranted. Additionally, an air emission analysis may be required for any new landing area at an airport in a community that does not meet National Ambient Air Quality Standards (NAAQS), as defined by the Environmental Protection Agency (EPA). Since L-UAS noise and emission levels are unknown, coordination with the manufacturer is likely needed to understand each aircraft's noise and emission profiles.

In many cases, L-UAS may provide an environmental benefit or have very little impact on noise or emissions because they do not use internal combustion engines or do not need the same amount of power required for air carrier-size aircraft. Approximately 32% of the L-UAS reviewed in the USRD were either fully or partially electric driven, so they result in little to no source emissions and are often quieter than traditional aircraft.

While there is a gap in information available to assess L-UAS aircraft in the AEDT model, the FAA Office of Environment and Energy can provide guidance to an AEDT user on an appropriate FAA-approved substitution aircraft. The process and information to make a request is provided in the FAA document "Guidance on Using the Aviation Environmental Design Tool (AEDT) to Conduct Environmental Modeling for FAA Actions Subject to NEPA."

Finally, some L-UAS are powered by lithium-based batteries, and a damaged battery almost always poses some type of hazard. If a lithium battery pack is damaged, hazardous waste may spill from the unit that should be treated with the proper relevant personal protective equipment and cleanup procedures. These spills may make their way to trench or slot drains or become part of stormwater runoff if appropriate measures are not put in place to address this possibility.

Gap Analysis and References

L-UAS aircraft are not available in FAA's AEDT model; therefore, studying the impacts from noise and emissions is not possible using the AEDT tool. The void in L-UAS aircraft in the tool represents a gap in available specifications that needs to be addressed.

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The following documents provide requirements and guidance for environmental analysis, modeling, and noise planning for traditional aircraft and are transferable to L-UAS with FAA consultation on the appropriate substitution aircraft for noise and air emission analysis (Table 3-10).

Airport Passenger Terminal

Impact Overview

A clear need for L-UAS to have access to a commercial airport terminal was not uncovered in this research; however, integrating them into the airport terminal could be a consideration. The need for terminal connectivity occurs mainly in the apron/ramp area for preflight and postflight procedures or loading and unloading. The USRD revealed only 14 L-UAS over 12,500 lb that have the potential for carrying large loads and may, therefore, need more maneuvering space for

Table 3-10. Environmental impacts.

AIRFIELD COMPONENT	ENVIRONMENTAL IMPACTS
Integration Potential	High.
Gap Areas	Noise and emission modeling of new L-UAS aircraft.
Potential Issues	Community concerns about noise and emission impacts of L-UAS, especially if operating on separate routes from other aircraft at an airport at low altitudes, like helicopter operations.
APPLICABLE STANDARD FOR GUIDANCE	**TITLE**
40 CFR Parts 1500-1508	*Regulations for Implementing the Procedural Provisions of the National Environmental Policy Action*
14 CFR Part 150	*Airport Noise Compatibility Planning*
14 CFR Part 161	*Notice and Approval of Airport Noise and Access Restrictions*
42 USC Chapter 85	*Air Pollution and Prevention Control (Clean Air Act)*
14 CFR Part 36	*Noise Standards: Aircraft Type and Airworthiness Certification*
14 CFR Part 34	*Fuel Venting and Exhaust Emissions Requirements for Turbine Engines Powered Airplanes*
DOT Order 5610.2(a)	*Department of Transportation Actions to Address Environmental Justice in Minority Populations and Low-Income Populations*
FAA Order 1050.1F	*Environmental Impacts: Policies and Procedures (In the process of being updated)*
FAA Order 5050.4B	*National Environmental Policy Act (NEPA) Implementing Instructions for Airport Actions*
FAA JO Order, Chapter 32	*Procedures for Handling Airspace Matters, Environmental Matters*
FAA Order 1050.1F Desk Reference	*1050.1F Desk Reference Version 2 —Office of Environment and Energy*
FAA Environmental Desk Reference for Airport Actions	*Environmental Desk Reference for Airport Actions—Office of Airport Planning and Programming—Airports Planning and Environmental Division, APP 400*
AC 36-1H	*Noise Levels for U.S. Certificated and Foreign Aircraft*
AC 36-3H	*Estimated Airplane Noise Levels in A-Weighted Decibels*
AC 150/5020-1	*Noise Control and Compatibility Planning for Airports*
AC 150/5200-3C	*Hazardous Wildlife Attractants on or Near Airports*
DOT-VNTSC-FAA-21-05	*Aviation Environmental Design Tool (AEDT) Version 3c User Guide*
DOT-VNTSC-FAA-21-06	*Aviation Environmental Design Tool (AEDT) Version 3c Technical Guide*
FAA Office of Environment and Energy Primer	*Aviation Emissions, Impacts & Mitigation: A Primer*

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loading and unloading; however, these aircraft have not typically been used for cargo. The listed uses for these 14 aircraft were as follows:

Communications	Patrol
Disaster response	Security
Electronic warfare	Search and rescue
Imaging	Survey
Intelligence	Mapping
Surveillance	Target acquisition
Reconnaissance	Attack
Logistics	Weapons delivery
Observation	Environmental research or monitoring

Key Considerations

Given that no L-UAS are being used for passenger travel, design guidelines for commercial terminals are not immediately needed. While the Boeing NeXt program was working on future uncrewed air mobility, this effort was "paused" in late 2020 due to the "pandemic-driven cash crisis" (Gates 2020). Aurora Flight Sciences, a subsidiary of Boeing, is working on an optionally piloted aircraft, the Centaur, which was developed from the Diamond DA42 twin-engine, four-passenger aircraft (Aurora Flight Sciences 2021). Airbus is also "introducing autonomous flight features to its helicopter Flightlab through a project code-named Vertex" (Airbus 2021) and recently demonstrated autonomous taxi, takeoff, and landing in an A250 (Pallini 2020). While Airbus, Boeing, and other large aircraft manufacturers are working with artificial intelligence for automating flights, the aircraft are traditional and are interacting with the airport terminal in the same way as before. The most likely future for large, passenger aircraft is for existing transport category aircraft to transition to autonomy, so the existing guidance found in FAA AC 150/5360-13, *Airport Terminal Planning*, is expected to continue to be applicable.

Both Xwing and Reliable Robotics have retrofitted Cessna Grand Caravans and other single-engine aircraft for autonomous flight (Black 2021). These aircraft also do not traditionally need a commercial airport terminal, but they have the potential to act as cargo feeder aircraft. Small eVTOL UAM aircraft (12,500 pounds and less) are also being developed and are getting the most attention in research and development across the globe. However, none have achieved FAA certification yet, and the frontrunners to certification will be piloted. These smaller aircraft would not initially use a commercial airport terminal but would likely transition through a fixed base operator (FBO) or cargo facility just like existing general aviation aircraft, whether they are single-engine piston aircraft or multi-engine jets. The terminal area needs for L-UAS referenced above are similar and, while the lift and thrust mechanisms may be unconventional, existing guidance applies to these aircraft as it relates to airport terminal planning. *ACRP Report 113: Guidebook on General Aviation Facility Planning*, provides guidance for planning airport facilities for similar general aviation aircraft (Sander et al. 2014).

Gap Analysis and References

While there is a gap in information for passenger-carrying UAS, this is primarily because there are no aircraft with FAA approval. Information does exist in terminal development related to traditional passenger aircraft that can provide guidance. This information should be paired with detailed aircraft specifications from UAS manufacturers for any future passenger-carrying UAS. (See Table 3-11.)

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Table 3-11. Airport terminal.

AIRFIELD COMPONENT	AIRPORT TERMINAL
Integration Potential	High.
Gap Areas	Aircraft specifications, as these aircraft have not yet been certified.
Potential Issues	None identified.
APPLICABLE STANDARD FOR GUIDANCE	**TITLE**
FAA AC 150/5360-13	*Airport Terminal Planning*
ACRP Report 11	*Guidebook on General Aviation Facility Planning*
ACRP Report 96	*Apron Planning and Design*
NFPA 415	*Standard on Airport Terminal Building, Fueling Ramp Drainage and Loading Walkways*

Air Traffic Control

Impact Overview

ATC facilities may include the air traffic control tower (ATCT), terminal radar approach control (TRACON) facilities, Department of Defense (DOD) radar approach control facilities and military ground-controlled approach (GCA) facilities, remote transmitter/receivers (RTR), and surveillance equipment. Facilities housing ATC operations are not traditionally the responsibility of the airport sponsor to install, operate, and maintain. The FAA or Contract Tower operator will be primarily responsible for identifying the specific needs to support L-UAS in the airspace.

As L-UAS technology advances and their integration into the NAS increases, additional ATC facilities supporting surveillance and communications should be anticipated. For example, the technology, policies, and operational protocols for unmanned aircraft system traffic management (UTM) are in research and development. It is recommended that airport sponsors and stakeholders review and understand UTM solutions when research and development are completed. As solutions materialize, airports will need to determine whether the airport requires additional ATC capability to support safe L-UAS operations.

L-UAS operations occurring at the stakeholder airports interviewed are required by FAA to operate L-UAS using the existing communication networks, surveillance technology, and ATC protocols traditionally piloted traffic are required to use. The airport sponsor and stakeholders may be affected by required investments to support L-UAS operations. The gaps in criteria related to ATC facilities may not be fully realized until FAA and the L-UAS industry have achieved a standard for an acceptable level of aircraft operations safety.

Key Considerations

According to the interviews conducted with airport managers and DOD L-UAS subject matter experts, UAS operations are integrated at military air bases and joint-use civilian airports with operating ATCTs with minimal impact during safe operating conditions. These operations are likely successful because of the close coordination between the airport, L-UAS operators, ATC, and the OEMs in some cases. The major considerations for airport sponsors are coordinating with the parties involved in the mission and determining whether additional ATC facilities are needed to achieve acceptable levels of safety. L-UAS may use various technologies and procedures to ensure safety; the lost-link procedures must be clearly defined and reviewed to understand additional considerations.

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When considering L-UAS operations at an airport with an operating ATCT, airport design principles, as detailed in AC 150/5300-13, *Airport Design*, and FAA Order 6480.4, *Air Traffic Control Tower Siting Process*, should be followed to ensure a proper line-of-sight is available to air traffic controllers to allow for safe separation of crewed aircraft and L-UAS. Existing communication and surveillance infrastructure typically operates on a standardized layout; land-use assessment should be completed for the ability to properly site each facility. Implementation of ATC facilities to support L-UAS integration should follow the same principles as the implementation of any other airfield infrastructure to ensure that existing facilities are not affected.

Gap Analysis and References

While no significant gaps exist in airport design criteria, gaps are expected in the ATC facility infrastructure and procedural requirements that potentially impact airport land-use and safety protocols (Table 3-12).

Aircraft Rescue and Firefighting Support

Impact Overview

The issues presented in this section are similar to those in the Fueling and Charging section. The standards set for aircraft rescue and firefighting vehicles, personnel, equipment, and training are all geared toward a response to aircraft that are typical in the existing NAS. UAS are not typical; in particular, some are powered by alternative fuels, such as batteries or hydrogen. The technology that goes into battery-powered aircraft produces conditions that burn much hotter than conventional fuels in most cases. In the past several years, the rapidly growing proliferation of battery-powered cars has produced documented cases of car fires causing more damage given their intense heat and propensity to reignite after being extinguished.

According to the U.S. Fire Administration (USFA), "safety issues with high-voltage, lithium-ion batteries used in electric vehicles first gained widespread attention when a Chevrolet Volt caught fire three weeks after a crash test in May 2011. Challenges for responders remain, due to

Table 3-12. Air traffic control facilities.

AIRFIELD COMPONENT	AIR TRAFFIC CONTROL FACILITIES
Integration Potential	Medium.
Gap Areas	Emerging ATC technology and procedures may introduce gaps when applied to conventional facilities based on airports to support traditional piloted flight. UTM systems, emerging DAA systems, and no standardized lost-link procedures are areas that could potentially introduce gaps between L-UAS operations and traditional piloted flights.
Potential Issues	Maintaining surveillance and communication capabilities with L-UAS may require additional infrastructure at an airfield. Proper line-of-sight between L-UAS and air traffic controllers. Additional infrastructure may be necessary for normal as well as degraded conditions like radio failure, transponder failure, part-time ATCT, or lost-link procedures.
APPLICABLE STANDARD FOR GUIDANCE	**TITLE**
AC 150/5300-13A	*Airport Design*
FAA Order 6580.3	*Remote Communication Facilities Installation Handbook*
FAA Order 6580.4	*Air Traffic Control Tower Siting Process*

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lack of standardized emergency response guidance for incidents involving electric vehicles with high-voltage batteries." (USFA 2021)

While not an extremely common fuel source, hydrogen is gaining attention as a sustainable energy source. Hydrogen fuel cells present another set of dangers to firefighters, especially because hydrogen is very light, ignites more easily, and a resultant fire is different from other fuel-based fires, which are heavier and pool close to the ground during a leak. Hydrogen rises rapidly, its flame is invisible, and even static electricity from a finger can set it off (Crowcon 2021).

Key Considerations

The categories of risks associated with battery fires are thermal runaway, stranded energy, and battery re-ignition. The NTSB has issued recommendations to the National Highway Traffic Safety Administration to develop guidance for emergency responders to address the implications of these risks for extinguishing fires, mitigating fire hazards, and safely storing batteries in cars. The implications for battery-powered L-UAS are like those for cars. Other risks associated with hydrogen fuel may require attention if this fuel source is used.

Gap Analysis and References

In reviewing the current standards for aircraft rescue and firefighting (ARFF), a gap exists specific to non-traditional aviation fuel sources: when ignited, these fuel sources act much differently than typical aviation fuels. Therefore, guidance specific to L-UAS and their internal safeguards, battery composition, and fuel sources is appropriate along with training all first responders on the firefighting techniques needed for any new fueling sources introduced by these aircraft (Table 3-13).

Table 3-13. Fire and rescue support.

AIRFIELD COMPONENT	FIRE AND RESCUE SUPPORT
Integration Potential	High. ARFF standards are specific to one essential function within airport operations. The addition of a different type of aircraft (different in the sense of pilotless) and potential fuel differences from conventional piloted aircraft is minor.
Gap Areas	Standards and firefighting training for any fuel or energy source specific to L-UAS and not a traditional aviation fuel.
Potential Issues	L-UAS covers hundreds of aircraft and several kinds of fuels. While Jet-A and AVgas are all consistent with piloted aircraft, battery-powered UAS might present different issues that ARFF personnel are not familiar with or trained to a standard on.
APPLICABLE STANDARD FOR GUIDANCE	**TITLE**
AC 150/5220-17B	*ARFF Training Facilities*
AC 150/5220-1E	*Guide Specification for ARFF Vehicles*
AC 150/5210-14B	*ARFF Equipment, Tools and Clothing*
AC 150/5210-6D	*Aircraft Fire Extinguishing Agents*
AC 150/5210-17C	*Programs for Training of Aircraft Rescue and Firefighting Personnel*
AC 150/5200-12C	*First Responders' Responsibility for Protecting Evidence at the Scene of an Aircraft Accident/Incident*
AC 150/5230-4B	*Aircraft Fuel Storage, Handling, Training, and Dispensing on Airports*
AC 150/5200-31C	*Airport Emergency Plan* (includes Change 2)
SAE J2990/1	*Hydrogen and Fuel Cell Vehicle First and Second Responder Recommended Practice*
NTSB	*Safety Risks to Emergency Responders from Lithium-Ion Battery Fires in Electric Vehicles*
H2 Hydrogen Tools	*Hydrogen Safety Training Materials*

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CHAPTER 4

Determining Airport Compatibility

This chapter guides the airport sponsor at the macro level by identifying areas across the airport ecosystem that may be of concern for L-UAS integration based on the vehicles' capabilities and the airport's overall environment. Airport sponsors have historically implemented some degree of integration and segregation for different types of aircraft and activities (e.g., fixed wing, helicopters, and skydivers) at their airports. L-UAS are no different. The introduction of these aircraft to the existing airport operational environment typically comes with a few common questions:

- How can the L-UAS become part of the ecosystem inhabited by crewed aircraft?
- Can the L-UAS integrate into existing airport operations?
- Do the L-UAS need to be segregated from the other airport facilities, occupants, and operators?

In view of these questions, this guide defines integration and segregation as follows:

- **Integration.** The permissible mix of crewed and uncrewed aircraft into airport operations while using existing airport infrastructure or using airport infrastructure compliant with existing airfield design criteria, allowing intermingled operations.
- **Segregation.** The physical separation of crewed and uncrewed aircraft on the airport surface or property, not allowing the mixing of aircraft and operations.

As this guide demonstrates, airport sponsors may not need to limit L-UAS operations to either one or the other situations listed above. Rather, they can determine whether the L-UAS fit into the traditional planning and design criteria applied to each specific airport component, and where they do not, determine what needs to be segregated.

Key Considerations

As detailed in Chapters 2 and 3, evaluating the design criteria applied to the existing facilities against the L-UAS platforms and capabilities will determine the level at which the L-UAS can be integrated into, or need to be isolated from, crewed aircraft operations. As previously detailed, careful consideration of the vehicle operational characteristics and system components (such as landing/takeoff mechanisms, vehicle type and dimensions, and command and control technology) is required because they affect the capabilities of an existing airport facility to accommodate an L-UAS.

Determining the extent to which L-UAS are integrated versus segregated will be driven by reviewing the key attributes of an airport, which are listed in Table 4-1, and their associated areas of impact. (Note: Chapter 3 of this guide provides information and impact considerations in greater detail for each specific airport component requiring review along with the criteria applicable to each.)

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Table 4-1. Key areas of impact.

ATTRIBUTE	FEATURE
Airfield surface infrastructure	Runways, taxiways, apron/ramp, configuration, pavements
Airfield navigation/data infrastructure	Light, markings, signage, NAVAIDs, communications, data
Planning and environmental	Fueling/recharging, noise, environmental
Auxiliary/support facility	Hangars, support facilities, terminal facilities
Airspace and air traffic	ATC facilities, obstacles, airspace
Emergency/safety operations	Fire and rescue support

The most common situations airport sponsors encounter regarding L-UAS fall into two scenarios:

- **Scenario 1.** Support of an L-UAS vehicle or type for an individual operator. This situation is straightforward, requiring information for only one vehicle and one mission type.
- **Scenario 2.** Support various L-UAS—acting as a hub or incubator for emerging aviation technology—potentially with the premise of attracting operators to become tenants. This situation is generally more complex because of the non-standardization of the aircraft, which may vary in type, size, support facility needs, and missions.

Regardless of the driving factors for accommodating L-UAS operations, the areas of impact that require consideration are the same. The major difference between the two scenarios is the level of review and planning when cross-referencing each airfield component/feature listed above to the L-UAS platforms and capabilities.

Segregation Versus Integration

There are numerous examples of airports across the country that accommodate multiple aircraft types and aviation activities. Existing facilities that accommodate airplanes, helicopters, gliders, seaplanes, banner towing, and skydiving demonstrate the integration of mixed aircraft types and activities into a singular facility or operation. While new, L-UAS are simply a different category of aircraft seeking a place to begin and end their missions safely and efficiently. A similar decision-making process will be used for determining how they will be mixed with the existing crewed aircraft operations as for gliders, banner towing, and other crewed aircraft types. The research on L-UAS provided in Chapter 2 demonstrates the variety of vehicle type, size, propulsion, and launch/recovery considerations critical for determining the impacts to airport infrastructure and the standards underlying airport design.

As identified in the case-study interviews with airports permitting L-UAS operations (Appendix B), many of these aircraft have been integrated with existing crewed aircraft operations. Despite being considered integrated at the macro level, each of the studied airports possessed unique levels of segregation and integration depending on the vehicle types, sizes, and performance characteristics, including variables introduced by specific airport configurations, existing airport users (e.g., military, commercial airlines, and flight training schools), and existing support facilities (e.g., ATC, hangars, and fueling). The remainder of this chapter covers the major considerations of integration versus segregation as revealed through these stakeholder interviews and independent research.

Guide to Airport Considerations for Segregation Versus Integration

Table 4-2 summarizes how an L-UAS characteristic or airport feature individually sways the ability to integrate or segregate the aircraft at an airport facility under current FAA design

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Table 4-2. Considerations for segregation versus integration.

L-UAS Characteristic	Scale of Segregation Vs. Integration	Brief Description of Challenges and Impacts
Fixed Wing 100% (HTOL)		L-UAS operating as fixed wing aircraft during both takeoff and landing vary in their ability to integrate. Their integration heavily depends on the vehicle size/weight and its need to use existing runway/taxiway surfaces for movement, with the heavier aircraft integrating more easily than the lighter ones (those similar in weight to an ultralight aircraft). The lighter ones will have more trouble mixing with traditional aircraft depending on their ability to withstand turbulence and crosswinds.
Rotorcraft 100% (VTOL)		Aircraft that operate with rotors during all phases of flight (i.e., they do not have wings or tilting rotors) do not need a runway for takeoff and landing. Their taxi capabilities vary, so integration with the runway and taxiway system may be limited. They may fly a traditional approach or segregate from fixed wing aircraft to a specific “helipad” on the airport. As for fixed wing L-UAS, weight, turbulence tolerance, and crosswind capability are also factors.
Hybrid (HTOL & VTOL)		Hybrid aircraft vary between fixed wing and rotorcraft during either takeoff or landing. Their need for a runway or ability to use a taxiway will impact their integration ability. The uniqueness of each vehicle will require the airport sponsor to understand its specific requirements to determine how existing infrastructure can support its operations.
Assisted Launch and Recovery Mechanism		These aircraft will not likely be able to use the existing runways, taxiways, and ramps because they require a special apparatus for launch and recovery. Segregation is probable with special attention to assure that their approach and departure paths do not cross existing runways and do not encroach on any other protected surfaces.
Propulsion System (Combustion or Turbojet)		As described in Chapter 2, various propulsion system types use liquid fuels and renewable energy. Aviation fuel has criteria established and poses no additional impacts. L-UAS that use other fuels like diesel and kerosene may require additional consideration to determine the appropriate safety measures to support these aircraft.

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Table 4-2. (Continued).

L-UAS Characteristic	Scale of Segregation Vs. Integration	Brief Description of Challenges and Impacts
Propulsion System (Electric)		The need to segregate electrically propelled aircraft is related to their charging characteristics vs. their flight or taxiing characteristics. Electrically propelled aircraft primarily impact airport infrastructure by their need for high-voltage electrical power and vehicle-specific charging systems. Additional risks are introduced by the hazardous materials their batteries contain, which can leak, explode, or catch fire. Caution must be taken in determining safety response for these aircraft and additional risk to other airport facilities.

Airport Feature	Scale of Segregation Vs. Integration	Brief Description of Challenges and Impacts
Airfield Configuration (Movement and Non-Movement Areas)		Each airport is unique in its location, configuration, and assets. Each airfield surface layout and L-UAS vehicle type poses unique issues for determining the level and locations of integration on the airport surface. Airports with multiple movement surfaces will increase integration potential by allowing the best option to be chosen for the mission of each user at the time.
Airfield Capacity		The capacity of an airport is defined by the number of operations it can handle during a specific period. If the airport demand is high and capacity is low, integrating UAS will be more difficult and segregation may be needed.
Airport Navigation and Guidance (Lighting, Signage, NAVAIDs, Marking)		Properly equipped L-UAS should be able to use an airport's existing devices to attain the situational awareness necessary to operate or have equivalent mechanisms for attaining it.
Environmental Impacts and Noise		The research yielded little additional information about impact on airports and their vicinities from the operations of L-UAS. However, very little is known about their noise signatures. If the L-UAS use electric propulsion, potential environmental impacts may arise from battery systems storage, disposal, fire, and hazardous materials.

(continued on next page)

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Table 4-2. (Continued).

Airport Feature	Scale of Segregation Vs. Integration	Brief Description of Challenges and Impacts
Airspace and Air Traffic Control (Obstacles, ATC Tower, Surveillance)		An L-UAS equipped with the proper avionics (e.g., VHS communication radio capability, transponder, ADS-B, GPS) should be able to operate and communicate with ATC analogous to crewed aircraft. The need for on-field ATC services depends on the level of crewed aviation service at each airport and is not driven by the L-UAS specifically. If the L-UAS do not have DAA capability, integration is easier with an ATCT that can provide aircraft separation services. No additional airspace protection or obstruction management was identified by interviewees as necessary for L-UAS to operate on and around airports; however, the airport's surveillance capability and the complexity of its airspace class will dictate the procedures under which they operate.

criteria. The objective is not to provide a recipe for integration or segregation, but to assist in the high-level decision-making process when considering L-UAS aircraft types, their support demands, planning and design criteria, and major impact areas.

The indicator gauge (middle column) associated with each item in Table 4-2 illustrates the probability of segregation versus integration as it applies to each characteristic or feature. The indicator gauges are a representation of the research and industry expertise applied and are not mathematically calculated. They are based on the variables of aircraft type, size, operating characteristics, and impacts to existing facility design from the L-UAS characteristic or airport feature. The location on the Scale of Segregation Vs. Integration gauges in Table 4-2 reflects the collective conclusions made from airport case-study interviews, technology research, and other existing literature identified during the research.

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APPENDIX A

Technology Review Inventory

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Table A-1 AUVSI USRD Database Summary (by Wingspan/Width)

Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Aurora Flight Sciences Corporation	Odysseus	Fixed Wing	243	2,000	-	-	-	Other	Runway, Wheeled	Electric Motor, Propeller
Boeing, Phantom Works	Phantom Eye	Fixed Wing	150	9,800	-	-	150	Runway, Wheeled	Runway, Wheeled	Other, Combustion Engine, Propeller
Aurora Flight Sciences Corporation	Orion	Fixed Wing	132	11,200	50	-	67	Runway, Wheeled	Runway, Wheeled	Turbo Engine, Combustion Engine, Propeller
Northrop Grumman Corporation	RQ-4 Block 20 Global Hawk	Fixed Wing	131	32,250	48	15	310	Runway, Wheeled	Runway, Wheeled	Jet Engine, Turbojet

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Northrop Grumman Corporation	RQ-4 Block 30 Global Hawk	Fixed Wing	131	32,250	48	15	310	Runway, Wheeled	Runway, Wheeled	Jet Engine, Turbojet
Northrop Grumman Corporation	RQ-4 Block 40 Global Hawk (EQ-4B / RQ-4D)	Fixed Wing	131	32,250	48	15	300	Runway, Wheeled	Runway, Wheeled	Jet Engine, Turbojet
Airbus Defense and Space (U.S.)	EURO HAWK	Fixed Wing	131	32,250	48	15	310	Runway, Wheeled	Runway, Wheeled	Jet Engine, Turbojet
Northrop Grumman Corporation	MQ-4C Triton BAMS	Fixed Wing	131	32,250	48	15	322	Runway, Wheeled	Runway, Wheeled	Jet Engine, Turbojet
Northrop Grumman Corporation	RQ-4A Global Hawk	Fixed Wing	116	25,600	44	14	345	Runway, Wheeled	Runway, Wheeled	Jet Engine, Turbojet
Northrop Grumman Corporation	NASA Global Hawk	Fixed Wing	116	26,700	44	15	-	Runway, Wheeled	Runway, Wheeled	Jet Engine, Turbojet
Northrop Grumman Corporation	RQ-4 Block 10 Global Hawk (BAMS-D)	Fixed Wing	116	26,700	44	14	343	Runway, Wheeled	Runway, Wheeled	Jet Engine, Turbojet

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Aurora Flight Sciences Corporation	SunLight Eagle	Fixed Wing	114	173	-	-	-	-	-	Electric Motor, Propeller
Airborne Systems North America	DragonFly ®	Other	110	10,000	-	-	-	Air Launch	Parachute	-
UAVOS, Inc. (U.S.)	APUS-30 Atmo-spheric Satellite (HAPS)	Fixed Wing	98	243	-	-	-	Runway, Wheeled	Runway, Wheeled	Electric Motor, Propeller
General Atomics Aeronautical Systems, Inc.	MQ-9B Sky-Guardian	Fixed Wing	79	12,500	38	-	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
General Atomics Aeronautical Systems, Inc.	Guardian	Fixed Wing	66	10,500	36	12	200	Runway, Wheeled	Runway, Wheeled	Turbo Engine, Combustion Engine, Propeller
NASA Armstrong Flight Research Center	Ikhana (Predator-B)	Fixed Wing	66	10,000	36	-	216	Runway, Wheeled	Runway, Wheeled	Turbo Engine, Combustion Engine, Propeller
General Atomics Aeronautical Systems, Inc.	MQ-9 Reaper (Predator-B)	Fixed Wing	66	10,500	36	12	200	Runway, Wheeled	Runway, Wheeled	Turbo Engine, Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
General Atomics Aeronautical Systems, Inc.	Predator C Avenger	Fixed Wing	66	18,200	44	-	350	Runway, Wheeled	Runway, Wheeled	Jet Engine, Turbojet
Northrop Grumman Corporation	Firebird	Fixed Wing	65	5,000	35	10	-	Runway, Wheeled	Runway, Wheeled	Turbo Engine, Combustion Engine, Propeller
Bell Helicopter Textron, Inc. (U.S.)	V-247 Vigilant	Hybrid	65	29,500	-	-	250	VTOL	VTOL	Combustion Engine, Propeller
Northrop Grumman Corporation	X-47B UCAS-D	Fixed Wing	62	44,000	38	10	460	Runway, Wheeled	Runway, Wheeled	Jet Engine, Turbojet
ShadowAir, Ltd.	Talon	Fixed Wing	59	2,315	28	-	145	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Airborne Systems North America	FIREFLY ®	Other	56	2,400	-	-	-	Air Launch	Parachute	-
General Atomics Aeronautical Systems, Inc.	MQ-1C Gray Eagle	Fixed Wing	56	3,600	28	-	130	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
General Atomics	MQ-1B Predator	Fixed Wing	55	2,250	27	-	70	Runway, Wheeled	Runway, Wheeled	Turbo Engine, Combustion

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Aeronautical Systems, Inc.										Engine, Propeller
General Atomics Aeronautical Systems, Inc.	MQ-1/RQ-1A Predator	Fixed Wing	55	2,550	27	-	70	Runway, Wheeled	Runway, Wheeled	Turbo Engine, Combustion Engine, Propeller
General Atomics Aeronautical Systems, Inc.	Predator XP	Fixed Wing	55	2,550	27	-	70	Runway, Wheeled	Runway, Wheeled	Turbo Engine, Combustion Engine, Propeller
ARCA Space Corporation (U.S.)	AirStrato Explorer	Fixed Wing	53	507	23	5	82	Launcher–Pneumatic, Launcher–Catapult, Launcher–General	Parachute	Electric Motor, Propeller
King Abdulaziz City for Science and Technology (Saudi Arabia)	Saker-1B	Fixed Wing	53	2,500	-	-	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
UAVOS, Inc. (U.S.)	ALBATROSS 2.2	Fixed Wing	49	1,873	22	6	108	Runway, Wheeled	Parachute, Runway, Wheeled	Combustion Engine, Propeller
Kaman Aerospace Corporation (U.S.)	K-MAX	Rotary Wing	48	12,000	52	-	80	VTOL	VTOL	Turbo Engine, Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Aurora Flight Sciences Corporation	AEH-1	Rotary Wing	48	10,500	57	15	-	VTOL	VTOL	Jet Engine, Propeller
NAVMAR Applied Sciences Corp	Teros	Fixed Wing	46	1,500	20	-	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Southwest Research Institute	HiSentinel 80	Other	46	1,116	200	46	18	VTOL	Parachute	Electric Motor, Propeller
Aurora Flight Sciences Corporation	Centaur (DA42 MPP), OPA	Fixed Wing	44	3,935	28	8	160	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Sikorsky Aircraft Corporation, a Lockheed Martin Company	Sikorsky Autono-mous Research Aircraft (SARA)	Rotary Wing	44	11,700	53	15	155	VTOL	VTOL	Turbo Engine, Combustion Engine, Propeller
Aerocross Systems, Inc. (U.S.)	Echo Hawk	Fixed Wing	42	1,475	20	-	100	Runway, Wheeled	Parachute, Runway, Wheeled	Combustion Engine, Propeller
Raspet Flight Research Laboratory-Mississippi State University	Owl / Ethereal	Fixed Wing	40	500	21	4	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
ARCA Space Corporation (U.S.)	AirStrato Pioneer	Fixed Wing	39	485	23	5	54	Launcher–Pneumatic, Launcher–Catapult, Launcher–General	Skid, Belly, Parachute	Electric Motor, Propeller
BAE Systems	QF-4 Phantom	Fixed Wing	38	61,796	30	16	-	Runway, Wheeled	Runway, Wheeled	Jet Engine
Sabrewing Aircraft Company (U.S.)	DRACO-2	Hybrid	38	3,400	28	7	-	VTOL	VTOL	Ducted Fan, Electric Motor
The Center for Interdisciplinary Remotely Piloted Aircraft Studies (U.S.)	Pelican / Skymaster, OPA	Fixed Wing	38	4,300	30	9	80	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Northrop Grumman Corporation	Fire-X / Unmanned 407X	Rotary Wing	37	6,000	35	11	-	VTOL	VTOL	Combustion Engine, Propeller
Northrop Grumman Corporation	MQ-8C Fire Scout	Rotary Wing	35	6,000	41	11	-	VTOL	VTOL	Turbo Engine, Combustion Engine, Propeller
Northrop Grumman Corporation	MQ-5B Hunter	Fixed Wing	34	1,950	23	7	60	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
VIMANA Global, Inc. (U.S.)	VIMANA VTOL AAV	Hybrid	33	2,300	23	-	-	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
Boeing Defense, Space and Security	QF-16	Fixed Wing	33	42,329	49	16	-	Runway, Wheeled	Runway, Wheeled	Jet Engine, Turbojet
Proxy Technologies, Inc.	SkyRaider, OPA	Fixed Wing	32	4,000	20	8	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
International Aviation Supply	Archimede (Sky Arrow U)	Fixed Wing	32	1,565	25	9	90	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Airborne Systems North America	MICRO-FLY®	Other	32	500	2	0	-	Air Launch	Parachute	-
NAVMAR Applied Sciences Corp	TigerShark Block 4 (24 hr loiter)	Fixed Wing	32	800	15	-	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Cubic Corporation (U.S.)	ISR-ONE	Fixed Wing	31	1,300	19	6	120	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
NAVMAR Applied Sciences Corp	TigerShark XP Long Range	Fixed Wing	30	800	-	-	80	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
UAV Solutions, Inc. (U.S.)	Intruder	Fixed Wing	30	850	16	-	75	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Pentagon Performance Inc. (U.S.)	SureCam	Fixed Wing	29	1,320	21	-	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Lockheed Martin Corporation	VARIOUS	Fixed Wing	29	8,300	22	8	160	VTOL	VTOL	Ducted Fan, Jet Engine
Yates Electrospace Corp. (U.S.)	Mercy	Fixed Wing	28	2,000	8	-	-	Air Launch	Skid, Belly	-
Yates Electrospace Corp. (U.S.)	Silent Arrow	Fixed Wing	28	2,000	13	-	-	Air Launch	Skid, Belly	-
Lockheed Martin Corporation	X-56A MUTT (Buckeye)	Fixed Wing	28	480	8	-	-	Runway, Wheeled	Parachute, Runway, Wheeled	Jet Engine, Turbojet
Northrop Grumman Corporation	MQ-8B Fire Scout	Rotary Wing	28	3,150	24	10	62	VTOL	VTOL	Combustion Engine, Propeller
The Boeing Company	Unmanned Little Bird (ULB) / MD-530F Little Bird	Rotary Wing	28	3,300	23	8	130	VTOL	VTOL	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Thorpe SEEOP Corporation	TS2000	Fixed Wing	27	1,000	17	-	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
API Technologies Corporation	PersiStar	Fixed Wing	26	1,322	18	7	-	Runway, Wheeled	Parachute	Combustion Engine, Propeller
Drone America, Inc.	Phoebus	Fixed Wing	26	500	13	4	90	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Frontline Aerospace, Inc.	V-STAR/ER	Fixed Wing	25	2,460	22	-	145	VTOL	VTOL	Ducted Fan, Combustion Engine, Propeller
MIT Lincoln Laboratory	Jungle Hawk Owl	Fixed Wing	24	150	-	-	-	Other	-	Combustion Engine, Propeller
Neany, Inc.	Arrow	Fixed Wing	23	1,350	19	7	60	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Textron Systems	NIGHT-WARDEN	Fixed Wing	23	750	-	-	-	Launcher–Catapult, Launcher–General	Parachute, Runway, Wheeled	Combustion Engine, Propeller
Textron Systems	Shadow M2	Fixed Wing	23	750	-	-	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Griffon Aerospace (U.S.)	Broad-Sword XL	Fixed Wing	23	549	14	-	75	Runway, Wheeled, Launcher–Catapult, Launcher–General	Skid, Belly, Runway, Wheeled	Combustion Engine, Propeller
Griffon Aerospace (U.S.)	MQM-171A Broad-Sword	Fixed Wing	23	551	15	-	75	Launcher–Catapult, Launcher–General	Skid, Belly	Combustion Engine, Propeller
Drone America, Inc.	Ariel	Fixed Wing	22	1,000	16	-	-	Other, Runway, Wheeled	Runway, Wheeled, Water Recovery	Combustion Engine, Propeller
NAVMAR Applied Sciences Corp	Arctic-Shark	Fixed Wing	22	650	14	-	60	-	-	Combustion Engine, Propeller
Joby Aviation	Lotus	Hybrid	22	275	-	-	-	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
NAVMAR Applied Sciences Corp	TigerShark XP	Fixed Wing	22	500	15	3	60	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
NAVMAR Applied Sciences Corp	TigerShark Block 4 (standard)	Fixed Wing	22	600	15	-	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Scion UAS, LLC.	SA-400 Jackal	Rotary Wing	21	1,200	19	7	60	VTOL	VTOL	Combustion Engine, Propeller
Textron Systems	RQ-7B Shadow 200 TUAS	Fixed Wing	20	467	12	-	90	Runway, Wheeled, Launcher–Catapult, Launcher–General	-	Combustion Engine, Propeller
Pegasus Aerospace (U.S.)	PHL-2 Perseus 200	Rotary Wing	20	2,200	25	6	-	VTOL	VTOL	Jet Engine, Propeller
Guided Systems Technologies, Inc.	SiCX-250	Rotary Wing	20	610	16	7	-	VTOL	VTOL	Combustion Engine, Propeller
Guided Systems Technologies, Inc.	SiCX-290	Rotary Wing	20	720	16	7	-	VTOL	VTOL	Combustion Engine, Propeller
UAV Solutions, Inc. (U.S.)	Talon 240	Fixed Wing	20	136	9	-	-	Launcher–Pneumatic, Launcher–Catapult, Launcher–General	Skid, Belly	Electric Motor, Combustion Engine, Propeller
UAV Solutions, Inc. (U.S.)	Talon 240-G	Fixed Wing	20	172	10	-	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
L3Harris Technologies	Viking 400	Fixed Wing	20	540	15	5	60	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
ARCTURUS UAV	JUMP 20	Hybrid	19	210	9	-	-	VTOL	VTOL	Combustion Engine, Propeller
PAE ISR	Resolute Eagle	Fixed Wing	18	185	10	-	100	VTOL, Launcher–Catapult	Skid, Belly, VTOL	Combustion Engine, Propeller
The Boeing Company	Cargo Aerial Vehicle (CAV)	Rotary Wing	18	700	15	4	-	VTOL	VTOL	Electric Motor, Propeller
Sonex Aircraft, LLC (U.S.)	SubSonex TwinJet UAS	Fixed Wing	18	1,100	17	-	-	Runway, Wheeled, Launcher–Catapult, Launcher–General	Runway, Wheeled	Jet Engine, Turbojet
Theiss UAV Solutions	Tarzan TD-1c	Fixed Wing	18	255	11	4	43	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
DroneTech UAV Corp	AV-1 Albatross	Fixed Wing	18	276	10	-	65	Runway, Wheeled	Runway, Wheeled	Electric Motor, Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
DroneTech UAV Corp	AV-1 Albatross VTOL	Hybrid	18	298	10	-	67	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
ARCTURUS UAV	T-20	Fixed Wing	18	185	9	-	45	VTOL, Launcher–Pneumatic, Launcher–Catapult, Launcher–General	Net, Skid, Belly, VTOL	Combustion Engine, Propeller
NAVMAR Applied Sciences Corp	TigerShark	Fixed Wing	18	318	14	4	56	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
L3Harris Technologies	Viking 300	Fixed Wing	18	318	14	4	56	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
American Aerospace Technologies, Inc.	RS-20	Fixed Wing	17	165	-	-	-	Launcher–Pneumatic, Launcher–Catapult, Launcher–General	Skid, Belly	Combustion Engine, Propeller
UAVOS, Inc. (U.S.)	SITARIA-E	Fixed Wing	17	86	9	-	43	Runway, Wheeled	Parachute, Runway, Wheeled	Electric Motor, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
GT Aeronautics, LLC. (U.S.)	Bandit	Fixed Wing	17	245	10	4	100	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
GT Aeronautics, LLC. (U.S.)	GT-1500	Fixed Wing	17	245	10	4	100	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Aero Composites Innovations (France)	V 500 / AS 350	Fixed Wing	17	330	13	4	81	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Pegasus Aerospace (U.S.)	PQV-5 Vulcan 5	Fixed Wing	17	265	6	3	37	Runway, Wheeled, Launcher–Catapult	Parachute, Runway, Wheeled	Electric Motor, Combustion Engine, Propeller
Frontline Aerospace, Inc.	V-STAR	Fixed Wing	17	2,360	22	-	288	VTOL	VTOL	Ducted Fan, Combustion Engine, Propeller
Georgia Tech Research Institute (U.S.)	GAUSS (GTRI Airborne Unmanned Sensor System)	Fixed Wing	16	180	9	-	70	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
BrockTek, LLC (U.S.)	HAVOC	Fixed Wing	16	175	11	-	50	Launcher–Pneumatic, Runway,	Skid, Belly, Runway, Wheeled	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
								Wheeled, Launcher–Catapult, Launcher–General		
Insitu, Inc.	Integrator / RQ-21A Blackjack	Fixed Wing	16	135	8	-	55	Launcher–Pneumatic, Launcher–Catapult, Launcher–General	Other	Combustion Engine, Propeller
Insitu, Inc.	Integrator Extended Range	Fixed Wing	16	145	8	-	55	Launcher–Pneumatic, Launcher–Catapult, Launcher–General	Other	Combustion Engine, Propeller
UAV Research Lab.	Merlin 200	Fixed Wing	16	161	11	4	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Griffon Aerospace (U.S.)	MQM-170C Outlaw G2	Fixed Wing	16	225	9	-	80	Runway, Wheeled, Launcher–Catapult, Launcher–General	Skid, Belly, Runway, Wheeled	Combustion Engine, Propeller
Griffon Aerospace (U.S.)	Outlaw SeaHunter	Fixed Wing	16	280	10	-	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
UAV Solutions, Inc. (U.S.)	Sidewinder	Fixed Wing	16	300	12	-	75	Runway, Wheeled	Runway, Wheeled	Jet Engine
L3 Latitude	Hybrid Quadrotor (HQ-90) / FVR-90	Hybrid	16	117	8	-	-	VTOL	VTOL	Combustion Engine, Propeller
SpektreWorks, Inc. (U.S.)	Cobalt 110 G-VTOL	Hybrid	16	110	8	-	50	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
Drone America, Inc.	HuginnAir SE	Hybrid	16	55	9	-	52	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
Drone America, Inc.	HuginnAir TE	Hybrid	16	80	9	-	52	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
Drone America, Inc.	MuninnAir SE	Fixed Wing	16	55	9	-	52	Launcher–General	Parachute	Combustion Engine, Propeller
Drone America, Inc.	MuninnAir TE	Fixed Wing	16	80	9	-	52	Launcher–General	Parachute	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Volans-i (U.S.)	Voly C-20	Hybrid	15	150	9	-	57	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
Volans-i (U.S.)	Voly M20	Hybrid	15	230	9	-	65	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
Northrop Grumman Corporation	Sandshark (formerly Sandstorm)	Fixed Wing	15	70	8	2	60	Runway, Wheeled	Runway, Wheeled	Electric Motor, Propeller
The Boeing Company	X-37B	Fixed Wing	15	11,001	29	10	-	Rocket-Assisted Takeoff (RATO)	Runway, Wheeled	Other
Lockheed Martin Corporation	Fury	Fixed Wing	14	300	7	-	65	Launcher–Pneumatic, Launcher–Catapult, Launcher–General	Net	Combustion Engine, Propeller
L3Harris Technologies	APEX (Airborne Pursuit & Exploita-tion)	Fixed Wing	14	62	6	2	-	Launcher–Catapult, Launcher–General	Parachute	Electric Motor, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Northrop Grumman Corporation	Bat 12+2	Fixed Wing	14	350	7	3	65	Launcher–General	Net	Combustion Engine, Propeller
Saxon Remote Systems (U.S.)	Viper M14	Fixed Wing	14	55	-	-	49	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Acuity Technologies	AT-3 Owl	Fixed Wing	14	410	10	-	88	Air Launch	-	Ducted Fan, Combustion Engine
Griffon Aerospace (U.S.)	MQM-170A Outlaw	Fixed Wing	14	120	8	2	51	Runway, Wheeled, Launcher–Catapult, Launcher–General	Skid, Belly, Runway, Wheeled	Combustion Engine, Propeller
Pegasus Aerospace (U.S.)	PQO-4 Odyssey 4	Fixed Wing	13	110	7	2	40	Runway, Wheeled, Launcher–Catapult, Launcher–General	Parachute, Runway, Wheeled	Electric Motor, Propeller
Martin UAV (U.S.)	Bat 4	Fixed Wing	13	125	8	3	40	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Dragonfly Pictures, Inc.	DP-14 Hawk	Rotary Wing	13	900	14	6	70	VTOL	VTOL	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Pegasus Aerospace (U.S.)	PHL-1 Asteria HL200	Rotary Wing	13	2,200	13	6	-	VTOL	VTOL	Electric Motor, Propeller
Insitu, Inc.	ScanEagle 3	Fixed Wing	13	80	8	-	50	Launcher–Pneumatic, Launcher–Catapult, Launcher–General	Other	Combustion Engine, Propeller
ARCTURUS UAV	T-16XL	Fixed Wing	13	85	7	-	50	VTOL, Launcher–Pneumatic, Launcher–General	Net, Skid, Belly, VTOL	Electric Motor, Combustion Engine, Propeller
American Aerospace Technologies, Inc.	RS-16	Fixed Wing	13	85	-	-	55	Launcher–Pneumatic, Launcher–Catapult, Launcher–General	Skid, Belly	Combustion Engine, Propeller
DRS Unmanned Technologies, Inc	Sentry / Sentry HP	Fixed Wing	13	420	11	-	75	Runway, Wheeled	Parachute, Runway, Wheeled	Combustion Engine, Propeller
L3 Latitude	Hybrid Quadrotor (HQ-60) / FVR	Hybrid	13	60	9	-	-	VTOL	VTOL	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
DroneTech UAV Corp	AV-2 / 25 HTOL DHP	Fixed Wing	12	55	6	2	40	Runway, Wheeled	Runway, Wheeled	Electric Motor, Combustion Engine, Propeller
DroneTech UAV Corp	AV-2 / 25 HTOL SFE	Fixed Wing	12	55	6	2	40	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
DroneTech UAV Corp	AV-2 / 25 VM4 VTOL DHP	Hybrid	12	55	6	2	40	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
DroneTech UAV Corp	AV-2 / 25 VM4 VTOL SFE	Hybrid	12	55	6	2	40	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
DroneTech UAV Corp	AV-2 / 25 VMX8 VTOL DHP	Hybrid	12	55	6	2	40	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
DroneTech UAV Corp	AV-2 / 25 VMX8 VTOL SFE	Hybrid	12	55	6	2	40	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
DroneTech UAV Corp	AV-2 / 32 HTOL DHP	Fixed Wing	12	71	6	2	44	Runway, Wheeled	Runway, Wheeled	Electric Motor, Combustion Engine, Propeller
DroneTech UAV Corp	AV-2 / 32 HTOL SFE	Fixed Wing	12	71	6	2	44	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
DroneTech UAV Corp	AV-2 / 32 VM4 VTOL DHP	Hybrid	12	71	6	2	44	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
DroneTech UAV Corp	AV-2 / 32 VM4 VTOL SFE	Hybrid	12	71	6	2	44	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
DroneTech UAV Corp	AV-2 / 32 VMX8 VTOL DHP	Hybrid	12	71	6	2	44	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
DroneTech UAV Corp	AV-2 / 32 VMX8 VTOL SFE	Hybrid	12	71	6	2	44	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Northrop Grumman Corporation	Bat 12	Fixed Wing	12	220	6	-	60	Launcher–Pneumatic, Launcher–General	Net	Combustion Engine, Propeller
Textron Systems	Aerosonde Mark 4.7	Fixed Wing	12	80	8	3	50	VTOL, Launcher–General	Net, VTOL	Combustion Engine, Propeller
UASTrakker LLC (U.S.)	ML-EX1	Hybrid	12	77	8	-	65	VTOL, VTOL	-	Electric Motor, Combustion Engine, Propeller
Joby Aviation	Lotus Demonstra-tor	Hybrid	11	55	-	-	62	VTOL	VTOL	Electric Motor, Propeller
Aero Surveillance, Inc.	ASV 150-EC	Rotary Wing	11	181	9	3	44	VTOL	VTOL	Combustion Engine, Propeller
UAV Research Lab.	Rotor Buzz II	Rotary Wing	11	265	12	3	-	VTOL	VTOL	Combustion Engine, Propeller
Dragonfly Pictures, Inc.	DP-5X Wasp	Rotary Wing	11	540	11	4	100	VTOL	VTOL	Combustion Engine, Propeller
Mississippi State University	Xawk 5, SN 001	Fixed Wing	11	55	8	2	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Mississippi State University	Xawk 5, SN 002	Fixed Wing	11	55	8	3	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Mississippi State University	Xawk X-4	Fixed Wing	11	55	5	1	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Kratos Defense and Security Solutions, Inc. (U.S.)	BQM-167A/i Skeeter (UTAP-22 Mako)	Fixed Wing	11	2,050	20	2	-	Rocket-Assisted Take Off (RATO)	Parachute	Jet Engine, Turbojet
UAV Research Lab.	Rotor Buzz	Rotary Wing	11	220	12	3	-	VTOL	VTOL	Combustion Engine, Propeller
Aeroscout GmbH (Switzerland)	Scout B1-100	Rotary Wing	11	170	11	4	-	VTOL	VTOL	Combustion Engine, Propeller
UAVOS, Inc. (U.S.)	Surveyor-H UVH-290E	Rotary Wing	11	236	8	3	38	VTOL	Parachute, VTOL	Combustion Engine, Propeller
Northrop Grumman Corporation	R-Bat	Rotary Wing	10	205	12	4	40	VTOL	VTOL	Combustion Engine, Propeller
Insitu, Inc.	ScanEagle 2	Fixed Wing	10	58	6	-	60	Launcher–Catapult, Launcher–General	Other	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Advanced Aircraft Company	Greased Lightning (GL-10)	Hybrid	10	62	-	-	-	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
Adaptive Flight, Inc.	Hornet Magnum	Rotary Wing	10	284	12	4	-	VTOL	VTOL	Combustion Engine, Propeller
Kratos Unmanned Aerial Systems, Inc.—A Kratos Company (U.S.)	MQM-107A Streaker	Fixed Wing	10	1,000	17	5	200	Rocket-Assisted Take Off (RATO), Launcher–Catapult, Launcher–General	Parachute	Jet Engine, Turbojet
Aero Surveillance, Inc.	ASV 150	Rotary Wing	10	154	9	3	32	VTOL	VTOL	Combustion Engine, Propeller
Volans-i (U.S.)	Voly C-10	Hybrid	10	55	7	-	43	VTOL	VTOL	Electric Motor, Propeller
SpektreWorks, Inc. (U.S.)	Cobalt 55 E-VTOL	Hybrid	10	55	6	-	50	VTOL	VTOL	Electric Motor, Propeller
SpektreWorks, Inc. (U.S.)	Cobalt 55 G-VTOL	Hybrid	10	55	6	-	50	VTOL	VTOL	Electric Motor, Combustion

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
										Engine, Propeller
Brandebury Tool Company, Inc.	NexDrone	Fixed Wing	10	79	8	-	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Acuity Technologies	AT-10 Responder	Fixed Wing	9	120	8	-	83	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
US Naval Research Laboratory	Dragon Warrior / Vantage	Rotary Wing	9	380	9	-	-	VTOL	VTOL	Combustion Engine, Propeller
Dragonfly Pictures, Inc.	DP-12 Rhino	Rotary Wing	9	394	7	5	54	VTOL	VTOL	Combustion Engine, Propeller
Martin UAV (U.S.)	V-BAT	Fixed Wing	9	82	8	2	45	VTOL	VTOL	Ducted Fan, Combustion Engine
GT Aeronautics, LLC. (U.S.)	GT-380	Fixed Wing	9	65	6	3	90	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Raytheon Company	Manta	Fixed Wing	9	61	6	2	45	Runway, Wheeled, Runway, Wheeled, Launcher–	Skid, Belly, Runway, Wheeled	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
								Catapult, Launcher–General		
UAVOS, Inc. (U.S.)	Surveyor-H UVH 170	Rotary Wing	9	99	11	3	32	VTOL	Parachute, VTOL	Combustion Engine, Propeller
UAVOS, Inc. (U.S.)	Surveyor-H UVH 25EL	Rotary Wing	9	55	9	2	24	VTOL	Parachute, VTOL	Electric Motor, Propeller
UAVOS, Inc. (U.S.)	Surveyor-H UVH-120E	Rotary Wing	9	76	9	3	32	VTOL	Parachute, VTOL	Combustion Engine, Propeller
Guided Systems Technologies, Inc.	SICX-75	Rotary Wing	8	170	8	3	-	VTOL	VTOL	Combustion Engine, Propeller
UAVOS, Inc. (U.S.)	Surveyor-H UVH-EL	Rotary Wing	8	66	7	2	32	VTOL	Parachute, VTOL	Electric Motor, Propeller
Harris Aerial LLC (U.S.)	Carrier H6 Hybrid HL	Rotary Wing	8	88	8	2	-	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
Mobile Recon Systems LLC (U.S.)	Dauntless	Rotary Wing	8	286	8	2	-	VTOL	VTOL	Electric Motor, Combustion

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
										Engine, Propeller
AeroVironment	Vapor 55	Rotary Wing	8	55	6	2	22	VTOL	VTOL	Electric Motor, Propeller
Accurate Automation Corporation	X-43A	Fixed Wing	7	246	12	-	-	Runway, Wheeled	Runway, Wheeled	Jet Engine
Black Peak Engineering, LLC	BP-210 Weasel	Rotary Wing	7	120	9	3	60	VTOL	VTOL	Jet Engine, Propeller
Copterworks Inc. (U.S.)	AF25B	Rotary Wing	7	67	6	2	-	VTOL	VTOL	Combustion Engine, Propeller
Kratos Defense and Security Solutions, Inc. (U.S.)	BQM-177A/i	Fixed Wing	7	1,400	17	-	-	Rocket-Assisted Take Off (RATO)	Parachute	Jet Engine, Turbojet
Northrop Grumman Corporation	BQM-74F	Fixed Wing	7	620	15	-	-	Launcher–General	Parachute	Jet Engine, Turbojet
Scion UAS, LLC.	SA-200 Weasel	Rotary Wing	7	120	7	3	60	VTOL	VTOL	Combustion Engine, Propeller
Kratos Defense and Security	MQM-178 Firejet	Fixed Wing	7	320	11	-	-	Launcher–Pneumatic	Parachute	Jet Engine, Turbojet

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Solutions, Inc. (U.S.)								Launcher–General		
Vanguard Defense Industries	Shadow-Hawk	Rotary Wing	6	70	7	2	19	VTOL	VTOL	Combustion Engine, Propeller
Reference Technologies, Inc. (U.S.)	HUM-MINGBIRD III	Rotary Wing	6	89	6	-	48	VTOL	Parachute, VTOL	Ducted Fan, Combustion Engine, Propeller
Harris Aerial LLC (U.S.)	Carrier Hx8 Power	Rotary Wing	6	75	6	3	-	VTOL	VTOL	Electric Motor, Propeller
Northrop Grumman Corporation	BQM-74C Chukar III	Fixed Wing	6	455	13	-	-	Rocket-Assisted Take Off (RATO), Launcher–General	Parachute	Jet Engine, Turbojet
Northrop Grumman Corporation	BQM-74E	Fixed Wing	6	455	13	-	-	Air Launch, Rocket-Assisted Take Off (RATO)	Parachute	Jet Engine, Turbojet
Reference Technologies, Inc. (U.S.)	HUM-MINGBIRD II	Rotary Wing	6	58	6	-	35	VTOL	Parachute, VTOL	Ducted Fan, Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Parallel Flight Technologies (U.S.)	Parallel Heavy Lift Platform	Rotary Wing	6	165	6	-	-	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
Homeland Surveillance & Electronics LLC (U.S.)	M6A PRO	Rotary Wing	5	76	5	-	-	VTOL	VTOL	Electric Motor, Propeller
Homeland Surveillance & Electronics LLC (U.S.)	AG-V8S	Rotary Wing	5	106	5	-	8	VTOL	VTOL	Electric Motor, Propeller
Top Flight Technologies (U.S.)	Airborg H8 10K	Rotary Wing	5	110	6	5	29	VTOL	VTOL	Electric Motor, Propeller
Homeland Surveillance & Electronics LLC (U.S.)	HERCULES HL6+	Rotary Wing	5	79	5	-	-	VTOL	Parachute, VTOL	Electric Motor, Propeller
Accurate Automation Corporation	LoFLYTE	Fixed Wing	5	70	8	-	-	Runway, Wheeled	Runway, Wheeled	Jet Engine, Turbojet
Accurate Automation Corporation	ALAMM (Air Launched Multi-	Fixed Wing	5	183	8	-	595	Air Launch	-	Jet Engine, Turbojet

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
	Mission), AT-1500									
Accurate Automation Corporation	ALAMM (Air Launched Multi-Mission), AT-1700	Fixed Wing	5	210	8	-	595	Air Launch	-	Jet Engine, Turbojet
Raytheon Company	MALD (Miniature Air Launched Decoy)	Fixed Wing	4	249	9	1	-	Air Launch	-	-
Magnumwings (U.S.)	A-8 Lapwing	Hybrid	4	88	-	-	-	VTOL	VTOL	Electric Motor, Propeller
FusionFlight Inc. (U.S.)	AB5 JetQuad	Other	4	130	4	2	261	VTOL	VTOL	Jet Engine
Beechcraft (U.S.)	AQM-37C/D	Fixed Wing	3	617	14	2	-	Air Launch	-	Other
Meggitt Defense Systems	GT-400	Fixed Wing	2	180	9	1	220	Air Launch	Parachute	-
BFD Systems	1200-S8	Rotary Wing	-	55	-	-	-	VTOL	VTOL	Electric Motor, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Airborne Systems North America	1T FireFly (1TFF®)	Other	-	99	-	-	-	Air Launch	Parachute	-
Airborne Systems North America	2K1T	Other	-	72	-	-	-	Air Launch	Parachute	-
Aerofex Inc. (U.S.)	Aerofex AG1	Rotary Wing	-	350	-	-	-	VTOL	VTOL	Ducted Fan, Combustion Engine
Aergility, LLC. (U.S.)	ATLIS	Rotary Wing	-	800	-	-	87	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
Advanced Tactics, Inc.	Black Knight TRANS-FORMER	Rotary Wing	-	4,400	31	8	-	VTOL	VTOL	Combustion Engine, Propeller
Flint Hills Solutions, LLC.	FH-620	Rotary Wing	-	55	7	3	-	VTOL	VTOL	Combustion Engine
Anduril Industries (U.S.)	Ghost	Rotary Wing	-	55	-	-	-	VTOL	VTOL	Electric Motor, Propeller
Center for Remote Sensing of Ice Sheets (U.S.)	GX1	Fixed Wing	-	85	-	-	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Aurora Flight Sciences Corporation	Lightning-Strike XV-24A Demonstra-tor	Hybrid	-	325	-	-	-	VTOL	VTOL	Ducted Fan, Electric Motor, Jet Engine
Watts Innovations (U.S.)	MFD 3000	Rotary Wing	-	55	-	-	-	VTOL	VTOL	Electric Motor, Propeller
Watts Innovations (U.S.)	MFD 5000	Rotary Wing	-	75	-	-	-	VTOL, VTOL	-	Electric Motor, Propeller
General Atomics Aeronautical Systems, Inc.	MQ-1C Gray Eagle Extended Range (GE-ER)	Fixed Wing	-	4,200	-	-	-	Runway, Wheeled	Runway, Wheeled	Combustion Engine, Propeller
Amazon	Prime Air (Prototype 2)	Hybrid	-	55	-	-	48	VTOL	VTOL	Electric Motor, Propeller
Integrity Applications Incorporated	REAP XLB (Rapidly Elevated Aerostat Platform)	Other	-	127	39	20	-	VTOL	VTOL	-
SHOTOVER (U.S.)	SHOT-OVER U1	Rotary Wing	-	73	-	-	-	VTOL	Parachute, VTOL	Electric Motor, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
Moog, Inc. (U.S.)	SureFly	Rotary Wing	-	1,500	-	-	-	VTOL	Parachute, VTOL	Electric Motor, Combustion Engine, Propeller
Malloy Aeronautics Ltd (United Kingdom)	T-80 / TRV-80 (Tactical Resupply Vehicle)	Rotary Wing	-	80	-	-	-	VTOL	VTOL	Electric Motor, Combustion Engine, Propeller
Piasecki Aircraft Corporation	Turais	Fixed Wing	-	772	-	-	-	Air Launch	Expendable, Parachute	Electric Motor, Combustion Engine, Propeller
VersaTOL (U.S.)	Venom	Rotary Wing	-	55	-	-	-	VTOL	VTOL	Electric Motor, Propeller
Volocopter (Germany)	Volodrone	Rotary Wing	-	1,764	-	-	43	-	-	Electric Motor, Propeller
Textron Systems	X5-55	Hybrid	-	75	-	-	-	VTOL	VTOL	Electric Motor, Propeller
xFold Rig (U.S.)	xFold Cinema X12	Rotary Wing	-	63	-	-	10	VTOL	VTOL	Electric Motor, Propeller

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Primary Organization	Model	Category	Wingspan/ Width (ft)	MTOW (lb)	Length (ft)	Height (ft)	Cruise (kts)	Launch	Recovery	Propulsion
xFold Rig (U.S.)	xFold Dragon x12	Rotary Wing	-	140	-	-	10	VTOL	VTOL	Electric Motor, Propeller
xFold Rig (U.S.)	xFold Dragon x8	Rotary Wing	-	115	-	-	10	VTOL	VTOL	Electric Motor, Propeller

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APPENDIX B

Stakeholder Interviews and Notes

Syracuse Hancock International Airport (SYR)

- Date of Meeting: September 2, 2020, and October 9, 2020
- Proponent Attendees: Jason Terreri – Airport Director
  John Carni – Airport Deputy Director
  Brian Dorman – Airport Operations
  Basil Yap – Hovecon
  Lt. Col. Aaron Brown – NY ANG 174th Attach Wing
  Sandra Gregoire – FAA ATC
- Research Team Attendees: Thomas Mackie
  Sheldon Menezes
  Maria Muia
  Robb Varani
  Dave Ramacorti
  Stephen Smith
  Dave Fleet

Background

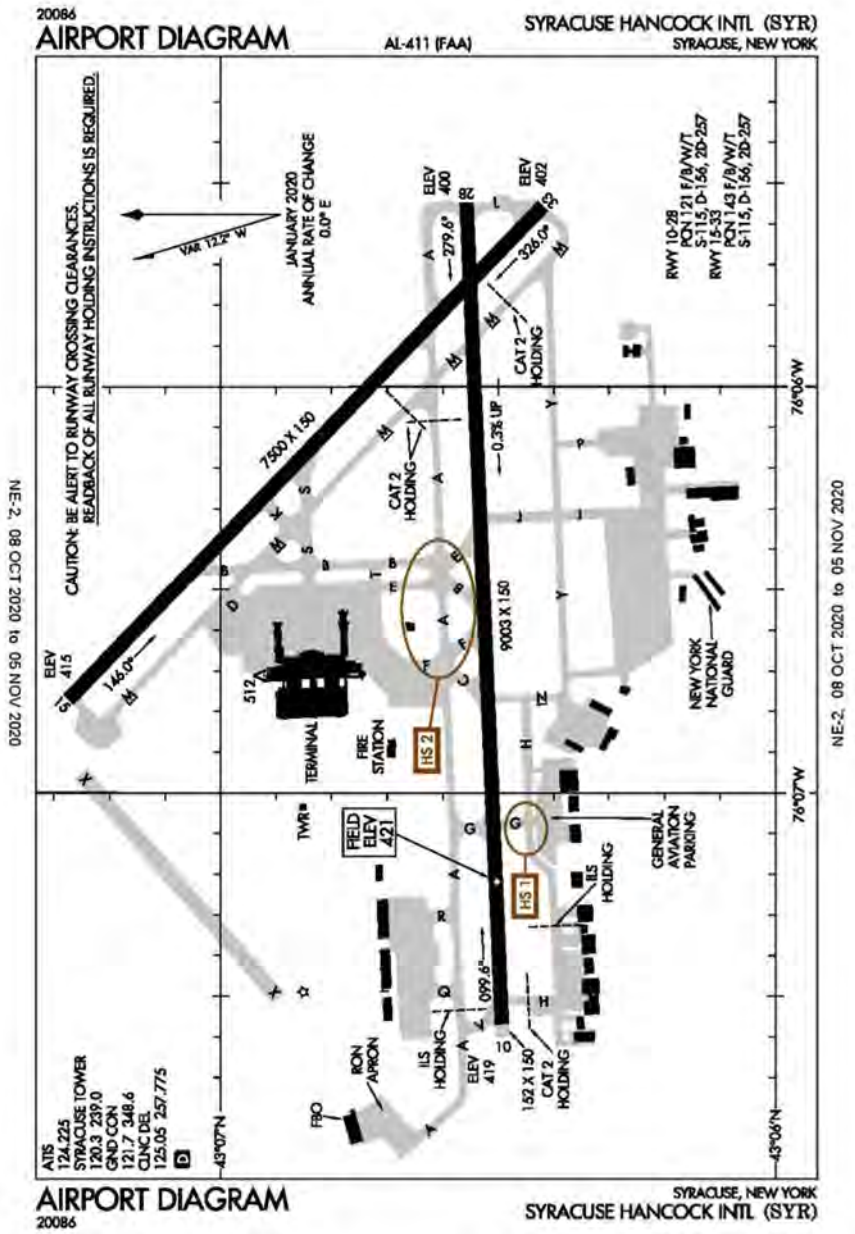
Syracuse Hancock International Airport (SYR) is a primary, small hub airport located five miles northeast of downtown Syracuse, in Onondaga County, NY. (See Figure B-1 for the FAA airport diagram of SYR.) SYR accommodates a diverse range of aeronautical activities, including scheduled passenger, cargo, general aviation, and military aircraft. Military operations are primarily associated with the Hancock Field Air National Guard (ANG) base, home to New York ANG's 174th Attack Wing, and are co-located with SYR. The ANG 174th Attack Wing initially operated F-16 aircraft at SYR. In 2010, F-16 aircraft operations were discontinued and replaced by the MQ-9 Reaper uncrewed combat aircraft by 2015.

SYR is associated with one of seven FAA-designated UAS test sites in the U.S. Managed by Northeast UAS Airspace Integration Research (NUAIR), the New York UAS test site consists of a 50-mile UAS Traffic Management Corridor spanning from Griffiss International Airport (RME) in Rome, NY to SYR. From 2015 to 2020, over 2,700 advanced UAS test flights have taken place at the NY UAS test site. The unique characteristics of operations at SYR and the surrounding airspace, which enable ongoing testing of uncrewed air traffic, make it a key stakeholder in this research.

Three MQ-9 Reaper UAS are based at SYR that conduct an average of nine operations per day. The ANG's 174th Attack Wing plans to add two more (for a total of five) MQ-9 Reaper L-UAS as they expand

their existing operations and the MQ-9 pilot training program. They train approximately 25% of all MQ-9 aircrew for the U.S. Air Force (over 150 per year).

Figure B-1 SYR airport diagram.



Source: http://aeronav.faa.gov/d-tpp/2010/05010AD.PDF, accessed 10-07-20.

Operational Impact

Initially, all MQ-9 Reaper uncrewed operations required the use of a crewed chase plane while operating in the local airspace. In addition to inducing operational costs, the use of a chase plane limited the frequency

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and duration of MQ-9 operations. Before departure, the chase plane needed to be airborne and circle near the airfield until the MQ-9 aircraft departed. The chase plane then joined the MQ-9 while maintaining Visual Flight Rules (VFR) separation under Visual Meteorological Conditions (VMC) until reaching the boundary of the Class A airspace above Flight Level (FL) 180 (18,000 ft. above mean sea level). Upon returning, the chase plane accompanied the MQ-9 as it descended below FL 180 until entering the local traffic pattern and landing at SYR.

Because the MQ-9 operations at SYR typically operate in Class A airspace and beyond line-of-sight, the MQ-9 is required to operate under an Instrument Flight Rules (IFR) flight plan. However, the MQ-9 must remain in VMC in flight until entering Class A airspace. At FL 180, the MQ-9 separates from the chase aircraft and continues with its mission. After its mission is completed, the MQ-9 rejoins the chase aircraft just as it exits Class A airspace. The chase aircraft observes the MQ-9 through landing or until a ground observer initiates visual contact and assumes responsibility.

In September 2019, the ANG installed a ground-based detect and avoid (GBDAA) radar system at SYR, which has obviated the need for chase aircraft. The GBDAA uses the LSTAR® Air Surveillance Radar system, which consists of a series of ground sensors to detect airborne traffic. The LSTAR system is also integrated with the Airport Surveillance Radar (ASR) system to extend its range. This system detects all fixed and rotary-wing aircraft, regardless of transponder equipage. It also detects small drones and remotely piloted aircraft. The LSTAR system provides a more robust awareness of the airspace than achieved with a chase plane.

The Certificate of Authorization (COA) the ANG operates under requires that an “emergency” be declared anytime the MQ-9 lands without the chase plane. Because of the VFR requirement, the ANG must declare an emergency about 15 to 20 times per year because VMC cannot be maintained, and the MQ-9 must leave the chase aircraft to get back to the airport faster than the chase aircraft can fly. While the term emergency is used, it is not treated like a true emergency other than the FAA filing paperwork that the MQ-9 landed without the chase aircraft. The emergency is declared because the MQ-9 must beat the weather and cannot wait for the chase aircraft. A weather emergency can also be called for high winds.

The MQ-9 is not equipped to deal with moisture, and is not ice-resistant; therefore, it must avoid water. Also, because of its electrical systems, it must avoid thunderstorms. Another operational limitation of the MQ-9 is its slow taxi speed. On the runway, the MQ-9 is limited to 20 knots taxi speed, and on taxiways, it is limited to 10 knots. This requires the ATCT to understand the speed limitation and take care in maintaining proper entrail separation due to the lack of visibility by the remote pilot. ATC can approve the MQ-9 for a visual approach with the caveats that it cannot be cleared to “follow” any other aircraft and ATC provides entrail separation. The MQ-9 also has no landing lights because it is equipped with a night vision camera.

Surface Infrastructure Impact

Several surface issues have been identified as part of the L-UAS operations associated with surveillance, taxi operations, and airfield lighting. The two GBDAAs used by the ANG (on and off the airport) are not considered part of the national airspace system infrastructure used by the FAA and available as a service to pilots. Therefore, the installation did not have to follow the FAA advisory circulars required for traditional radar. The FAA does not classify them as necessary for the separation of traffic. They supplant the pilot’s visual acuity and therefore do not have to meet the radar/NAVAIDs siting requirements.

Because of the slow taxi speed of the MQ-9, operational delays on the runway may result if runway exits are not optimally located. Additionally, on some of the taxiways at SYR, the taxiway centerlines do not go exactly where the MQ-9 needs to go. The remote pilot has been able to compensate for the lack of specific marking. Finally, the airfield lighting does not impact the infrared cameras of the MQ-9, but they only see in black and white, albeit extremely high definition. Even with the black-and-white limitation, colored airfield markings or signage has not presented any problems at SYR.

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Airspace

The LSTAR installation at SYR has some operational limitations. The system can accurately detect other airborne aircraft for a range of 22-nautical miles from the ASR site. However, there is an area directly above the installation referred to as the “cone of silence.” This portion of the airspace is directly above the ASR site and impairs the ability to detect aircraft directly above. This cone expands uniformly to an overall width of 5-nautical miles at FL 180. The volume of airspace located between the outer edge of the cone of silence to the outer range limit of 22-nautical miles is referred to as the Service Volume.

Only one GBDAA system is installed at the airport. Due to the limitation of this system, a cone of silence exists directly above the radar and is splayed to 5 nautical miles at FL 180 (Class A airspace). (See Figure B-2 for the FAA sectional chart for SYR.) Therefore, the MQ-9s operate at a lower altitude to a point 10 nautical miles away and then climb in a spiral fashion to FL180 (Class A airspace) within a 2-nautical mile radius cylinder while staying visible to the GBDAA system the entire time. These predetermined 2-nautical mile cylinders or “elevator points” were not a limitation of the MQ-9 but allowed for more predictable standard procedures for handling the aircraft by the FAA ATC. With the addition of a second GBDAA, the overlap eliminates the cone of silence and the associated operational limitations within the local terminal airspace.

Figure B-2 SYR on sectional.



Source: https://aeronav.faa.gov/content/aeronav/sectional_files/PDFs/New_York_101_P.pdf, accessed 10-11-20.

Master Planning Considerations

As described previously, the MQ-9 has an extremely slow taxi speed, limited to 20 knots on the runway and 10 knots on the taxiway. If the runway being utilized by the MQ-9 is in high demand, the integration

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of the MQ-9 has the potential to cause delays. Additional runway to taxiway exits and/or independent taxiways intended for aircraft with similar performance as the MQ-9 may be warranted for study.

Environmental Considerations

There are no available environmental reports or analyses available related to the MQ-9 Reaper operations at SYR. The MQ-9 Reaper L-UAS replaced F-16 operations, and as a result, noise levels were significantly reduced and no additional environmental concerns at SYR were identified at the time the mission changed from F-16 to MQ-9 Reaper aircraft.

Cost Considerations

Because the ANG was previously operating F-16 aircraft at SYR, much of the infrastructure needed to support MQ-9 operations is in place. No airfield or facility modifications were necessary. However, the installation of the LSTAR RADAR system required additional infrastructure investment, installed and funded by an on-airport aerospace company (Saab Sensis). The only other foreseeable cost consideration for integrating the MQ-9 is if delays were incurred by other aircraft, but this has not been an issue at SYR.

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Grand Sky–Grand Forks Air Force Base (RDR)

- Date of Meeting: September 22, 2020
- Proponent Attendees: Thomas Swoyer – Director, Grand Sky
  Jennifer Eckman – Northern Plains UAS Test Site
- Research Team Attendees: Thomas Mackie
  Sheldon Menezes
  Dave Ramacorti
  Dave Fleet

Background

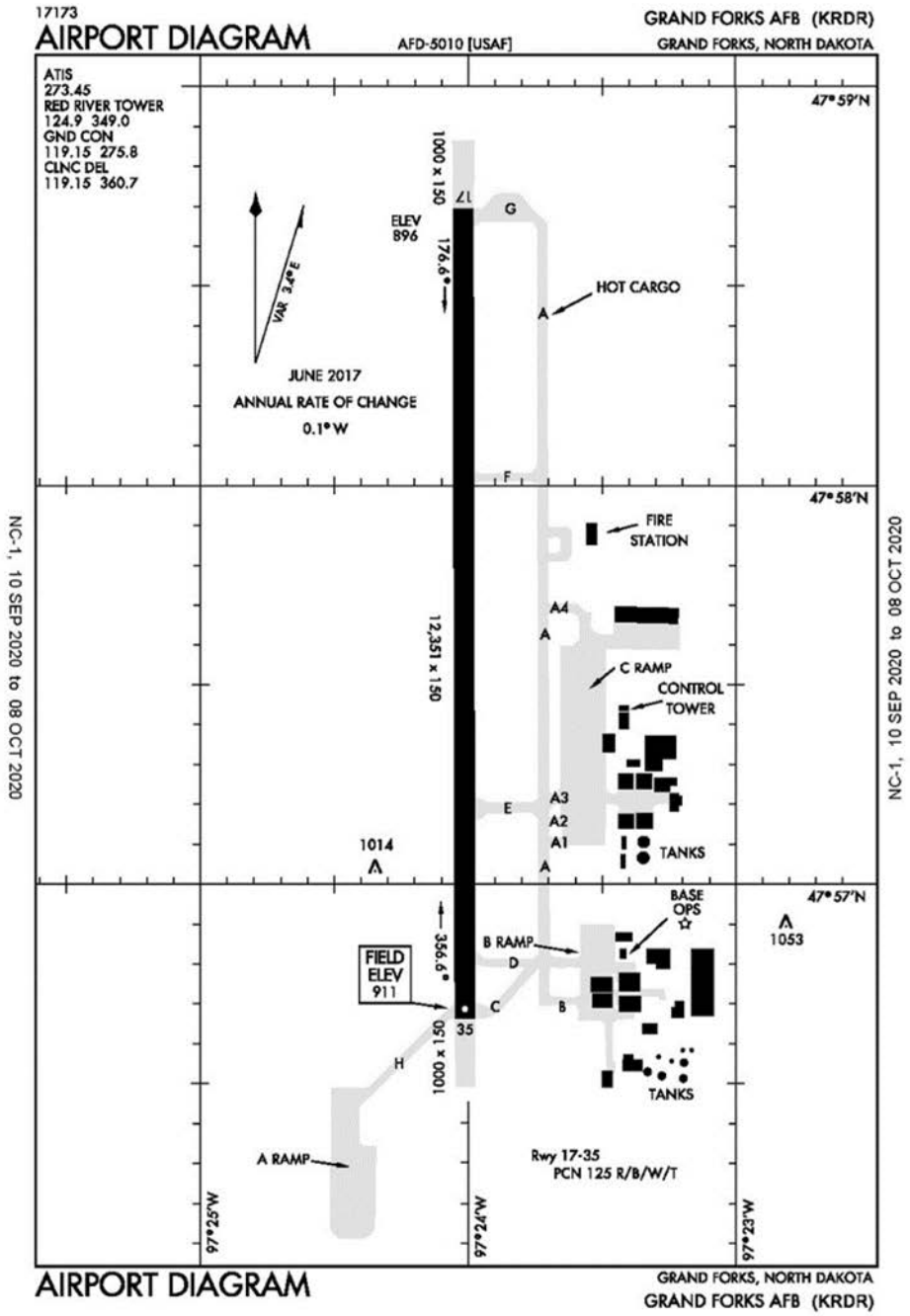
Grand Forks Air Force Base (RDR) is a military airport with no commercial or private crewed aircraft operations. (See Figure B-3 for the FAA airport diagram of RDR.) It is located approximately 13 nautical miles west of Grand Forks, North Dakota. It is home to the 319th Reconnaissance Wing of the USAF. RDR was originally designed as a B-52 base, providing sufficient land, pavement, and support facilities to enable the placement of large UAS operations at the airbase. Grand Sky is the UAS management and land developer team overseeing the commercial UAS on the airfield.

The mission of the 319th Reconnaissance Wing is to provide a decisional advantage to U.S. warfighters and national leaders through support of the Nation's Global Hawk High Altitude Intelligence, Surveillance, and Reconnaissance (ISR) Mission. The RQ-4 Global Hawk is a high altitude, long-endurance, remotely piloted aircraft capable of all-weather day and night ISR missions. The U.S. Customs and Border Protection (CBP) Air and Marine Operations (AMO) and National Air Security Operations Center–Grand Forks (NASOC-GF) are also located at RDR and include L-UAS operations. MQ-9 Bs are the current L-UAS operated by the CBP/AMO.

RDR is unique in that it is an operational U.S. Air Force base with specially approved L-UAS operations. There are no crewed commercial or general aviation operations in the field, but student pilots periodically conduct instrument approaches training missions at RDR without landing. General Atomics and Northrop Grumman are located in the Grand Sky operating area conducting flights, training, and testing of L-UAS.

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Figure B-3 RDR airport diagram.



Source: http://aeronav.faa.gov/d-tpp/2010/05010AD.PDF, accessed 10-07-20.

Operational Impact

The different L-UAS flying at RDR operate with a chase aircraft when in civilian airspace in a similar fashion as described for SYR. As in the other case studies, the only true emergency that has occurred at RDR was not with the L-UAS, but with the chase aircraft, which had smoke in the flight deck. Also, like other case studies, there have been no lost-link incidents reported to date. ARFF functions to support the L-UAS operations are the responsibility of the L-UAS operators and are not provided by the Air Force.

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This is accomplished through equipment investment from Grand Sky; the personnel and associated training certifications are the responsibility of the L-UAS operators.

Operations consist of both remotely piloted and autonomous L-UAS missions. The MQ-9 is remotely piloted, and the RQ-4 operates autonomously. The autonomous L-UAS are extremely precise but operate with a pilot on standby to take control of the L-UAS in case of an emergency. Because of the precise accuracy of the landings (i.e., same spot each time), the autonomous L-UAS have started using "randomizers" for landings to spread the impact over different areas of the pavement to reduce wear on the runway.

Grand Sky invested in the installation of a fiber network for direct access to the USAF ASR-11, which was already on site at the time of the facility deployment. There is a real-time feed from the USAF ASR-11 used for BVLOS operations under an approved COA from the FAA. (Note: This is the last ASR-11 radar operated by the USAF.)

Surface Infrastructure Impact

Approximately $20 million in improvements were made at RDR to support L-UAS operations. This included the purchase of a 1,500-gallon foam-capable aircraft rescue and firefighting (ARFF) truck to support the non-military operations. The foam is necessary because of the carbon fiber composition of the L-UAS, which burns extremely hot. Also, given the altitudes flown, their engines operate at higher-than-normal temperatures. All the hangars are equipped with foam fire suppression systems, while the offices have traditional sprinklers.

Approximately $9 million of the $20 million was invested in fiber cabling to handle all the data required to remotely pilot the L-UAS and process the data collected during missions. Power modifications, called "N+2 power redundancy," were also required to provide system resilience. A satellite dish was installed for global communications, which required clear siting like that required for radar.

An additional $9 million of the improvement money was invested in concrete to access the existing airfield and enable segregation of L-UAS operations on the ground. Additionally, the investment included the installation of a 175-foot-wide security gate controlled by ATC across a taxiway for the USAF, which allows access to the runway for L-UAS. This gate can accommodate the Global Hawk.

Some of the L-UAS operate via autonomous taxi operations; therefore, the path to and from the runway must be kept clear. The purpose of the procedural separation is to provide a layer of safety in mitigating incursions between autonomous and remotely piloted aircraft.

NOTE: The USAF provided the specifications for the highly reflective paint used on the airfield. This is an important point when considering local PIC for L-UAS during taxi operations.

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Figure B-4 RDR on FAA sectional chart.



Source: https://www.faa.gov/air_traffic/flight_info/aeronav/digital_products/vfr/, accessed 10-07-20.

Airspace

In general, the management of aircraft on the ground is more challenging than managing aircraft in the airspace. (See Figure B-4 for the FAA for RDR on the FAA sectional chart.) The USAF ATC manages all aircraft, crewed and uncrewed. Taxi operations are the most difficult to manage given there is a mix of autonomous taxi operations with remotely piloted (i.e., camera-controlled) taxi operations. Autonomous L-UAS taxi operations are conducted via GPS waypoints established by the L-UAS operators. Every operator sets their waypoints, although the airport can and sometimes does provide them.

Master Planning Considerations

The current configuration of the airfield complies with Airport Reference Code E-V design standards. This is adequate to serve aircraft with a wingspan of 214 ft. and an approach speed of 166 knots or more. The runway is 150 ft wide, and the taxiways are 75 ft wide. The lateral separation between the runway centerline and parallel taxiway centerline is 1,000 ft.

Environmental Considerations

An environmental assessment was prepared owing to noise concerns and a land-use plan for property development. This required coordination with 42 Native American tribes but did not present any issues. ARFF and fuel runoff were not noted as issues during the EA process, and the contamination found was below limits for mitigation. In summary, there were no special considerations or analyses conducted for the L-UAS that differed from those of other fixed wing aircraft operations.

Deconflicting radio frequencies required extensive study due to numerous L-UAS operating at RDR, resulting in a “communications master plan,” which was needed to receive Federal Communications Commission (FCC) approval.

Wildlife is an issue, as the airbase is in a flyway for several migratory bird species; however, there are no additional wildlife hazards resulting from L-UAS operations. The grass is kept from exceeding 11 inches

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to discourage nesting. One bird strike has occurred thus far, and they do issue bird alerts during high migratory times.

Economic Impact

A study of the added L-UAS operations showed $19.3 million in total economic impact: $11.5 million in wages, $4.9 million in support, and $3.3 million in ancillary facilities.

The airport will support the FAA UAS test site program, established under the FAA Modernization and Reform Act of 2012 (FMRA 2012) to establish six UAS test sites in support integration of L-UAS into the NAS. With more L-UAS support in sight, the need to increase data transmission, processing, and storage capability is perceived as never ending by RDR. The facility is anticipating large UAS cargo operations, with the mature facility established.

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Ellington Airport (EFD)

- Date of Meeting: October 6, 2020
- Proponent Attendees: Arturo Machuca – Airport Director
  Josh Finch
  Ray Harper
  Sandra Favela
  Pete Fress
  Charles Kirkman
- Research Team Attendees: Thomas Mackie
  Sheldon Menezes
  Maria Muia
  Robb Varani
  Stephen Smith
  Dave Ramacorti
  Dave Fleet

Background

Ellington Field Airport (EFD) is a general aviation reliever airport located approximately seven miles east of the William P. Hobby Airport. EFD supports general aviation, NASA for astronaut training, and the Ellington Field Joint Reserve Base for the 147th Reconnaissance Wing, the 147th Air Support Operations Squadron, and the 272nd Engineering Installation Squadron. Additionally, EFD is home to the Houston Spaceport, the Lone Star Flight Museum, the U.S. Coast Guard, and several general aviation tenants including air ambulance services.

EFD has three runways with the longest measuring 9,001 ft by 150 ft (see Figure B-5 for the FAA airport diagram of EFD), which are served by an instrument landing system. The facility has a 24-hour ATCT, is certified as a CFR Part 139 Class IV airport, and provides ARFF Index A response.

EFD consists of a diverse mixture of aeronautical activity including L-UAS, NASA training, military, corporate general aviation (GA), and limited recreational. Aircraft types range from light GA and heavy jets to supersonic military aircraft and helicopters. Most L-UAS flight missions at EFD are conducted over the Gulf of Mexico, but they also conduct numerous touch-and-go (T&G) operations with the MQ-9. No other L-UAS operators are at EFD.

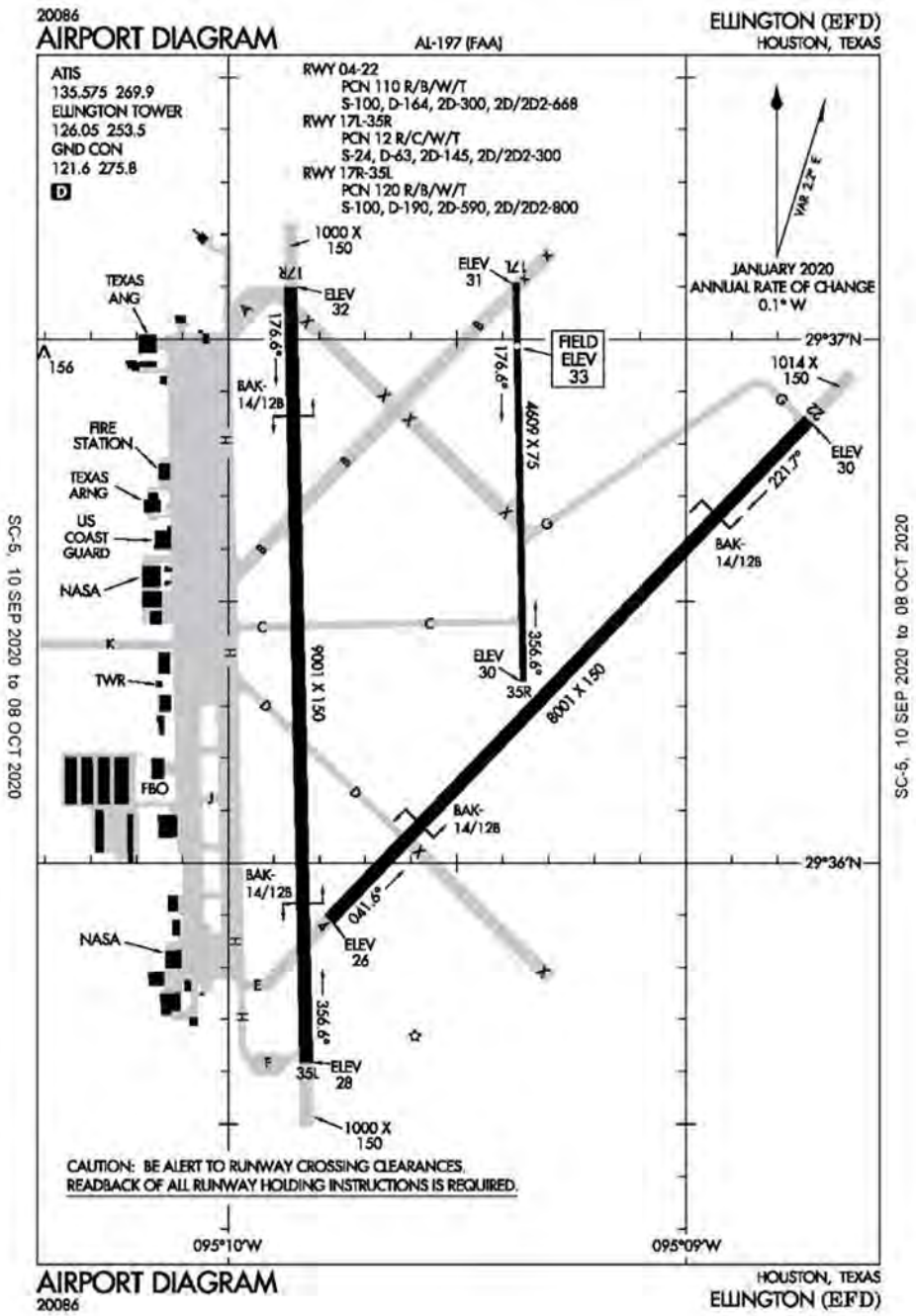
The ANG performs remotely piloted aircraft missions from this site with four MQ-9 Reaper UAS. Operations are regularly conducted three times per week. The ANG is located on the northwest side of the airport just west of Taxiway A. It also has a control station located on the north side of the airfield that communicates with the MQ-9s via C-Band frequencies. The ANG is self-sufficient, and the airport does not provide services besides airfield access.

The ANG MQ-9 needs a minimum of 5,000 ft of runway length, but this may vary based on the weight being carried. The ANG prefers longer runways so they can carry different payloads, including ammunition. The heaviest takeoff weight for the MQ-9 is 11,500 lb and the heaviest landing weight is 10,500 lb; however, they typically operate at approximately 8,500 lb.

L-UAS operations are conducted exclusively on Runway 17R-35L. Runway 4-22 is capable of handling L-UAS operation but must interact with the Class B airspace for Hobby Airport. Runway 17L-35R is too short (4,609 ft $\times$ 75 ft).

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Figure B-5 EFD airport diagram.



Source: http://aeronav.faa.gov/d-tpp/2010/00197AD.PDF, accessed 10-07-20.

Operational Impact

Initially, the FAA required L-UAS operations to be segregated from crewed aircraft operations at EFD. During this time, ATC was required to hold all other traffic (sterilize the airspace) when the MQ-9 was operating. This prevented the life-saving Coast Guard rescue and the hospital air ambulance rotor aircraft from departing and landing expeditiously. It took approximately six months for the FAA to allow the first integrated operation at EFD and almost another year for full integration (which occurred two years ago).

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This reflects the time needed for the airport and FAA to conduct collaborative discussions and design of the L-UAS operation, establish operational procedures, and obtain a COA before the first flights could occur.

The MQ-9 is operated via a C-Band link. The C-Band link capabilities can be disturbed by large aircraft or buildings. Theoretically, there is a maximum number of MQ-9s that can operate at one time due to the C-Band frequency spectrum limitation. There can be potential signal overlap, but ANG does not see this as a significant limitation. The ANG antennae are located on the north side of the airport and the aircraft access the runway by using Taxiways A, H, and F. The MQ-9s typically only use Runway 17R-35L because the airport falls under the Class B airspace associated with Hobby Airport. They appear to operate as traditional aircraft with the only issue being limitations in their field of view when panning up with the two cameras installed in the front. These cameras have 30-degree fields of vision providing visibility to the right, left, and down, but up can be an issue. Other than the panning-up issue, the MQ-9s see better in hazy conditions than the naked eye.

The ANG is mandated by their COA to use a chase aircraft; this activity is performed by the Civil Air Patrol using a Cessna 208. Because of the chase aircraft, the ATC must accommodate a two-aircraft bubble for aircraft separation.

ATC has the MQ-9 operate at 1,600 ft above ground level (AGL) while in the local traffic pattern. Since this is also the altitude for overhead flights, aircraft are not allowed to transition through the Class D airspace while the MQ-9 is in operation. Unless conducting T&Gs at EFD, most L-UAS operations are within the local airspace for approximately 15 minutes.

The MQ-9 is operated under a COA restricted to VMC. The aircraft is not instrument equipped, but it does have GPS positioning, and the ANG can create their own approach in an emergency. Eventually, the ANG MQ-9 will auto takeoff and land, but this is still a few years away for EFD. Because of the limited field of view provided by the 30-degree cameras, ATC must provide turn instructions while operating in the local traffic pattern. While this increases controller workload, they also must do this for several of their training aircraft.

The MQ-9 is built out of composites not traditionally found in aircraft. However, they are not treated any differently for ARFF except in understanding what the composites are and how they burn.

Surface Infrastructure Impact

There appear to be no surface infrastructure impacts associated with the operation of the MQ-9 at EFD outside of what is discussed under line-of-sight in the master planning considerations section. The operation does require a significant amount of data transfer capability and redundant backup power systems. Most of the related equipment and infrastructure (control center, antennae, and satellite) are mobile units, so they can reposition to another airport or installation.

Airspace

The MQ-9s typically only use Runway 17R-35L because the airport falls under Class B airspace. (See Figure B-6 for EFD on the FAA sectional chart.) Use of Runway 14-32 enters the Class B airspace, so it is only used in the case of an emergency.

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Figure B-6 EFD on FAA sectional chart.



Source: https://www.faa.gov/air_traffic/flight_info/aeronav/digital_products/vfr/, accessed 10-07-20.

The MQ-9 operates almost entirely like a normal aircraft in the airspace. The ANG flies a typical departure within their FAA COA. If they have a lost-link event, the aircraft already has a planned route, which ATC will follow. The ANG has not experienced any lost-link events since starting operations at EFD. For redundancy, the ANG also operates ground-based radios. If this happens, they squawk a lost communication code and report the lost link to ATC via the ground radio. The ANG supervisor on duty is also on the radio if needed. Before COVID-19, this person was in the ATCT.

Master Planning Considerations

The planning consideration necessary for the operation of the MQ-9 falls mostly within the signal blocking category. The MQ-9 aircraft must operate with a "clear line-of-sight to the ground data terminal antenna, which provides line-of-sight communications for takeoff and landing." Therefore, no buildings or large aircraft parking should be planned that may interfere with the line-of-sight between the ground antenna location and the MQ-9's operating path to and from the runway for takeoff and landing.

The MQ-9 can operate on runways with a width of 75 ft and taxiways with a width of 50 ft. The current configuration of the airfield complies with Airport Reference Code E-V design standards. This is adequate to serve aircraft with a wingspan of 214 ft and an approach speed of 166 knots or more. The runway is 150 ft wide, and the taxiways are 75 ft wide. The lateral separation between the runway centerline and parallel taxiway centerline is 750 ft.

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Environmental Considerations

No environmental or noise studies were conducted before conducting L-UAS operations at EFD. Houston Airport System (HAS) conducted extensive community outreach events before starting L-UAS operations. They also offer facility tours to the public to foster a better understanding of the L-UAS operations.

Cost Considerations

There appear to be no cost considerations for the airport to address related to the operation of the MQ-9 at EFD, as most of the equipment and infrastructure is provided by the military. However, there were some limited enhancements to the power and data infrastructure to support L-UAS operations.

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Eastern Oregon Regional Airport (PDT)

- Date of Meeting: November 30, 2020
- Proponent Attendees: Steve Chrisman – Airport Director
  Darrell Abline – PUR Range Manager
  Wayne Greene – Airfield Engineer
- Research Team Attendees: Thomas Mackie
  Sheldon Menezes
  Maria Muia
  Drishti Valecha
  Robb Varani
  Stephen Smith
  Dave Ramacorti
  Dave Fleet

Background

Pendleton is in northern Umatilla County, approximately 209 miles east of Portland on U.S. Interstate 84 (I-84), the main east–west travel route across northern Oregon. Eastern Oregon Regional Airport (PDT) and the adjacent City of Pendleton Airport Industrial Park are located approximately three miles northwest of downtown Pendleton, within the Pendleton city limits. Eastern Oregon Regional Airport is the largest public airport in northeast Oregon and the only airport with scheduled passenger air service in eastern Oregon. There are 12 public-use airports located within 60 nautical (air) miles of Pendleton, including two airports—Tri-Cities Airport (Pasco) and Walla Walla Regional Airport—that also provide commercial air service.

Eastern Oregon Regional Airport accommodates a wide variety of aeronautical activity, including small single- and multi-engine aircraft, business class turbine aircraft (business jets and turboprops), civilian helicopters, military fixed wing aircraft and helicopters, and UAS. In addition to scheduled passenger service, the airport has several commercial tenants providing aerial application, aircraft maintenance, fueling, flight training, and other services that generate local flight activity and attract itinerant users. The airport also accommodates the Oregon Army National Guard aviation facility and is the designated airport for the Pendleton Unmanned Aerial Systems Test Range.

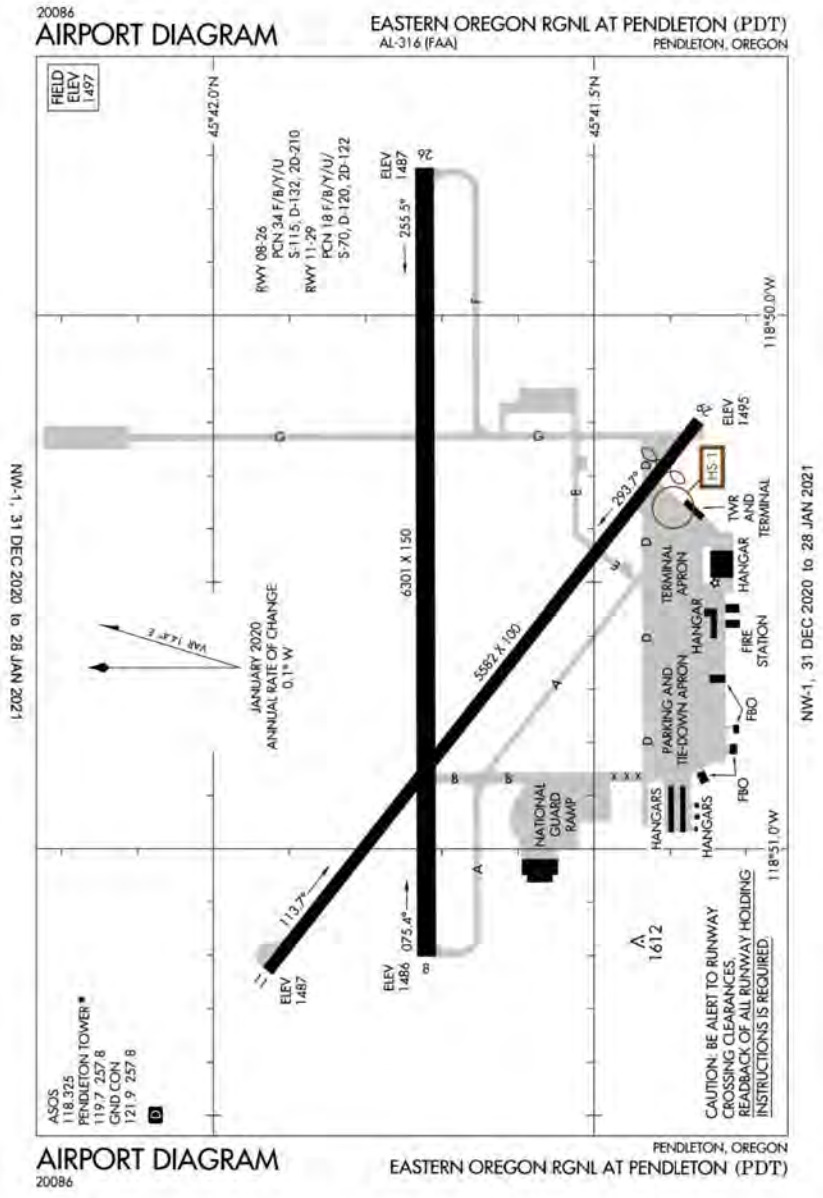
The 2002 Airport Master Plan estimated that Eastern Oregon Regional Airport had 97 based aircraft, 34,537 aircraft operations, and 1,407 enplaned passengers in 1999. The 1999 aircraft operations consisted of 7,155 commercial, 26,132 general aviation, and 1,250 military operations. The FAA Airport Record Form (5010-1) lists 51 based aircraft and 14,638 aircraft operations (takeoffs and landings) for the 12 months ending in January 2014. The Pendleton air traffic control tower recorded 15,387 aircraft operations in 2013. An updated based aircraft count provided by airport management in February 2015 lists a total of 67 aircraft.

PDT completed their current Master Plan in December of 2016, previously updated in 2002. The airport and vicinity have seen conditions change, triggering the need to update long-term planning. Additionally, trends in the aviation industry and additional UAS activity were requested to be integrated into the updated airport planning. When considering all UAS (small and large), PDT forecasted the increase of UAS operations through 2035 at a 10% annual increase from their current annual average of 500 operations. PDT considered all types of UAS, their airfield infrastructure, and support equipment needs with the goal of attracting tenants and operators (Figure B-7).

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They work with the University of Alaska under University of Alaska's test site. The UAS aircraft are grouped under local civil GA. They have four categories: less than 5 lb, 5 to 55 lb, 56 to 300 lb, and over 300 lb. They also have an X band portable radar truck that can be put on an Internet feed if needed, but few users have requested this. They also engage locally with support from the city to educate local hobbyists and the recreational flying community on UAS safety and airspace protection.

Figure B-7 PDT airport diagram.



Source: https://aeronav.faa.gov/d-tpp/2014/00316AD.PDF.

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Operational Impact

Overall, PDT experiences little complication with UAS operations at the airport, where UAS are fully integrated with crewed aircraft. Fluent and nearly issue-free operations occur at PDT because of the following focuses: primarily procedurally separated, well-planned, and coordinated UAS operations and consolidated ground operations.

In 2013 PDT designed 16 UAS test pads and provided utilities to support a wide range of potential operators wishing to conduct testing (long- or short-term) or maintain continual presence and operations. PDT installed power, water, and fiber to each test pad and strategically located them along the Taxiway G. The UAS primarily launch off Taxiway G when needed; VTOL systems primarily use the test pads. When necessary and required for launch or recovery, the primary runway has been used.

The UAS that have conducted flights at PDT range from 1.5 lb Styrofoam to 16,000 lb tricycle conventional landing aircraft. Some are VTOL. All the UAS aircraft operating at the airport are new uncrewed aircraft; nothing has been retrofitted. In October 2020 they had 200 missions, but this does not equate to the number of operations. They anticipate over 3,000 UAS operations this year.

The pads were instrumental in attracting users to PDT. They did not design pads around any specific aircraft. Rather, they were designed around the typical operations needed by the UAS users/manufacturers. They located them where they were easy to access for non-pilots in the non-movement areas, not around other aircraft. Many operations are located on the west side of the airport, and the UAS operators must taxi to the east side where the pads are. This does present occasional challenges.

The tower has indicated that there is an area of the UAS pad area that is hard to see, and a change would be helpful. When they are using the taxiway as a launchpad, that operationally impacts other use of the taxiway by traditional aircraft.

Airspace

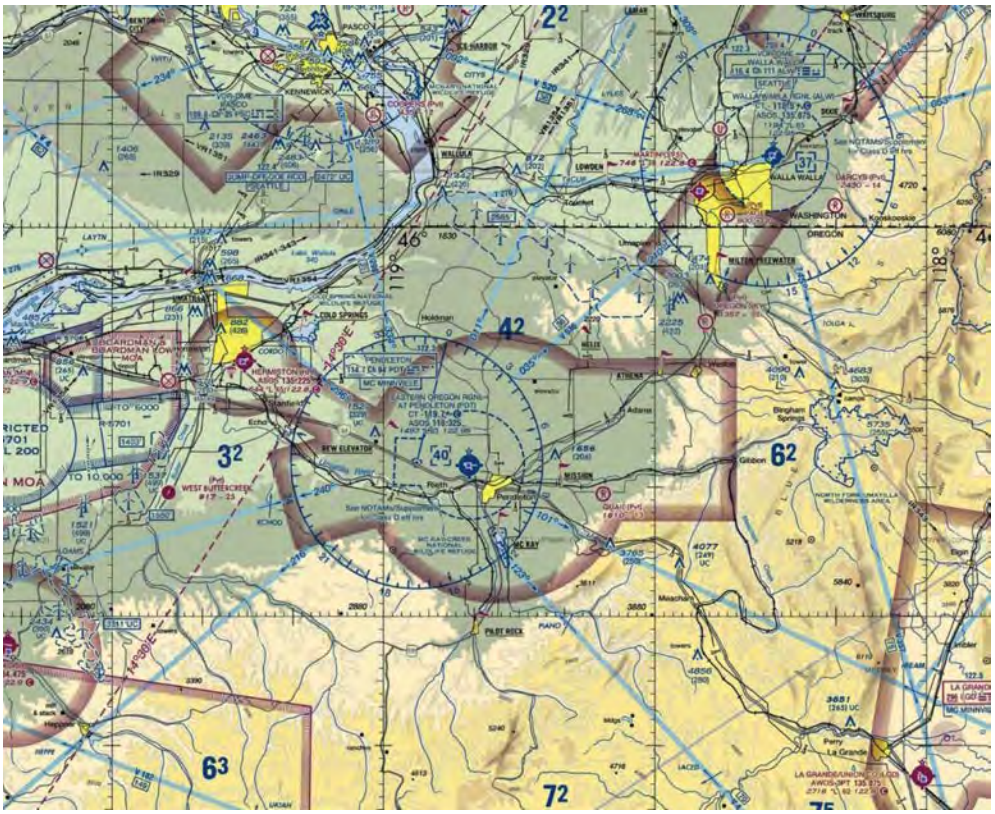
Aircraft operations would not have gotten off the ground at PDT without close collaboration with the FAA and local ATC; all parties were very cooperative. Depending on the aircraft operating, various safety procedures are in place to maintain separation and enable safe ground operations. Approach speeds vary, but due to the relatively light traffic, ATC can confidently maintain safe approach and departure operations (Figure B-8).

PDT does not have on-field surveillance capable of providing independent operations; they require chase plane support to maintain visual line-of-sight. Alternatively, PDT and ATC have protocols to provide forward observers, and/or a daisy chain observer, to maintain visibility and support communications to the pilot.

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Figure B-8 PDT on FAA sectional chart.



Source: http://vfrmap.com/?type=vfrc&lat=45.695&lon=118.843&zoom=10&api_key=763xxE1MJHyhr48DlAP2qQ.

Master Planning Considerations

The airport attributes their growth in UAS activities and their record of safe UAS operations to their planning efforts, addressed specifically for UAS facilities in the 2016 master plan. The development of the test pads, operational protocols, and ATC coordination were the key elements considered before initiating the development of UAS facilities. They have not used direct AIP grant dollars to date on the development of the test pads or any other improvements to support UAS operations.

The airport seeks to grow from its success by developing facilities to attract tenants and users of the UAS test pads/sites. The pads were instrumental in users coming to PDT. They are developing a new industrial area on the north side, which will be for research and development and commercial use. It will be located on the other end of Taxiway Golf (former north–south runway). Tenants on the opposite side of the taxiway would be ancillary businesses. This park is envisioned to be segregated from but likely will commingle with traditional aircraft simply due to the need for space in general. Currently, they are out of hangar space.

They are adding a section to the north side of the airport that will be an industrial park designed around the UAS initiative. They fortunately have a Level 3 fiber trunk line near the airport. They ran a 10-gigabyte loop through the airport.

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Environmental Considerations

The airport has reported that their UAS operations have virtually no environmental impacts. Various UAS systems have conducted flights from PDT over the past 5 years and have seldom had impacts. Several years ago, they had the "loudest UAS known to man" operate at the airport. It was an ArcticShark model and flew under the callsign *Chainsaw* because it was so loud. While they got several calls about it, most of the other aircraft are quiet. That aircraft did not have a muffler, supposedly because it would reduce engine performance. They have not done acoustic tests for noise as it relates to the airport, but their users are doing this to determine the characteristics of their aircraft.

Cost Considerations

The only addition to the airfield PDT has developed is a 16-test-pad campus. The engineering and construction were completed by the airport and the city, all with public works staff and equipment. The only cost presented to the airport was $125,000 spent on materials.

The impact on the airport business and the community has been very positive. The UAS operations have brought a lot of people onto the airport who would not otherwise be there, because of the R&D focus. The UAS initiative provides about 75 jobs now, and they have 20 to 30 open positions. The initiative also attracts business for local hotels and restaurants. The local population is small, so they do not anticipate much activity related to passengers or cargo; they anticipate continuing their focus on R&D.

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United Parcel Service (UPS)—Flight Forward

- Date of Meeting: March 3, 2021
- Proponent Attendees: Jeff Luckett–UAS Strategy & RDT&E
  Art Saylar – Network Planner
  Jared Navert – Technical Rep
  Paige Demetriades – Data Analysis/Tracking
- Research Team Attendees: Thomas Mackie
  Sheldon Menezes
  Maria Muia
  Drishti Valecha

Background

UPS Flight Forward is a subsidiary of the United Parcel Service (UPS) established to plan, design, and implement the carrier service company's UAS delivery services for integration into their existing transfer and delivery routes, and enabling unused airports. (https://www.ups.com/us/en/services/shipping-services/flight-forward-drones.page). To date, most of the UPS's focus has been on the "last-mile" step in the delivery process utilizing a combination of infrastructure and technology. The solutions they are developing are a combination of existing warehousing and ground infrastructure, surface vehicles and labor, small UAS (with appropriately sized payloads), and control/network systems necessary to plan, deploy, and control their vision of the entire "last mile" segment of the delivery process. Furthermore, a heavy focus has been put on the safety and regulatory components to enable the technology and processes developed to complete their mission. Demonstrating their efforts to develop a feasible solution, UPS has received their FAA Part 135 certification through the UAS Integration Pilot Program (IPP), becoming the first P-135 certified UAS delivery team.

As an operator of the aircraft rather than an airport facility, their perspectives on impacts to airfield infrastructure and associate design criteria are important to understand and plan for. UPS has primarily been focused on small UAS operations and their efforts to migrate to L-UAS operations have begun but are not as mature in the formulation. UPS is a commercial business with the following outlook on their priorities for implementing UAS into their operational ecosystem:

- Short-term Focus: small UAS – Campus-scale and "Last-Mile" environments.
  Example provided: Wilmington, NC facility to hospital/business/resident location.
- Mid-term Focus: L-UAS (55 – 1,000 lb) – Flights between UPS airport facilities.
  Example provided: Raleigh, NC facility to Wilmington, NC facility.
- Long-term Focus: L-UAS (>1,000 lb) – Longer range transport (200 miles and up) providing service to their current "feeder routes" and to connect with jet airway routes.
  Example provided: Louisville, KY facility to Raleigh, NC facility.

Operational Impact

Operational impacts to UPS are not focused on the airport surface infrastructure as much as on their operational fleet and the routing. The input on operational impacts from UPS was in reference to their flight program operations and not those on the airport surface specifically. The current plan shared with the

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research team on their plans for utilizing L-UAS (remotely piloted or autonomous) was conveyed in two ways:

- Fly current feeder routes currently completed by primarily Class 1 aircraft, but also includes some Class 2, 3, and 4 aircraft.
- Begin air service to airports/communities UPS does not currently provide that level of service to. Essentially replacing ground service in some locations with air service results in time savings, increases in their employee safety, and more job opportunities in those communities.

The types of aircraft to be used were not conveyed and considered confidential, but they are considering a mixture of retrofitted existing aircraft and future newly designed L-UAS aircraft. They are working with vendors developing control systems to retrofit, as well as discussing potential investment with L-UAS aircraft manufacturers.

Surface Infrastructure Impact

UPS has plans for using their existing ground and airport infrastructure including parking, hangars, load/unloading facilities, and maintenance. Based on the forecast for the fleet, they see little impact on existing airport infrastructure, but admittedly have not considered what standards possess gaps for the introduction of L-UAS. However, UPS's perspective regarding airport design criteria impacted by the integration of L-UAS is that airport infrastructure modification or improvements will be driven by the needs of the individual aircraft they consider using for uncrewed and/or autonomous piloting operations. Despite not having a strong perspective on the effects their future L-UAS will have on currently designed airfields, they do not feel there will be a substantial issue as they plan on using aircraft that comply with the current ADG and AAC criteria.

UPS is focused on addressing surface infrastructure issues relating to the power needs to fuel aircraft and ground service vehicles and support their service vendors (when applicable). In summary, they will be focused on ensuring that sufficient and reliable power sources will be available to service their vehicles and support teams.

Airspace

As UPS L-UAS operations mature, a concern raised is the need to ensure instrument flight procedures (IFP) are established at airports that do not have them, particularly at smaller or municipal airports they do not operate in/out of and that they seek to add to their network. Will there be an opportunity for airports to obtain an IFP level of service in the future?

They did not have specific experience or references to address concerns related to communications or surveillance capabilities, but they recognize that planning to provide the necessary levels of both those capabilities will be important to deploy a fleet of L-UAS. They are also paying close attention to advances in sense-and-avoid technology, as they recognize the criticality of reliability before deploying. They are preparing for a layered approach to safely operating in the NAS, as well as paying close attention to future modifications and the introduction of new federal regulations.

Master Planning Considerations

Related to their business planning for using L-UAS, UPS feels there is limited reliable and accurate forecasting in L-UAS and the eVTOL industry. As they look to expand their network with additional air service to airports used, or new airports, UPS staff think it is important to work with airports they foresee as potential locations and collaborate on their potential aircraft and support needs.

Regarding the challenges of operating L-UAS in the NAS, UPS feels a potential solution could be using short takeoff and landing (STOL) aircraft to access airfields with limited runway surfaces. UPS does not consider this a better solution to L-UAS, but rather an alternative if necessary.

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Airfield Design for Large Unmanned Aircraft Systems—A Guide

Environmental Considerations

The conversation mainly addressed reduced environmental impacts due to UPS's use of UAS, both small UAS and L-UAS. As they continue to expand their "last-mile" delivery model, it is obvious the removal of vehicles from the road networks decreases emissions directly by using fewer trucks for delivery.

Additionally, the use of electrically propelled L-UAS is forecast to produce few emissions at the airport by use of less fossil fuel, or more efficient ground operations when coupled with improved navigation and flight planning.

Cost Considerations

The impacts on the cost of UPS operations were not conveyed (as considered confidential) in the context of our topic. However, it was universally agreed there will be increased cost efficiency across their network due to the use of L-UAS. They do not anticipate significant investment needs by an airport to accommodate future L-UAS and operations because the aircraft will be able to use the existing surface infrastructure. UPS will collaborate directly with airport sponsors on studying any gaps in infrastructure, technology, or procedures their specific L-UAS may introduce, an approach UPS already takes, even when considering crewed aircraft operations.

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APPENDIX C

Abbreviations

AAC	Aircraft approach category
AAM	Advanced air mobility
AC	Advisory circular
ACRP	Airport Cooperative Research Program
ADG	Aircraft design group
ADS-B	Automatic dependent surveillance—broadcast
AEDT	Aviation Environmental Design Tool
AGL	Above ground level
AIP	Airport Improvement Program
ALP	Airport layout plan
ALS	Approach lighting system
ARFF	Aircraft rescue and firefighting
ASR	Air Surveillance Radar
ATC	Air traffic control
ATCT	Air traffic control tower
AUVSI	Association for Unmanned Vehicle Systems International
BVLOS	Beyond visual line-of-sight
CFR	Code of Federal Regulations
CPDLC	Controller pilot data link communication
DAA	Detect and avoid
DOD	Department of Defense
EB	Engineering Brief
EMAS	Engineered Materials Arresting Systems
ESS	Energy storage system
eVTOL	Electric vertical takeoff and landing
FAARFIELD	FAA rigid and flexible iterative elastic layered design
FATO	Final approach and takeoff area
FBO	Fixed base operator
FEMA	Federal Emergency Management Agency
GIS	Geographical information systems
GPS	Global positioning system
GVGI	Generic visual glideslope indicators
HTOL	Horizontal takeoff and landing

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IFC	International Fire Code
IFR	Instrument flight rules
ILS	Instrument landing system
JO	Joint Order
MTOW	Maximum certified takeoff weight
NAAQS	National Ambient Air Quality Standards
NAS	National airspace system
NAVAIDs	Navigational aids
NDB	Non-directional beacon
NEC	National Electric Code
NEPA	National Environmental Policy Act
NFPA	National Fire Protection Association
NGS	National Geodetic Survey
OEM	Original equipment manufacturer
OSHA	Occupational Safety and Health Administration
PAPI	Precision approach path indicator
PIC	Pilot in command
RATO	Rocket-assisted takeoff
RDC	Runway design code
RID	Remote identification rule
RTR	Remote transmitter/receivers
STOL	Short takeoff and landing
TDG	Taxiway design group
TERPS	Terminal instrument procedures
TLOF	Touchdown and liftoff area
TRACON	Terminal radar approach control
UAM	Urban air mobility
UAS	Unmanned aircraft systems
UPS	United Parcel Service
USFA	U.S. Fire Administration
USRD	Unmanned Systems and Robotics Database
UTM	Unmanned aircraft system traffic management
VASI	Visual approach slope indicator
VFR	Visual Flight Rules
VTOL	Vertical takeoff and landing
V_{REF}	Approach speed or landing speed
V_{SO}	Stall speed

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APPENDIX D

Glossary

Aircraft Approach Category (AAC)	FAA's grouping of aircraft based on aircraft approach speed or V_{REF}.
Aircraft Design Group (ADG)	Classification of aircraft based on wingspan and tail height.
Crosswind Component	The crosswind component of wind direction and velocity is the resultant vector that acts at a right angle to the runway. It is equal to the wind velocity multiplied by the trigonometric sine of the angle between the wind direction and the runway direction.
FAARFIELD	FAARFIELD, which stands for FAA Rigid and Flexible Iterative Elastic Layered Design, is software used for the design and evaluation of airport pavements.
Fixed Wing Large UAS	Similar to traditional-looking aircraft and equipped with a fuselage, wings, propulsion source, and stabilizers.
Hybrid Large UAS	These are a combination of fixed wing and rotary wing UAS that have rotors for landing and takeoff and wings to sustain flight.
Integration	The permissible mix of crewed and uncrewed aircraft into airport operations while using existing airport infrastructure or using airport infrastructure compliant with existing airfield design criteria, allowing intermingled operations.
Large UAS (L-UAS)	Unmanned aircraft systems weighing 55 pounds or more.
Rotary Wing Large UAS	Rotary Wing UAS provide VTOL, like helicopters, and have a broad spectrum of rotor configurations.
Runway Design Code	The AAC, ADG, and available instrument approach procedure visibility minimums together make up what is known as the runway design code (RDC) for each runway at an airport.
Small UAS (UAS)	Unmanned aircraft systems weighing less than 55 pounds.
Segregation	The physical separation of crewed and uncrewed aircraft on the airport surface or property, not allowing the mixing of aircraft and operations.
Taxiway Design Group (TDG)	A classification of airplanes based on outer-to-outer main gear width (MGW) and cockpit to main gear distance (CMG).
Vertiport/vertipad	Identifiable ground or elevated area, including any buildings or facilities thereon, used for the takeoff and landing of tilt-rotor aircraft and rotorcraft.

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Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
GHSA	Governors Highway Safety Association
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

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ISBN 978-0-309-68741-6



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